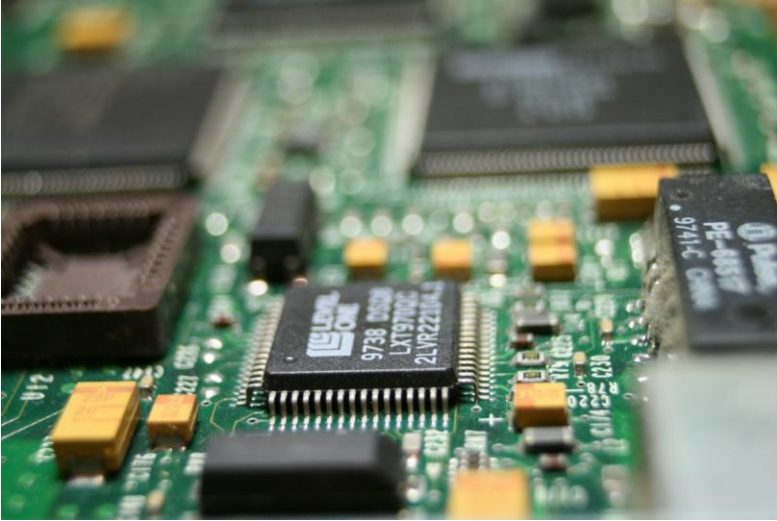




ELECTRONICS BASICS

By Great Scott

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Video 01 : Electronic Basics #1: The Multimeter

List of Components Used

1. Multimeter
 2. Resistor
 3. LED (Light Emitting Diode)
 4. Cables/Wires
 5. Fuse (Inside Multimeter)
-

Detailed Breakdown of Each Component

1. Multimeter

• What is it?

A multimeter is an electronic measuring instrument that combines several measurement functions in one unit. It can measure voltage, current, and resistance.

• Structure:

Typically, a multimeter has a digital display, a rotary switch to select measurement modes, and three input jacks:

- **COM (Common):** Connects to the black probe.
- **V Ω mA:** Connects to the red probe for measuring voltage, resistance, and current up to a certain limit (usually 200mA).
- **10A:** Connects to the red probe for measuring current up to 10A

• What is it used for?

Used to measure electrical parameters:

- **Voltage (V):** Measures the potential difference between two points.
- **Current (A):** Measures the flow of electric charge.
- **Resistance (Ω):** Measures how much a component resists the flow of current.
- **Continuity Test:** Checks if there is a complete path for current flow

• In the project:

The multimeter is the central tool demonstrated throughout the video. It is used to:

- Measure the resistance of a resistor.
 - Check continuity in cables.
 - Measure voltage across a battery and an LED circuit.
 - Measure current in a circuit and demonstrate the importance of using the correct input jack to prevent blowing a fuse .
-

2. Resistor**• What is it?**

A resistor is a passive electrical component that limits or regulates the flow of electrical current in a circuit.

• Structure:

Resistors are typically cylindrical with two leads (pins) for connection. They have colored bands that indicate their resistance value.

• What is it used for?

Used to control current flow, divide voltages, and protect components by limiting current.

• In the project:

A 68 Kilo Ohm resistor is used to demonstrate how to measure resistance with a multimeter. The video also shows how measuring resistance incircuit can give inaccurate results due to parallel paths.

3. LED (Light Emitting Diode)**• What is it?**

An LED is a semiconductor device that emits light when an electric current passes through it.

• Structure:

LEDs have two leads:

- **Anode (+):** The longer lead, connected to the positive voltage.
- **Cathode (-):** The shorter lead, connected to the negative voltage.

• What is it used for?

Used as indicator lights, in displays, and for lighting.

• In the project:

An LED is connected to a power supply, and the video demonstrates how to measure voltage across the LED. It also shows troubleshooting when the LED doesn't light up, leading to checking the continuity of the connecting cables.

4. Cables/Wires

- **What is it?**

Conductive materials used to connect different components in a circuit.

- **Structure:**

Typically consist of a metal conductor (like copper) surrounded by insulating material.

- **What is it used for?**

Used to establish electrical connections between components.

- **In the project:**

Cables are used to connect the LED to the power supply. The video demonstrates using the multimeter's continuity function to check if the cables are intact or broken.

5. Fuse (Inside Multimeter)

- **What is it?**

A fuse is a safety device that protects electrical circuits by breaking the connection if the current exceeds a certain level.

- **Structure:**

A thin wire enclosed in a casing; when excessive current flows, the wire melts, breaking the circuit.

- **What is it used for?**

Protects devices from overcurrent conditions.

- **In the project:**

The video shows that if the multimeter's fuse blows (due to measuring current beyond its capacity), it needs to be replaced. It demonstrates opening the multimeter, locating the fuse, and replacing it with one of the same rating.

Summary

Provides an introduction to using a digital multimeter (DMM), an essential tool for measuring electrical properties in circuits. It covers the basics of measuring voltage, current, and resistance, explaining how to select the appropriate settings and connect the multimeter to components. The video also touches on advanced features like auto-ranging and continuity testing, which simplify the measurement process.

Video 02 : Electronic Basics #2: Dimming all kinds of LEDs!?

List of Components Used

1. Standard LED (Green, 3.2V Forward Voltage)
 2. Resistors (Various Values: 10K Ω , etc.)
 3. Potentiometer (For Adjustable Brightness Control)
 4. Arduino Nano (For PWM Signal Generation)
 5. 555 Timer IC (For PWM Without Microcontroller)
 6. MOSFET (For High-Power LED Control)
 7. Oscilloscope (For Signal Analysis)
 8. LED Strips (For High-Power Applications)
 9. Bench Power Supply (For Voltage Testing)
 10. PWM Signal Generator (For Frequency Control)
-

Detailed Breakdown of Each Component

1. Standard LED (Green, 3.2V Forward Voltage)

- **What is it?**
A light-emitting diode used for brightness control testing.
 - **Structure:**
 - Two terminals: Anode (+) and Cathode (-)
 - **What is it used for?**
 - Demonstrating PWM-based dimming
 - **In the project:**
 - Used to visualize brightness changes with PWM signals
-

2. Resistors (Various Values: 10K Ω , etc.)

- **What are they?**
Components that limit current flow and set reference values.
- **Structure:**
 - Fixed resistance values
- **What are they used for?**
 - Preventing excessive current flow in LED circuits
- **In the project:**

- Used to stabilize LED brightness control
-

3. Potentiometer (For Adjustable Brightness Control)

- **What is it?**
A variable resistor used to fine-tune circuit parameters.
 - **Structure:**
 - Three terminals
 - Adjustable resistance
 - **What is it used for?**
 - Modifying PWM duty cycle dynamically
 - **In the project:**
 - Used to adjust LED brightness via PWM control
-

4. Arduino Nano (For PWM Signal Generation)

- **What is it?**
A compact microcontroller used for digital control.
 - **Structure:**
 - 14 digital I/O pins
 - **What is it used for?**
 - Generating PWM signals for LED dimming
 - **In the project:**
 - Used to automate LED brightness control
-

5. 555 Timer IC (For PWM Without Microcontroller)

- **What is it?**
A circuit that generates pulse-width modulation signals.
 - **Structure:**
 - Adjustable frequency output
 - **What is it used for?**
 - Creating PWM signals for LED dimming without a microcontroller
 - **In the project:**
 - Used to control LED brightness with a simple circuit
-

6. MOSFET (For High-Power LED Control)

- **What is it?**
A transistor used for fast electronic switching.
 - **Structure:**
 - Gate, Drain, and Source terminals
 - **What is it used for?**
 - Controlling high-power LEDs efficiently
 - **In the project:**
 - Used to switch LED strips with PWM signals
-

7. Oscilloscope (For Signal Analysis)

- **What is it?**
A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing PWM signal behavior
 - **In the project:**
 - Used to analyze frequency and duty cycle variations
-

8. LED Strips (For High-Power Applications)

- **What are they?**
Arrays of LEDs used for large-scale lighting applications.
 - **Structure:**
 - Multiple LEDs connected in series or parallel
 - **What are they used for?**
 - Demonstrating PWM-based dimming on larger loads
 - **In the project:**
 - Used to test MOSFET-controlled PWM dimming
-

9. Bench Power Supply (For Voltage Testing)

- **What is it?**
A regulated power source for electronic circuits.
- **Structure:**
 - Adjustable voltage output
- **What is it used for?**
 - Providing stable voltage for LED dimming tests
- **In the project:**

- Used to verify LED brightness at different voltages

10. PWM Signal Generator (For Frequency Control)

- **What is it?**
A circuit that generates pulse-width modulation signals.
- **Structure:**
 - Adjustable frequency output
- **What is it used for?**
 - Creating precise timing signals for LED dimming
- **In the project:**
 - Used to control LED brightness with variable duty cycles



Summary

This video explores **PWM fundamentals**, covering **LED dimming**, **duty cycle control**, and **frequency modulation**. It demonstrates how **PWM signals adjust brightness**, how **MOSFETs enable high-power LED control**, and how **555 timer circuits provide microcontroller-free solutions**. The **Arduino Nano** generates **PWM signals**, while **oscilloscopes analyze frequency behavior**.

Video 03 : Electronic Basics #3: Programming an Attiny+Homemade Arduino Shield



List of Components Used

1. ATtiny85 Microcontroller
 2. Arduino Uno (Used as a Programmer)
 3. Resistors (Various Values: 10KΩ, etc.)
 4. Capacitors (10μF for Reset Stabilization)
 5. LED (For Blink Test)
 6. Push Button (For Input Testing)
 7. WS2801 LED Strip (For SPI Communication Test)
 8. PCB Board (For Custom Programming Shield)
 9. Male & Female Header Pins (For Easy Connections)
 10. Bit-Banged SPI Library (For ATtiny85 Compatibility)
-

Detailed Breakdown of Each Component

1. ATtiny85 Microcontroller

- **What is it?**
A small, affordable microcontroller with limited I/O pins.
 - **Structure:**
 - 8-pin package
 - 5 I/O pins
 - **What is it used for?**
 - Running simple programs with minimal hardware
 - **In the project:**
 - Used to control an LED strip and handle button inputs
-

2. Arduino Uno (Used as a Programmer)

- **What is it?**
A widely used microcontroller board with built-in USB programming capability.
 - **Structure:**
 - ATmega328P microcontroller
 - 14 digital I/O pins
 - **What is it used for?**
 - Programming the ATtiny85 using Arduino IDE
 - **In the project:**
 - Used to upload code to the ATtiny85 via ISP
-

3. Resistors (Various Values: 10K Ω , etc.)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Stabilizing signals and preventing floating inputs
 - **In the project:**
 - Used to ensure proper operation of the ATtiny85
-

4. Capacitors (10 μ F for Reset Stabilization)

- **What are they?**
Components that store and release electrical energy.
 - **Structure:**
 - Electrostatic field storage
 - **What are they used for?**
 - Preventing unwanted resets during programming
 - **In the project:**
 - Used between Arduino reset and ground for stable programming
-

5. LED (For Blink Test)

- **What is it?**
A light-emitting diode used to visualize microcontroller output.
 - **Structure:**
 - Two terminals: Anode (+) and Cathode (-)
 - **What is it used for?**
 - Testing basic output functionality
 - **In the project:**
 - Used to verify successful programming of the ATtiny85
-

6. Push Button (For Input Testing)

- **What is it?**
A mechanical switch used to trigger digital inputs.
 - **Structure:**
 - Two terminals for signal activation
 - **What is it used for?**
 - Sending input signals to the microcontroller
 - **In the project:**
 - Used to cycle through LED animations on the ATtiny85
-

7. WS2801 LED Strip (For SPI Communication Test)

- **What is it?**
An addressable LED strip controlled via SPI communication.
- **Structure:**
 - Individual LEDs with built-in controllers
- **What is it used for?**
 - Displaying animations controlled by the ATtiny85
- **In the project:**
 - Used to test SPI communication with the ATtiny85

8. PCB Board (For Custom Programming Shield)

- **What is it?**
A printed circuit board used for structured wiring.
 - **Structure:**
 - Copper traces for electrical connections
 - **What is it used for?**
 - Creating a dedicated programming shield for the ATtiny85
 - **In the project:**
 - Used to simplify programming connections
-

9. Male & Female Header Pins (For Easy Connections)

- **What are they?**
Electrical connectors used for interfacing components.
 - **Structure:**
 - Metal pins with plastic housing
 - **What are they used for?**
 - Establishing connections between the ATtiny85 and Arduino
 - **In the project:**
 - Used to create a reusable programming shield
-

10. Bit-Banged SPI Library (For ATtiny85 Compatibility)

- **What is it?**
A software-based SPI implementation for microcontrollers without hardware SPI.
 - **Structure:**
 - Emulates SPI communication using digital I/O pins
 - **What is it used for?**
 - Enabling SPI functionality on the ATtiny85
 - **In the project:**
 - Used to control the WS2801 LED strip with the ATtiny85
-

 Summary

This video explores **ATTiny85 programming**, covering **Arduino ISP setup**, **SPI communication**, and **LED control**. It demonstrates how **small microcontrollers handle animations**, how **custom shields simplify programming**, and how **bit-banged SPI enables LED strip control**. The **Arduino Uno** acts as a **programmer**, while **oscilloscopes analyze signal behavior**.

Video 04 : Electronic Basics #4: Arduino+Bluetooth+Android=Awesome

 List of Components Used

1. **HC-05 Bluetooth Module**
2. **Arduino Nano (For Microcontroller Control)**
3. **Resistors (Voltage Divider: $2K\Omega$ & $4.7K\Omega$)**
4. **RGB LED (Common Anode Type)**
5. **Android Smartphone (For Bluetooth Communication)**
6. **S2 Terminal App (For Sending Commands via Bluetooth)**
7. **Oscilloscope (For Signal Analysis)**
8. **Bench Power Supply (For Voltage Testing)**
9. **TX & RX Connections (For Serial Communication)**
10. **Code Uploading Setup (For Arduino Programming)**

 Detailed Breakdown of Each Component

1. HC-05 Bluetooth Module

- **What is it?**
A wireless communication module used for Bluetooth connectivity.
- **Structure:**
 - Four pins: VCC, GND, TX, RX
- **What is it used for?**
 - Sending and receiving data wirelessly
- **In the project:**
 - Used to control an RGB LED via Bluetooth commands

2. Arduino Nano (For Microcontroller Control)

- **What is it?**
A compact microcontroller used for digital control.
 - **Structure:**
 - ATmega328P microcontroller
 - 14 digital I/O pins
 - **What is it used for?**
 - Processing Bluetooth commands and controlling LEDs
 - **In the project:**
 - Used to receive Bluetooth signals and execute LED control
-

3. Resistors (Voltage Divider: 2K Ω & 4.7K Ω)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Converting 5V logic to 3.3V for Bluetooth module compatibility
 - **In the project:**
 - Used to prevent damage to the HC-05 module
-

4. RGB LED (Common Anode Type)

- **What is it?**
A multi-color LED with separate cathodes for red, green, and blue.
 - **Structure:**
 - Four terminals: Anode (+) and three cathodes (-)
 - **What is it used for?**
 - Displaying different colors based on received commands
 - **In the project:**
 - Used to demonstrate Bluetooth-controlled lighting
-

5. Android Smartphone (For Bluetooth Communication)

- **What is it?**
A mobile device used to send commands via Bluetooth.
- **Structure:**
 - Wireless communication interface
- **What is it used for?**
 - Controlling Arduino-based projects remotely
- **In the project:**

- Used to send commands to the HC-05 module
-

6. S2 Terminal App (For Sending Commands via Bluetooth)

- **What is it?**
A mobile application used for Bluetooth serial communication.
 - **Structure:**
 - ASCII-based command interface
 - **What is it used for?**
 - Sending text-based commands to the Arduino
 - **In the project:**
 - Used to control LED colors via Bluetooth
-

7. Oscilloscope (For Signal Analysis)

- **What is it?**
A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing Bluetooth signal behavior
 - **In the project:**
 - Used to analyze TX/RX communication signals
-

8. Bench Power Supply (For Voltage Testing)

- **What is it?**
A regulated power source for electronic circuits.
 - **Structure:**
 - Adjustable voltage output
 - **What is it used for?**
 - Providing stable voltage for Bluetooth module operation
 - **In the project:**
 - Used to verify HC-05 module functionality
-

9. TX & RX Connections (For Serial Communication)

- **What are they?**
Transmission and reception lines for serial data exchange.
 - **Structure:**
 - TX (Transmit) and RX (Receive) pins
 - **What are they used for?**
 - Sending and receiving data between Arduino and Bluetooth module
 - **In the project:**
 - Used to establish communication between HC-05 and Arduino
-

10. Code Uploading Setup (For Arduino Programming)

- **What is it?**
The process of transferring code to the Arduino microcontroller.
 - **Structure:**
 - USB connection to Arduino IDE
 - **What is it used for?**
 - Programming the Arduino to process Bluetooth commands
 - **In the project:**
 - Used to upload LED control logic to the Arduino
-



Summary

This video explores **Bluetooth module integration**, covering **serial communication, voltage level conversion, and LED control**. It demonstrates how **HC-05 modules receive commands**, how **Arduino processes Bluetooth signals**, and how **RGB LEDs display colors based on received data**. The **S2 Terminal app sends commands**, while **oscilloscopes analyze TX/RX signals**.

Video 05 : Electronic Basics #5: How to Multiplex



List of Components Used

1. **Arduino Nano**
2. **TLC5940 LED Driver**
3. **F9540N P-Channel MOSFETs (×5)**
4. **2K Resistor (×1)**
5. **1K Resistors (×5)**
6. **RGB LEDs (for matrix or cube)**

Detailed Breakdown of Each Component

1. Arduino Nano

- **What is it?**
A small, versatile microcontroller based on the ATmega328P, ideal for compact projects.
 - **Structure:**
 - 14 digital I/O pins (6 PWM)
 - 8 analog inputs
 - Operates at 5V
 - **What is it used for?**
 - Controlling LED matrices or cubes with multiplexing
 - Running custom animation or lighting sequences
 - **In the project:**
 - Used to control LEDs while minimizing I/O usage
-

2. TLC5940 LED Driver

- **What is it?**
A 16-channel PWM LED driver with adjustable current control.
 - **Structure:**
 - 16 output channels
 - SPI-based communication with microcontroller
 - Adjustable current setting via external resistor
 - **What is it used for?**
 - Controlling multiple LEDs with PWM signals
 - Managing brightness levels for individual LEDs
 - **In the project:**
 - Used to drive the LED matrix and cube efficiently
-

3. F9540N P-Channel MOSFETs

- **What is it?**
A type of transistor used for switching high-current loads.
- **Structure:**
 - Gate, source, and drain terminals
 - Low on-resistance for efficient switching
 - Works with pull-up resistors to control state

- **What is it used for?**
 - Switching LED rows in multiplexing setups
 - Handling higher current loads than a microcontroller can directly manage
 - **In the project:**
 - Used to control LED matrix rows through multiplexing
-

4. 2K Resistor

- **What is it?**

A resistor that limits current and sets reference values.
 - **Structure:**
 - Fixed resistance value ($2K\Omega$)
 - Standard axial-lead package
 - **What is it used for?**
 - Setting the constant current for the TLC5940
 - **In the project:**
 - Ensures a stable 20mA current per LED
-

5. 1K Resistors

- **What are they?**

Resistors used as pull-up components to prevent unwanted state changes.
 - **Structure:**
 - Fixed resistance value ($1K\Omega$)
 - Standard axial-lead package
 - **What are they used for?**
 - Keeping MOSFET gates stable in multiplexing setups
 - **In the project:**
 - Ensures proper MOSFET switching behavior
-

6. RGB LEDs (Matrix/Cube)

- **What are they?**

Light-emitting diodes capable of producing different colors by mixing red, green, and blue.
- **Structure:**
 - Three individual LED chips (Red, Green, Blue) in a single package
 - Controlled via multiplexing and PWM signals
- **What are they used for?**
 - Dynamic lighting effects

- Creating visual patterns in LED cubes or matrices
 - **In the project:**
 - Used for custom animations and text displays
-

Summary

This video explains how to efficiently control a large number of LEDs using an Arduino Nano, TLC5940 LED driver, and MOSFET multiplexing. The TLC5940 helps manage brightness, while MOSFETs enable row-by-row activation of LEDs to minimize I/O pin usage. Multiplexing allows an entire matrix or cube to display images and animations without requiring individual control of each LED simultaneously. The result is a visually engaging project with minimal hardware complexity.

Video 06 : Electronic Basics #6: Standalone Arduino Circuit

List of Components Used

1. **ATmega328P Microcontroller**
 2. **16 MHz Clock Crystal**
 3. **22 Picofarad Capacitors (×2)**
 4. **10 Kiloohm Resistor**
 5. **FTDI Chip (for USB to Serial Conversion)**
-

Detailed Breakdown of Each Component

1. ATmega328P Microcontroller

- **What is it?**

The main processing unit used in Arduino boards, capable of running standalone circuits.
- **Structure:**
 - 28-pin IC
 - Requires external clock for precise timing
 - Operates at 5V
- **What is it used for?**
 - Running embedded projects without an Arduino board
 - Controlling circuits with custom programming
- **In the project:**
 - Used as the core processor for the LED color organ circuit

2. 16 MHz Clock Crystal

- **What is it?**
A timing component that provides a stable clock signal for the microcontroller.
 - **Structure:**
 - Two pins for connection
 - Works with capacitors to stabilize frequency
 - **What is it used for?**
 - Ensuring accurate timing for microcontroller operations
 - **In the project:**
 - Connected to ATmega328P to provide an external clock signal
-

3. 22 Picofarad Capacitors

- **What are they?**
Small capacitors used to stabilize the clock crystal.
 - **Structure:**
 - Two terminals
 - Connected between crystal pins and ground
 - **What are they used for?**
 - Reducing noise and improving clock signal stability
 - **In the project:**
 - Used alongside the clock crystal for reliable timing
-

4. 10 Kiloohm Resistor

- **What is it?**
A resistor used to prevent unwanted resets of the microcontroller.
 - **Structure:**
 - Fixed resistance value (10K Ω)
 - Connected between reset pin and 5V
 - **What is it used for?**
 - Ensuring the microcontroller does not reset unexpectedly
 - **In the project:**
 - Keeps the ATmega328P running without interruptions
-

5. FTDI Chip (USB to Serial Converter)

- **What is it?**
A chip that allows programming the microcontroller via USB.
 - **Structure:**
 - RX and TX pins for serial communication
 - USB interface for connection to a computer
 - **What is it used for?**
 - Uploading code to the ATmega328P without an Arduino board
 - **In the project:**
 - Used as an alternative to the Arduino board for programming
-

Summary

This video explains how to remove the ATmega328P microcontroller from an Arduino board and embed it into a standalone circuit. By adding a clock crystal, capacitors, and a pull-up resistor, the microcontroller can function independently. Programming can be done using an FTDI chip or by temporarily placing the ATmega328P back into an Arduino board. This method allows for compact and efficient circuit designs without relying on bulky Arduino hardware.

VIDEO 7 : Electronic Basics #7: 7 Segment Display

List of Components Used

1. **7-Segment Display (LTS546A)**
 2. **SN74LS247 BCD to 7-Segment Display Driver**
 3. **SN74290 4-Bit Binary Counter**
 4. **SAA1064 LED Display Driver**
 5. **BC337 NPN Transistors (×2)**
 6. **2.2nF Capacitor (×1)**
 7. **4.7KΩ Resistor (×1)**
 8. **Arduino (for microcontroller-based control)**
-

Detailed Breakdown of Each Component

1. 7-Segment Display (LTS546A)

- **What is it?**
A numeric display consisting of seven individual LED segments plus a decimal point.
- **Structure:**

- Common anode configuration
 - Pins labeled A-G for segment control
 - Decimal point labeled DP
 - **What is it used for?**
 - Displaying numbers and letters in clocks, counters, and measurement devices
 - **In the project:**
 - Used to display time, sensor readings, or voltage values
-

2. SN74LS247 BCD to 7-Segment Display Driver

- **What is it?**

A chip that converts binary-coded decimal (BCD) inputs into signals for a 7-segment display.
 - **Structure:**
 - Active low output for common anode displays
 - Four input pins (A, B, C, D) for BCD values
 - **What is it used for?**
 - Simplifies control of 7-segment displays by handling segment activation
 - **In the project:**
 - Converts binary input into readable digits on the display
-

3. SN74290 4-Bit Binary Counter

- **What is it?**

A counter IC that increments in binary-coded decimal format.
 - **Structure:**
 - Four output pins (QA, QB, QC, QD)
 - Clock input for counting pulses
 - **What is it used for?**
 - Counting events, such as button presses or sensor triggers
 - **In the project:**
 - Used to increment displayed numbers without a microcontroller
-

4. SAA1064 LED Display Driver

- **What is it?**

A specialized IC for controlling multiple 7-segment displays via I²C communication.
- **Structure:**
 - Supports up to 16 digits
 - Multiplexes two digits at a time

- **What is it used for?**
 - Efficiently driving multiple 7-segment displays with minimal microcontroller processing
 - **In the project:**
 - Used to control four-digit displays with Arduino
-

5. *BC337 NPN Transistors*

- **What are they?**

General-purpose transistors used for switching and amplification.
 - **Structure:**
 - Three terminals: Base, Collector, and Emitter
 - Used for multiplexing display digits
 - **What are they used for?**
 - Switching between display digits in multiplexed setups
 - **In the project:**
 - Helps control multiple digits efficiently
-

6. *2.2nF Capacitor*

- **What is it?**

A capacitor used to set the multiplexing speed for the display driver.
 - **Structure:**
 - Two terminals
 - Connected to the display driver IC
 - **What is it used for?**
 - Stabilizing multiplexing timing
 - **In the project:**
 - Ensures smooth display transitions
-

7. *4.7K Ω Resistor*

- **What is it?**

A pull-up resistor used in I²C communication.
- **Structure:**
 - Fixed resistance value (4.7K Ω)
 - Connected between I²C lines and 5V
- **What is it used for?**
 - Ensuring stable communication between the Arduino and display driver
- **In the project:**

- Helps maintain reliable data transmission
-

8. Arduino (for microcontroller-based control)

- **What is it?**
A microcontroller used for programming and controlling electronic circuits.
 - **Structure:**
 - Digital and analog I/O pins
 - Supports I²C communication
 - **What is it used for?**
 - Running display animations and real-time clock functions
 - **In the project:**
 - Used to control the 7-segment display efficiently
-

Summary

This video explains how to use 7-segment displays with and without an Arduino. The SN74LS247 driver simplifies display control, while the SN74290 counter allows number increments without a microcontroller. The SAA1064 IC enables efficient multiplexing for multi-digit displays using I²C communication. Transistors help switch between digits, and additional resistors and capacitors ensure stable operation.

VIDEO 8 : Electronic Basics #8: Everything about LEDs and current limiting resistors

List of Components Used

1. **LEDs (Various Types)**
 2. **Resistors (Various Values: 29Ω, 130Ω, 300Ω, etc.)**
 3. **LM317 Voltage Regulator**
 4. **TL5940 Constant Current LED Driver**
 5. **Power Sources (9V Battery, 12V Acid Battery, Button Cells)**
-

Detailed Breakdown of Each Component

1. LEDs (Various Types)

- **What are they?**
Light-emitting diodes that convert electrical energy into light.
 - **Structure:**
 - Two terminals: Anode (+) and Cathode (-)
 - Forward voltage varies (typically 3.2V for blue LEDs)
 - **What are they used for?**
 - Indicator lights, displays, and decorative lighting
 - **In the project:**
 - Used to demonstrate proper LED handling techniques
-

2. Resistors (Various Values: 29Ω, 130Ω, 300Ω, etc.)

- **What are they?**
Components that limit current flow to protect LEDs.
 - **Structure:**
 - Fixed resistance values
 - Power rating ($\frac{1}{4}$ W resistors used in the video)
 - **What are they used for?**
 - Preventing excessive current that could damage LEDs
 - **In the project:**
 - Used to calculate proper LED current flow using Ohm's Law
-

3. LM317 Voltage Regulator

- **What is it?**
An adjustable voltage regulator used to maintain a constant current.
 - **Structure:**
 - Three terminals: Input, Output, and Adjustment
 - Works with a resistor to set output current
 - **What is it used for?**
 - Providing stable current to LEDs
 - **In the project:**
 - Used to demonstrate constant current LED driving
-

4. TL5940 Constant Current LED Driver

- **What is it?**
A specialized IC for controlling multiple LEDs with constant current.
- **Structure:**
 - 16 output channels

- Adjustable current settings
 - **What is it used for?**
 - Ensuring consistent brightness across multiple LEDs
 - **In the project:**
 - Mentioned as an alternative for efficient LED control
-

5. Power Sources (9V Battery, 12V Acid Battery, Button Cells)

- **What are they?**

Electrical sources used to power LEDs.
 - **Structure:**
 - Varying voltage levels
 - Direct current (DC) output
 - **What are they used for?**
 - Supplying energy to LED circuits
 - **In the project:**
 - Used to demonstrate different LED configurations
-

Summary

This video explains the correct way to use LEDs, covering basic resistor calculations, voltage considerations, and advanced techniques like constant current driving. It highlights the importance of using resistors to prevent LED damage and explores alternatives like the LM317 regulator and TL5940 driver for efficient LED control.

VIDEO 9 : Electronic Basics #9: Diodes & Bridge Rectifiers

List of Components Used

1. 1N4007 Diode
2. Bridge Rectifier (Made from Four Diodes)
3. Capacitor (For Smoothing DC Output)
4. Transformer (220V to 15V AC)
5. Oscilloscope (For Waveform Analysis)

Detailed Breakdown of Each Component

1. 1N4007 Diode

- **What is it?**
A general-purpose silicon rectifier diode.
 - **Structure:**
 - Standard DO-41 axial package
 - Two terminals: Anode (+) and Cathode (-)
 - Max repetitive peak reverse voltage: 1000V
 - Forward current: 1A typical
 - Forward voltage drop: ~0.7V
 - **What is it used for?**
 - Allows current to flow only in one direction, protecting circuits from reverse polarity and rectifying AC to DC.
 - **In the project:**
 - Used to prevent damage by blocking reverse polarity.
 - Forms part of the bridge rectifier to convert AC to DC.
-

2. Bridge Rectifier

- **What is it?**
A configuration of four diodes arranged to convert AC input into pulsating DC output.
 - **Structure:**
 - Four diodes connected in a bridge arrangement
 - Two AC input terminals and two DC output terminals (+ and -)
 - **What is it used for?**
 - Converts both halves of an AC sine wave into a unidirectional DC output.
 - **In the project:**
 - Demonstrates how the bridge rectifier outputs a unidirectional current regardless of input polarity, improving DC power supply quality.
-

3. Capacitor (For Smoothing DC Output)

- **What is it?**
An electronic component that stores and releases electrical energy.
- **Structure:**
 - Usually an electrolytic capacitor with two leads (positive and negative)

- Rated for voltage higher than the rectified output voltage
 - **What is it used for?**
 - To smooth out the pulsating DC voltage from the rectifier by charging and discharging to reduce voltage ripple.
 - **In the project:**
 - Added after the bridge rectifier to reduce voltage ripple and produce a more stable DC output.
-

4. Transformer (220V to 15V AC)

- **What is it?**

An electromagnetic device that steps down high voltage AC to a lower AC voltage.
 - **Structure:**
 - Primary winding (high voltage input)
 - Secondary winding (low voltage output)
 - **What is it used for?**
 - To safely reduce mains AC voltage to a lower AC voltage suitable for electronic circuits.
 - **In the project:**
 - Supplies a lower AC voltage to the bridge rectifier for conversion to DC.
-

5. Oscilloscope (For Waveform Analysis)

- **What is it?**

A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - Probes for measuring circuit signals
 - **What is it used for?**
 - Observing AC and DC voltage waveforms
 - Measuring voltage fluctuations in rectifier circuits
 - **In the project:**
 - Used to analyze AC sine waves and rectified DC output.
-

Summary

This video explains the importance of diodes in electronics, particularly in rectification and circuit protection. It demonstrates how a **bridge rectifier** converts AC to DC and how a **capacitor** smooths the output for a more stable power supply. The **oscilloscope** helps visualize voltage waveforms, showing the impact of rectification and filtering.

VIDEO 10 : Electronic Basics #10: Digital to Analog Converter (DAC)

List of Components Used

1. Resistor Ladder (10K Ω & 20K Ω Resistors)
 2. Operational Amplifier (Voltage Follower)
 3. DAC0800 IC (Digital-to-Analog Converter)
 4. PCF8591 8-Bit DAC
 5. MCP4725 12-Bit DAC
 6. Arduino Nano (For Digital Signal Generation)
 7. Speaker (For Audio Output)
 8. Capacitors (For Filtering PWM Signals)
-

Detailed Breakdown of Each Component

1. Resistor Ladder (10K Ω & 20K Ω Resistors)

- **What is it?**
A simple DAC circuit using resistors to create voltage division.
 - **Structure:**
 - 10K Ω and 20K Ω resistors arranged in a binary-weighted fashion
 - Connected to digital output pins of Arduino
 - **What is it used for?**
 - Converting digital signals into analog voltage levels
 - **In the project:**
 - Used to create a basic DAC for waveform generation
-

2. Operational Amplifier (Voltage Follower)

- **What is it?**
An amplifier that maintains voltage stability when driving a load.
- **Structure:**
 - Input, output, and power terminals
 - High input impedance, low output impedance
- **What is it used for?**

- Preventing voltage collapse when connecting a speaker
 - **In the project:**
 - Ensures stable analog output for audio signals
-

3. DAC0800 IC (*Digital-to-Analog Converter*)

- **What is it?**

A dedicated IC for converting digital signals into precise analog voltages.
 - **Structure:**
 - 8-bit resolution
 - Requires external reference voltage
 - **What is it used for?**
 - Generating smooth analog waveforms
 - **In the project:**
 - Alternative to resistor ladder DAC for better precision
-

4. PCF8591 8-Bit DAC

- **What is it?**

A DAC with integrated ADC functionality, commonly used in embedded systems.
 - **Structure:**
 - I²C communication interface
 - 8-bit resolution
 - **What is it used for?**
 - Converting digital signals into analog voltages
 - **In the project:**
 - Used for simple waveform generation
-

5. MCP4725 12-Bit DAC

- **What is it?**

A high-resolution DAC with I²C communication.
- **Structure:**
 - 12-bit resolution
 - Built-in EEPROM for storing settings
- **What is it used for?**
 - Generating precise analog signals for audio and waveform applications
- **In the project:**
 - Recommended for higher-quality DAC applications

6. Arduino Nano (For Digital Signal Generation)

- **What is it?**
A compact microcontroller used for generating digital signals.
 - **Structure:**
 - 14 digital I/O pins
 - Supports PWM and I²C communication
 - **What is it used for?**
 - Controlling DAC circuits and generating waveforms
 - **In the project:**
 - Used to send digital values to DACs for conversion
-

7. Speaker (For Audio Output)

- **What is it?**
A device that converts electrical signals into sound waves.
 - **Structure:**
 - Coil and diaphragm mechanism
 - Requires stable voltage input
 - **What is it used for?**
 - Playing generated analog signals as sound
 - **In the project:**
 - Used to test DAC-generated waveforms
-

8. Capacitors (For Filtering PWM Signals)

- **What are they?**
Components used to smooth out voltage fluctuations.
 - **Structure:**
 - Two terminals
 - Connected in low-pass filter configurations
 - **What are they used for?**
 - Filtering PWM signals to create smoother analog output
 - **In the project:**
 - Used to convert PWM signals into usable analog voltages
-

Summary

This video explains how **Digital-to-Analog Converters (DACs)** work, covering resistor ladder DACs, dedicated DAC ICs, and filtering techniques. It demonstrates how to generate waveforms using an **Arduino Nano**, convert them into analog signals using **DAC0800, PCF8591, and MCP4725**, and stabilize the output using an **operational amplifier**. The final output can be used for **audio applications**, waveform generation, and signal processing.

VIDEO 11 : Electronic Basics #11: Sending SMS with Arduino || TC 35 GSM Module

List of Components Used

1. TC35 GSM Module
 2. Arduino Uno
 3. FTDI Breakout Board
 4. SIM Card (Prepaid Recommended)
 5. MAX232 IC (RS232 Communication)
 6. Jumper Wires
 7. 5V Power Supply
-

Detailed Breakdown of Each Component

1. TC35 GSM Module

- **What is it?**
A GSM communication module used for sending SMS and making calls.
 - **Structure:**
 - Requires a SIM card for network connectivity
 - Communicates via RS232 or TTL serial interface
 - Operates at 3.3V logic levels
 - **What is it used for?**
 - Sending SMS messages
 - Automating mobile communication in embedded projects
 - **In the project:**
 - Used to send SMS notifications via Arduino
-

2. Arduino Uno

- **What is it?**
A microcontroller board based on the ATmega328P, used for controlling electronic circuits.
 - **Structure:**
 - 14 digital I/O pins
 - Supports serial communication
 - **What is it used for?**
 - Interfacing with the TC35 GSM module
 - Automating SMS sending
 - **In the project:**
 - Used to trigger SMS sending via AT commands
-

3. FTDI Breakout Board

- **What is it?**
A USB-to-serial converter used for debugging and communication.
 - **Structure:**
 - TX and RX pins for data transmission
 - USB interface for connection to a computer
 - **What is it used for?**
 - Communicating with the TC35 module via serial commands
 - **In the project:**
 - Used to test AT commands before integrating with Arduino
-

4. SIM Card (Prepaid Recommended)

- **What is it?**
A mobile network card required for GSM communication.
 - **Structure:**
 - Stores network credentials
 - Enables mobile connectivity
 - **What is it used for?**
 - Sending and receiving SMS messages
 - **In the project:**
 - Inserted into the TC35 module for network access
-

5. MAX232 IC (RS232 Communication)

- **What is it?**
A voltage level converter for RS232 serial communication.

- **Structure:**
 - Converts TTL signals to RS232 levels
 - Requires external capacitors for operation
 - **What is it used for?**
 - Ensuring proper voltage levels for serial communication
 - **In the project:**
 - Used for interfacing the TC35 module with external devices
-

6. Jumper Wires

- **What are they?**

Electrical wires used for connecting components.
 - **Structure:**
 - Flexible insulated conductors
 - Various lengths and connector types
 - **What are they used for?**
 - Establishing connections between Arduino and TC35 module
 - **In the project:**
 - Used to automate the module's startup process
-

7. 5V Power Supply

- **What is it?**

A regulated power source for electronic circuits.
 - **Structure:**
 - Provides stable 5V output
 - Can be battery or adapter-based
 - **What is it used for?**
 - Powering the TC35 GSM module and Arduino
 - **In the project:**
 - Ensures stable operation of the GSM module
-

Summary

This video explains how to set up and use the **TC35 GSM module** with an **Arduino Uno** to send SMS messages. It covers **powering the module**, **connecting a SIM card**, **using AT commands**, and **automating the startup process**. The **FTDI breakout board** helps debug serial communication, while the **MAX232 IC** ensures proper voltage levels. The final setup allows for **automated SMS notifications**, which can be adapted for alarm systems or other applications.

VIDEO 12 : Electronic Basics #12: Coils / Inductors (Part 1)

List of Components Used

1. Inductor (Coil)
2. Electromagnet (Iron Core Coil)
3. Relay (Electromagnetic Switch)
4. Transformer (Two Coils for Voltage Conversion)
5. DC Motor (Uses Coils for Rotation)
6. AC Motor (Uses Coils for Induction)
7. Flyback Diode (For Protection Against Voltage Spikes)
8. Boost Converter Circuit (Uses Inductor for Voltage Step-Up)
9. Step-Down Converter (Uses Inductor for Voltage Regulation)
10. RLC Meter (For Measuring Inductance)

Detailed Breakdown of Each Component

1. Inductor (Coil)

- **What is it?**
A passive electronic component that stores energy in a magnetic field when current flows through it.
- **Structure:**
 - Wire wound into a coil
 - Can have an air or iron core
- **What is it used for?**
 - Filtering signals in circuits
 - Energy storage in power supplies
- **In the project:**
 - Demonstrates how inductors resist changes in current flow

2. Electromagnet (Iron Core Coil)

- **What is it?**
A coil wrapped around a ferromagnetic core to enhance its magnetic field.
- **Structure:**
 - Coil wound around an iron core
 - Generates a strong magnetic field when powered
- **What is it used for?**

- Used in relays and motors for switching and movement
 - **In the project:**
 - Shows how inductors can be used to create electromagnets
-

3. Relay (Electromagnetic Switch)

- **What is it?**

A switch that uses an electromagnet to control high-power circuits.
 - **Structure:**
 - Coil, armature, and contacts
 - Activated by an electric current
 - **What is it used for?**
 - Switching high-current loads using low-power signals
 - **In the project:**
 - Demonstrates how inductors are used in relays
-

4. Transformer (Two Coils for Voltage Conversion)

- **What is it?**

A device that transfers electrical energy between circuits using electromagnetic induction.
 - **Structure:**
 - Primary and secondary coils
 - Works on AC voltage
 - **What is it used for?**
 - Stepping up or stepping down voltage levels
 - **In the project:**
 - Used to show AC voltage transformation
-

5. DC Motor (Uses Coils for Rotation)

- **What is it?**

A motor that converts electrical energy into mechanical motion using coils.
- **Structure:**
 - Rotor with coils
 - Requires a power source
- **What is it used for?**
 - Driving mechanical systems in robotics and automation
- **In the project:**
 - Used to demonstrate electromagnetic induction in motion

6. AC Motor (Uses Coils for Induction)

- **What is it?**
A motor that operates on alternating current using electromagnetic induction.
 - **Structure:**
 - Stator and rotor with coils
 - Works with AC power
 - **What is it used for?**
 - Industrial applications requiring continuous rotation
 - **In the project:**
 - Used to show how AC signals interact with inductors
-

7. Flyback Diode (For Protection Against Voltage Spikes)

- **What is it?**
A diode used to protect circuits from voltage spikes caused by inductors.
 - **Structure:**
 - Connected across an inductor or motor
 - Allows current to flow in reverse when needed
 - **What is it used for?**
 - Preventing damage to transistors and switches
 - **In the project:**
 - Used to protect circuits from inductor-generated voltage spikes
-

8. Boost Converter Circuit (Uses Inductor for Voltage Step-Up)

- **What is it?**
A circuit that increases voltage using an inductor.
 - **Structure:**
 - Inductor, switch, and capacitor
 - Converts low voltage to higher voltage
 - **What is it used for?**
 - Powering devices that require higher voltage than the battery provides
 - **In the project:**
 - Used to demonstrate energy storage and conversion
-

9. Step-Down Converter (Uses Inductor for Voltage Regulation)

- **What is it?**
A circuit that reduces voltage using an inductor.
 - **Structure:**
 - Inductor, capacitor, and switching transistor
 - Converts high voltage to lower voltage
 - **What is it used for?**
 - Providing stable voltage for sensitive electronics
 - **In the project:**
 - Used to show how inductors regulate voltage
-

10. RLC Meter (For Measuring Inductance)

- **What is it?**
A device used to measure resistance, inductance, and capacitance.
 - **Structure:**
 - Digital display
 - Connects to components for measurement
 - **What is it used for?**
 - Determining the inductance value of coils
 - **In the project:**
 - Used to measure inductance in different coil setups
-

Summary

This video explains the fundamentals of **inductors**, their role in **electromagnetic fields**, and their applications in **power supplies, motors, and signal filtering**. It covers **boost converters, step-down regulators, and flyback diodes** for circuit protection. The demonstration highlights how inductors resist changes in current flow and store energy for efficient power conversion.

VIDEO 13: Electronic Basics #13: Coils / Inductors (Part 2) **|| Reactance**

List of Components Used

1. **Inductor (Coil)**
2. **LED**

3. **230V to 15V RMS Transformer**
 4. **Resistors (Various Values: 33Ω , 100Ω , 10Ω , $5K\Omega$)**
 5. **Function Generator**
 6. **LTSpice Simulation Software**
 7. **MOSFET (N-Channel)**
 8. **Microwave Motor (Inductive Load)**
 9. **Transistor Tester (For Measuring Inductance & Resistance)**
-

Detailed Breakdown of Each Component

1. Inductor (Coil)

- **What is it?**
A passive electronic component that stores energy in a magnetic field when current flows through it.
 - **Structure:**
 - Wire wound into a coil
 - Can have an air or iron core
 - **What is it used for?**
 - Filtering signals in circuits
 - Energy storage in power supplies
 - **In the project:**
 - Demonstrates how inductors resist changes in current flow and affect AC signals
-

2. LED

- **What is it?**
A light-emitting diode that converts electrical energy into light.
 - **Structure:**
 - Two terminals: Anode (+) and Cathode (-)
 - Requires a current-limiting resistor
 - **What is it used for?**
 - Indicator lights, displays, and decorative lighting
 - **In the project:**
 - Used to demonstrate how inductors affect current flow in AC circuits
-

3. 230V to 15V RMS Transformer

- **What is it?**
A device that steps down high-voltage AC to a lower AC voltage.
 - **Structure:**
 - Primary and secondary coils
 - Works on AC voltage
 - **What is it used for?**
 - Providing a safe AC voltage for experiments
 - **In the project:**
 - Supplies AC voltage for testing inductors and reactance
-

4. Resistors (Various Values: 33Ω, 100Ω, 10Ω, 5KΩ)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - Power rating varies
 - **What are they used for?**
 - Controlling current in circuits
 - Creating frequency filters
 - **In the project:**
 - Used to compare inductive reactance with pure resistance
-

5. Function Generator

- **What is it?**
A device that generates different types of electrical waveforms.
 - **Structure:**
 - Adjustable frequency and amplitude
 - Outputs sine, square, and triangle waves
 - **What is it used for?**
 - Testing circuit response to different frequencies
 - **In the project:**
 - Used to analyze how inductors affect AC signals at various frequencies
-

6. LTSpice Simulation Software

- **What is it?**
A circuit simulation tool used for analyzing electronic designs.
- **Structure:**

- Graphical interface for circuit modeling
 - Supports AC and DC analysis
 - **What is it used for?**
 - Simulating circuit behavior before physical testing
 - **In the project:**
 - Used to model high-pass and low-pass filters with inductors
-

7. MOSFET (N-Channel)

- **What is it?**

A type of transistor used for switching and amplification.
 - **Structure:**
 - Three terminals: Gate, Drain, and Source
 - High-speed switching capability
 - **What is it used for?**
 - Controlling high-frequency signals in circuits
 - **In the project:**
 - Used to detect high-frequency audio signals in a filter circuit
-

8. Microwave Motor (Inductive Load)

- **What is it?**

A motor that operates using electromagnetic induction.
 - **Structure:**
 - Stator and rotor with coils
 - Works with AC power
 - **What is it used for?**
 - Driving mechanical systems in appliances
 - **In the project:**
 - Used to demonstrate phase shift in inductive loads
-

9. Transistor Tester (For Measuring Inductance & Resistance)

- **What is it?**

A device used to measure resistance, inductance, and capacitance.
- **Structure:**
 - Digital display
 - Connects to components for measurement
- **What is it used for?**
 - Determining the inductance value of coils

- **In the project:**
 - Used to measure inductance in different coil setups
-

Summary

This video explores **inductive reactance**, **phase shift**, and **frequency filtering** using inductors. It demonstrates how inductors resist changes in current flow, affect AC signals, and store energy in magnetic fields. The experiments highlight **reactive power**, **high-pass and low-pass filters**, and **phase shifts in inductive loads**. The **transistor tester** provides an affordable way to measure inductance, making it a useful tool for electronics enthusiasts.

VIDEO 14: Electronic Basics #14: Capacitors

List of Components Used

1. **Aluminum Sheet (For DIY Capacitor)**
 2. **Electrolytic Capacitor**
 3. **Resistors (Various Values: 33Ω, 100Ω, 5KΩ)**
 4. **Distilled Water (As Dielectric Material)**
 5. **5V Power Supply**
 6. **555 Timer IC**
 7. **Microwave Motor (Inductive Load)**
 8. **Function Generator**
 9. **LTSpice Simulation Software**
-

Detailed Breakdown of Each Component

1. Aluminum Sheet (For DIY Capacitor)

- **What is it?**
A conductive material used to create a basic capacitor.
- **Structure:**
 - Two metal plates positioned close together
 - Electrostatic field forms between them
- **What is it used for?**
 - Demonstrating capacitor fundamentals
- **In the project:**
 - Used to build a simple capacitor and measure its capacitance

2. Electrolytic Capacitor

- **What is it?**
A capacitor with a high capacitance value, commonly used in power supplies.
 - **Structure:**
 - Metal films with dielectric material
 - Polarized terminals (Anode and Cathode)
 - **What is it used for?**
 - Filtering voltage fluctuations
 - Energy storage in circuits
 - **In the project:**
 - Used to demonstrate real-world capacitor behavior
-

3. Resistors (Various Values: 33Ω , 100Ω , $5K\Omega$)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - Power rating varies
 - **What are they used for?**
 - Controlling current in circuits
 - Creating frequency filters
 - **In the project:**
 - Used to compare capacitive reactance with pure resistance
-

4. Distilled Water (As Dielectric Material)

- **What is it?**
A non-conductive liquid used to enhance capacitance.
 - **Structure:**
 - H_2O dipoles align with electrostatic fields
 - Increases force on electrons
 - **What is it used for?**
 - Improving capacitor efficiency
 - **In the project:**
 - Used to boost capacitance in a DIY capacitor
-

5. 5V Power Supply

- **What is it?**
A regulated power source for electronic circuits.
 - **Structure:**
 - Provides stable 5V output
 - Can be battery or adapter-based
 - **What is it used for?**
 - Powering capacitors and circuits
 - **In the project:**
 - Used to charge capacitors and observe their behavior
-

6. 555 Timer IC

- **What is it?**
A versatile integrated circuit used for timing applications.
 - **Structure:**
 - Three modes: Astable, Monostable, Bistable
 - Generates square wave signals
 - **What is it used for?**
 - Creating oscillating signals
 - Timing delays in circuits
 - **In the project:**
 - Used to demonstrate capacitor charging and discharging
-

7. Microwave Motor (Inductive Load)

- **What is it?**
A motor that operates using electromagnetic induction.
 - **Structure:**
 - Stator and rotor with coils
 - Works with AC power
 - **What is it used for?**
 - Driving mechanical systems in appliances
 - **In the project:**
 - Used to demonstrate phase shift and reactive power compensation
-

8. Function Generator

- **What is it?**
A device that generates different types of electrical waveforms.
 - **Structure:**
 - Adjustable frequency and amplitude
 - Outputs sine, square, and triangle waves
 - **What is it used for?**
 - Testing circuit response to different frequencies
 - **In the project:**
 - Used to analyze how capacitors affect AC signals at various frequencies
-

9. LTSpice Simulation Software

- **What is it?**
A circuit simulation tool used for analyzing electronic designs.
 - **Structure:**
 - Graphical interface for circuit modeling
 - Supports AC and DC analysis
 - **What is it used for?**
 - Simulating circuit behavior before physical testing
 - **In the project:**
 - Used to model high-pass and low-pass filters with capacitors
-

Summary

This video explores **capacitor fundamentals**, **reactance**, and **frequency filtering** using capacitors. It demonstrates how capacitors store energy, regulate voltage, and filter signals in circuits. The experiments highlight **capacitive reactance**, **phase shift compensation**, and **high-pass/low-pass filters**. The **555 Timer IC** is used to showcase capacitor charging behavior, while the **function generator** helps analyze frequency response.

VIDEO 15: Electronic Basics #15 : Temperature Measurement (Part 1) || NTC, PT100, Wheatstone Bridge

List of Components Used

1. NTC Thermistor (Negative Temperature Coefficient)
2. PT100 RTD (Resistance Temperature Detector)

3. **LM317 Voltage Regulator**
 4. **Wheatstone Bridge Circuit**
 5. **Operational Amplifier (Op-Amp for Signal Processing)**
 6. **Pre-Made Temperature Transmitter**
 7. **24V Power Supply**
 8. **250 Ω Resistor (For Current Measurement)**
 9. **16x2 LCD Display**
 10. **Microcontroller (Arduino or Similar)**
 11. **LM35 Temperature Sensor**
 12. **DS18B20 Digital Temperature Sensor**
-

Detailed Breakdown of Each Component

1. NTC Thermistor (Negative Temperature Coefficient)

- **What is it?**
A resistor that decreases its resistance as temperature increases.
 - **Structure:**
 - Available in different resistance values (1K Ω , 10K Ω , 100K Ω)
 - Non-linear response curve
 - **What is it used for?**
 - Measuring temperature in consumer electronics and industrial applications
 - **In the project:**
 - Used to demonstrate basic temperature measurement principles
-

2. PT100 RTD (Resistance Temperature Detector)

- **What is it?**
A temperature sensor with a nominal resistance of 100 Ω at 0°C.
 - **Structure:**
 - Linear resistance-temperature relationship
 - Can measure up to 850°C
 - **What is it used for?**
 - Industrial temperature measurement with high accuracy
 - **In the project:**
 - Used for precise temperature readings in the DIY thermometer
-

3. LM317 Voltage Regulator

- **What is it?**
A voltage regulator used to provide a stable current source.
 - **Structure:**
 - Adjustable output voltage
 - Requires external resistors for setting current
 - **What is it used for?**
 - Supplying a low constant current for PT100 measurement
 - **In the project:**
 - Ensures accurate resistance measurement without excessive heating
-

4. Wheatstone Bridge Circuit

- **What is it?**
A circuit used to measure unknown resistance values precisely.
 - **Structure:**
 - Four resistors arranged in a bridge configuration
 - Measures voltage difference between two points
 - **What is it used for?**
 - Eliminating offset voltage in PT100 measurements
 - **In the project:**
 - Used to improve accuracy in temperature readings
-

5. Operational Amplifier (Op-Amp for Signal Processing)

- **What is it?**
An amplifier used to process small voltage signals.
 - **Structure:**
 - High input impedance
 - Can be configured as a differential amplifier
 - **What is it used for?**
 - Amplifying small temperature-related voltage changes
 - **In the project:**
 - Used to process PT100 signals before sending them to the microcontroller
-

6. Pre-Made Temperature Transmitter

- **What is it?**
A ready-made module that converts temperature readings into a standard signal.
- **Structure:**
 - Outputs a current signal (4mA to 20mA)

- Can be used with two-wire or three-wire configurations
 - **What is it used for?**
 - Simplifying temperature measurement without complex circuits
 - **In the project:**
 - Used to provide an easy-to-read temperature signal
-

7. 24V Power Supply

- **What is it?**

A regulated power source for industrial sensors.
 - **Structure:**
 - Provides stable 24V output
 - Can be battery or adapter-based
 - **What is it used for?**
 - Powering the temperature transmitter
 - **In the project:**
 - Ensures stable operation of the PT100 measurement system
-

8. 250Ω Resistor (For Current Measurement)

- **What is it?**

A resistor used to convert current signals into voltage readings.
 - **Structure:**
 - Fixed resistance value (250Ω)
 - Used in series with the transmitter
 - **What is it used for?**
 - Converting 4-20mA signals into voltage for microcontroller input
 - **In the project:**
 - Helps translate temperature readings into usable voltage values
-

9. 16x2 LCD Display

- **What is it?**

A small screen used to display temperature readings.
- **Structure:**
 - Two rows of 16 characters
 - Controlled via microcontroller
- **What is it used for?**
 - Showing real-time temperature values
- **In the project:**

- Displays processed temperature data from the microcontroller
-

10. Microcontroller (Arduino or Similar)

- **What is it?**
A programmable device used to process sensor data.
 - **Structure:**
 - Digital and analog I/O pins
 - Supports ADC for voltage measurement
 - **What is it used for?**
 - Handling temperature calculations and display output
 - **In the project:**
 - Converts voltage readings into temperature values
-

11. LM35 Temperature Sensor

- **What is it?**
A simple analog temperature sensor with a linear voltage output.
 - **Structure:**
 - Outputs 10mV per °C
 - Operates from 0°C to 150°C
 - **What is it used for?**
 - Measuring temperature with minimal external components
 - **In the project:**
 - Alternative to PT100 for simpler applications
-

12. DS18B20 Digital Temperature Sensor

- **What is it?**
A digital temperature sensor with a one-wire interface.
 - **Structure:**
 - Provides direct temperature readings
 - Operates from -55°C to 125°C
 - **What is it used for?**
 - High-accuracy temperature measurement with minimal wiring
 - **In the project:**
 - Alternative to analog sensors for easy integration
-

Summary

This video explores **temperature measurement techniques**, covering **NTC thermistors**, **PT100 RTDs**, and **digital sensors**. It demonstrates how to build a **DIY thermometer** using a **microcontroller**, **LCD display**, and **temperature transmitter**. The **Wheatstone bridge** and **operational amplifier** help improve accuracy, while **pre-made sensors** like **LM35** and **DS18B20** simplify the process.

VIDEO 16: Electronic Basics #16: Resistors

List of Components Used

1. Resistors (Various Values: 524Ω, 680Ω, 10KΩ, 50KΩ, etc.)
2. LED (5mm Standard LED)
3. Power Source (Battery or DC Supply)
4. Potentiometer (Variable Resistor)
5. Arduino Nano
6. Push Button (For Digital Input)
7. MOSFET (For Switching Applications)
8. Differential Amplifier (For Current Measurement)
9. PT100 Temperature Sensor
10. AC Signal Generator (For Frequency Testing)

Detailed Breakdown of Each Component

1. Resistors (Various Values: 524Ω, 680Ω, 10KΩ, 50KΩ, etc.)

- **What are they?**
Components that limit current flow and set reference values.
- **Structure:**
 - Fixed resistance values
 - Power rating varies
- **What are they used for?**
 - Controlling current in circuits
 - Creating voltage dividers
- **In the project:**
 - Used for LED protection, pull-up/pull-down configurations, and current sensing

2. LED (5mm Standard LED)

- **What is it?**
A light-emitting diode that converts electrical energy into light.
 - **Structure:**
 - Two terminals: Anode (+) and Cathode (-)
 - Requires a current-limiting resistor
 - **What is it used for?**
 - Indicator lights, displays, and decorative lighting
 - **In the project:**
 - Used to demonstrate resistor-based current control
-

3. Power Source (Battery or DC Supply)

- **What is it?**
A regulated power source for electronic circuits.
 - **Structure:**
 - Provides stable voltage output
 - Can be battery or adapter-based
 - **What is it used for?**
 - Powering resistors and circuits
 - **In the project:**
 - Supplies energy for LED and resistor experiments
-

4. Potentiometer (Variable Resistor)

- **What is it?**
A resistor with an adjustable resistance value.
 - **Structure:**
 - Three terminals
 - Movable wiper for resistance adjustment
 - **What is it used for?**
 - Adjusting voltage levels dynamically
 - **In the project:**
 - Used to modify input voltage for microcontroller applications
-

5. Arduino Nano

- **What is it?**
A compact microcontroller used for digital and analog control.
 - **Structure:**
 - 14 digital I/O pins
 - Supports PWM and ADC
 - **What is it used for?**
 - Processing sensor inputs and controlling circuits
 - **In the project:**
 - Used to read resistor-based voltage dividers and control LEDs
-

6. Push Button (For Digital Input)

- **What is it?**
A simple switch used to trigger digital signals.
 - **Structure:**
 - Two terminals
 - Mechanical press activation
 - **What is it used for?**
 - Sending high/low signals to microcontrollers
 - **In the project:**
 - Used with pull-down resistors to stabilize input signals
-

7. MOSFET (For Switching Applications)

- **What is it?**
A transistor used for high-speed switching.
 - **Structure:**
 - Three terminals: Gate, Drain, Source
 - Low resistance when activated
 - **What is it used for?**
 - Controlling high-power loads efficiently
 - **In the project:**
 - Used with pull-up resistors for stable gate control
-

8. Differential Amplifier (For Current Measurement)

- **What is it?**
An amplifier used to measure voltage differences.
- **Structure:**
 - Two input terminals

- High precision gain control
 - **What is it used for?**
 - Measuring voltage drop across resistors to calculate current
 - **In the project:**
 - Used to monitor current flow in circuits
-

9. PT100 Temperature Sensor

- **What is it?**

A resistance-based temperature sensor.
 - **Structure:**
 - 100Ω resistance at 0°C
 - Linear temperature response
 - **What is it used for?**
 - Measuring temperature in industrial applications
 - **In the project:**
 - Used to demonstrate resistor-based temperature sensing
-

10. AC Signal Generator (For Frequency Testing)

- **What is it?**

A device that generates AC waveforms.
 - **Structure:**
 - Adjustable frequency and amplitude
 - Outputs sine, square, and triangle waves
 - **What is it used for?**
 - Testing resistor behavior in AC circuits
 - **In the project:**
 - Used to analyze parasitic inductance and capacitance effects
-



Summary

This video explores **resistor fundamentals**, **voltage dividers**, **pull-up/pull-down configurations**, and **current sensing**. It demonstrates how resistors regulate current, stabilize signals, and interact with AC frequencies. The **Arduino Nano** processes resistor-based inputs, while the **MOSFET** and **differential amplifier** enable advanced applications.

VIDEO 17: Electronic Basics #17: Oscillators || RC, LC, Crystal

List of Components Used

1. RC Oscillator (Resistor-Capacitor Circuit)
2. 555 Timer IC
3. LC Tank Circuit (Inductor-Capacitor Resonator)
4. NPN Transistor (For Amplification)
5. Crystal Oscillator (Piezoelectric Resonator)
6. LEDs (For Visualizing Oscillations)
7. Capacitors (Various Values: 100nF, etc.)
8. Resistors (Various Values: 680Ω, 150KΩ, etc.)
9. Potentiometer (For Frequency Adjustment)
10. Microcontroller (For Clock Signal Processing)

Detailed Breakdown of Each Component

1. RC Oscillator (Resistor-Capacitor Circuit)

- **What is it?**
A simple oscillator that generates periodic signals using resistors and capacitors.
- **Structure:**
 - Uses charging and discharging cycles
 - Produces square or triangle waveforms
- **What is it used for?**
 - Low-frequency signal generation
 - Timing circuits
- **In the project:**
 - Used to demonstrate basic oscillation principles

2. 555 Timer IC

- **What is it?**
A versatile integrated circuit used for generating precise timing signals.
- **Structure:**
 - Two comparators, flip-flop, and logic gates
 - Adjustable frequency output

- **What is it used for?**
 - Creating stable square wave signals
 - Pulse-width modulation (PWM)
 - **In the project:**
 - Used to generate adjustable oscillations
-

3. LC Tank Circuit (Inductor-Capacitor Resonator)

- **What is it?**

A circuit that oscillates at a specific resonance frequency using inductors and capacitors.
 - **Structure:**
 - Stores energy in magnetic and electrostatic fields
 - Produces sine wave signals
 - **What is it used for?**
 - High-frequency oscillation
 - Radio frequency applications
 - **In the project:**
 - Used to generate stable sine waves
-

4. NPN Transistor (For Amplification)

- **What is it?**

A transistor used to amplify weak signals.
 - **Structure:**
 - Three terminals: Base, Collector, Emitter
 - Controls current flow
 - **What is it used for?**
 - Boosting oscillator signals
 - Maintaining stable oscillations
 - **In the project:**
 - Used to sustain LC circuit oscillations
-

5. Crystal Oscillator (Piezoelectric Resonator)

- **What is it?**

A highly stable oscillator using mechanical vibrations of a crystal.
- **Structure:**
 - Quartz crystal with electrodes
 - Generates precise clock signals
- **What is it used for?**

- Microcontroller clock generation
 - Radio frequency applications
 - **In the project:**
 - Used to provide a stable 16 MHz clock signal
-

6. LEDs (For Visualizing Oscillations)

- **What are they?**
Light-emitting diodes used to indicate circuit activity.
 - **Structure:**
 - Two terminals: Anode (+) and Cathode (-)
 - Requires a current-limiting resistor
 - **What are they used for?**
 - Displaying oscillation cycles visually
 - **In the project:**
 - Used to show waveform transitions in RC circuits
-

7. Capacitors (Various Values: 100nF, etc.)

- **What are they?**
Components that store and release electrical energy.
 - **Structure:**
 - Two terminals
 - Electrostatic field storage
 - **What are they used for?**
 - Timing circuits
 - Frequency control in oscillators
 - **In the project:**
 - Used in RC and LC oscillators
-

8. Resistors (Various Values: 680Ω, 150KΩ, etc.)

- **What are they?**
Components that limit current flow and set reference values.
- **Structure:**
 - Fixed resistance values
 - Power rating varies
- **What are they used for?**
 - Controlling capacitor charge rates
 - Adjusting oscillator frequency

- **In the project:**
 - Used in RC and LC oscillators
-

9. Potentiometer (For Frequency Adjustment)

- **What is it?**
A variable resistor used to fine-tune circuit parameters.
 - **Structure:**
 - Three terminals
 - Adjustable resistance
 - **What is it used for?**
 - Modifying oscillator frequency dynamically
 - **In the project:**
 - Used to adjust 555 timer output frequency
-

10. Microcontroller (For Clock Signal Processing)

- **What is it?**
A programmable device used to process electronic signals.
 - **Structure:**
 - Digital and analog I/O pins
 - Supports external clock sources
 - **What is it used for?**
 - Synchronizing operations in embedded systems
 - **In the project:**
 - Used to process oscillator-generated clock signals
-

Summary

This video explores **oscillator fundamentals**, covering **RC oscillators**, **LC tank circuits**, and **crystal oscillators**. It demonstrates how oscillators generate **square, sine, and triangle waves**, and how they are used in **microcontrollers**, **timing circuits**, and **radio communication**. The **555 timer IC** provides adjustable oscillations, while **LC circuits** enable high-frequency stability.

VIDEO 18: Electronic Basics #18: : DC & Brushless DC Motor + ESC

List of Components Used

1. **Brushless DC Motor (BLDC)**
 2. **Electronic Speed Controller (ESC)**
 3. **Permanent Magnets (Rotor)**
 4. **Coils (Stator)**
 5. **MOSFETs (For Switching Control)**
 6. **Tachometer (For RPM Measurement)**
 7. **7.4V & 11.1V Batteries**
 8. **PWM Signal Generator**
-

Detailed Breakdown of Each Component

1. Brushless DC Motor (BLDC)

- **What is it?**
A motor that operates without brushes, using electronic commutation.
 - **Structure:**
 - Rotor with permanent magnets
 - Stator with multiple coils
 - **What is it used for?**
 - High-efficiency motion control in drones, electric vehicles, and hard drives
 - **In the project:**
 - Used to drive an electric longboard
-

2. Electronic Speed Controller (ESC)

- **What is it?**
A circuit that controls the speed and direction of a BLDC motor.
 - **Structure:**
 - MOSFET-based switching
 - PWM signal input
 - **What is it used for?**
 - Adjusting motor speed based on input signals
 - **In the project:**
 - Used to regulate the motor's RPM
-

3. Permanent Magnets (Rotor)

- **What are they?**
Magnets that create the rotating magnetic field in a BLDC motor.
 - **Structure:**
 - Alternating polarity arrangement
 - **What are they used for?**
 - Generating motion through electromagnetic interaction
 - **In the project:**
 - Used to drive rotation in the motor
-

4. Coils (Stator)

- **What are they?**
Wound copper wires that generate electromagnetic fields.
 - **Structure:**
 - Arranged in multiple phases
 - **What are they used for?**
 - Creating rotating magnetic fields to drive the rotor
 - **In the project:**
 - Used to control motor movement
-

5. MOSFETs (For Switching Control)

- **What are they?**
Transistors used for high-speed switching in ESCs.
 - **Structure:**
 - P-channel and N-channel configurations
 - **What are they used for?**
 - Controlling current flow to motor coils
 - **In the project:**
 - Used to regulate motor speed and torque
-

6. Tachometer (For RPM Measurement)

- **What is it?**
A device used to measure rotational speed.
- **Structure:**
 - Optical or mechanical sensing
- **What is it used for?**
 - Determining motor speed in real-time
- **In the project:**

- Used to verify motor RPM calculations
-

7. 7.4V & 11.1V Batteries

- **What are they?**
Power sources for the BLDC motor and ESC.
 - **Structure:**
 - Lithium-ion or lithium-polymer cells
 - **What are they used for?**
 - Providing energy for motor operation
 - **In the project:**
 - Used to test motor performance at different voltages
-

8. PWM Signal Generator

- **What is it?**
A circuit that generates pulse-width modulation signals.
 - **Structure:**
 - Adjustable duty cycle
 - **What is it used for?**
 - Controlling motor speed via ESC
 - **In the project:**
 - Used to regulate motor RPM
-



Summary

This video explores **brushless DC motors**, **electronic speed controllers**, and **PWM-based motor control**. It demonstrates how **MOSFETs regulate current**, how **permanent magnets drive rotation**, and how **ESCs adjust speed dynamically**. The **KV rating** determines RPM per volt, while **tachometers verify speed measurements**.

VIDEO 19: Electronic Basics #19: I2C and how to use it



List of Components Used

1. **Arduino Nano (Master Device)**
2. **TEA5767 FM Radio IC (Slave Device)**
3. **Stripboard (For Breakout Board)**
4. **Male Header Pins (For Connections)**

5. **Silver Wire (For Circuit Traces)**
 6. **10K Ω Resistors (Pull-Up Resistors for I²C Lines)**
 7. **Oscilloscope (For Signal Analysis)**
 8. **Amplifier & Speaker (For Audio Output)**
 9. **Antenna (For FM Signal Reception)**
 10. **Headphones (For Direct Audio Output)**
-

Detailed Breakdown of Each Component

1. Arduino Nano (Master Device)

- **What is it?**
A compact microcontroller used for digital and analog control.
 - **Structure:**
 - 14 digital I/O pins
 - Supports I²C communication
 - **What is it used for?**
 - Sending and receiving data via I²C protocol
 - **In the project:**
 - Acts as the master device controlling the TEA5767 FM radio IC
-

2. TEA5767 FM Radio IC (Slave Device)

- **What is it?**
An integrated circuit used for FM radio reception.
 - **Structure:**
 - Supports I²C communication
 - Outputs audio signals
 - **What is it used for?**
 - Receiving FM radio signals and sending audio output
 - **In the project:**
 - Controlled via I²C to tune into specific radio frequencies
-

3. Stripboard (For Breakout Board)

- **What is it?**
A perforated board used for prototyping circuits.
- **Structure:**

- Copper traces for electrical connections
 - **What is it used for?**
 - Mounting and wiring the TEA5767 IC
 - **In the project:**
 - Used to create a breakout board for easier connections
-

4. Male Header Pins (For Connections)

- **What are they?**

Electrical connectors used for interfacing components.
 - **Structure:**
 - Metal pins with plastic housing
 - **What are they used for?**
 - Establishing connections between the breakout board and Arduino
 - **In the project:**
 - Used to connect the TEA5767 IC to the Arduino Nano
-

5. Silver Wire (For Circuit Traces)

- **What is it?**

Conductive wire used for making electrical connections.
 - **Structure:**
 - Thin metallic strands
 - **What is it used for?**
 - Creating circuit paths on the breakout board
 - **In the project:**
 - Used to wire the TEA5767 IC to the header pins
-

6. 10K Ω Resistors (Pull-Up Resistors for I²C Lines)

- **What are they?**

Resistors used to stabilize I²C communication lines.
 - **Structure:**
 - Fixed resistance value (10K Ω)
 - **What are they used for?**
 - Ensuring proper voltage levels for I²C signals
 - **In the project:**
 - Used to pull the SDA and SCL lines to 5V
-

7. Oscilloscope (For Signal Analysis)

- **What is it?**
A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing I²C communication signals
 - **In the project:**
 - Used to analyze data transmission between Arduino and TEA5767
-

8. Amplifier & Speaker (For Audio Output)

- **What are they?**
Devices used to amplify and play audio signals.
 - **Structure:**
 - Amplifier boosts signal strength
 - Speaker converts electrical signals into sound
 - **What are they used for?**
 - Enhancing FM radio audio output
 - **In the project:**
 - Used to make the received radio signal audible
-

9. Antenna (For FM Signal Reception)

- **What is it?**
A conductive element used to receive radio waves.
 - **Structure:**
 - Metal wire or rod
 - **What is it used for?**
 - Capturing FM radio signals
 - **In the project:**
 - Connected to the TEA5767 IC for better reception
-

10. Headphones (For Direct Audio Output)

- **What are they?**
Audio devices used for private listening.
- **Structure:**
 - Speakers enclosed in ear cups

- **What are they used for?**
 - Listening to FM radio output directly
 - **In the project:**
 - Used to test the TEA5767 IC's audio output
-

Summary

This video explores **I²C communication**, covering **master-slave interactions, pull-up resistors, and data transmission**. It demonstrates how **Arduino Nano controls the TEA5767 FM radio IC**, how **oscilloscopes analyze signal timing**, and how **audio output is processed through amplifiers and speakers**.

VIDEO 20: Electronic Basics #20: Thyristor, Triac || Phase Angle Control

List of Components Used

1. Thyristor (TYN604)
 2. Triac (For AC Control)
 3. MOSFET (For Switching)
 4. LED (For Load Testing)
 5. Light Bulb (Higher Current Load)
 6. Heat Sink (For Thermal Management)
 7. Auto Transformer (For Safe AC Testing)
 8. Full Bridge Rectifier (For AC Signal Processing)
 9. Optocoupler (For Zero Crossing Detection)
 10. Arduino Nano (For Microcontroller Control)
 11. Potentiometer (For Adjustable Delay)
-

Detailed Breakdown of Each Component

1. Thyristor (TYN604)

- **What is it?**

A semiconductor switch that allows current to flow in one direction when triggered.
- **Structure:**
 - Four-layer semiconductor

- Gate terminal for activation
 - What is it used for?
 - Controlling AC power in circuits
 - In the project:
 - Used to regulate current flow in an experimental phase angle control circuit
-

2. Triac (For AC Control)

- What is it?

A bidirectional semiconductor switch used for AC power control.
 - Structure:
 - Two thyristors in inverse parallel configuration
 - What is it used for?
 - Controlling both halves of an AC waveform
 - In the project:
 - Used to regulate AC power for the light bulb
-

3. MOSFET (For Switching)

- What is it?

A transistor used for fast electronic switching.
 - Structure:
 - Gate, Drain, and Source terminals
 - What is it used for?
 - Interrupting current flow to turn off the thyristor
 - In the project:
 - Used to demonstrate thyristor turn-off behavior
-

4. LED (For Load Testing)

- What is it?

A light-emitting diode used to visualize circuit behavior.
- Structure:

- Two terminals: Anode (+) and Cathode (-)
 - What is it used for?
 - Indicating thyristor activation
 - In the project:
 - Used to test low-current switching behavior
-

5. Light Bulb (Higher Current Load)

- What is it?

A resistive load used to test power regulation.
 - Structure:
 - Filament-based incandescent bulb
 - What is it used for?
 - Demonstrating thyristor behavior under higher current
 - In the project:
 - Used to show heat dissipation and power control
-

6. Heat Sink (For Thermal Management)

- What is it?

A component used to dissipate excess heat.
 - Structure:
 - Metal fins for heat dissipation
 - What is it used for?
 - Preventing overheating in high-power circuits
 - In the project:
 - Used to cool the thyristor under high current loads
-

7. Auto Transformer (For Safe AC Testing)

- What is it?

A transformer used to step down AC voltage for safer experiments.
- Structure:

- Adjustable voltage output
 - What is it used for?
 - Providing a lower AC voltage for testing
 - In the project:
 - Used to safely experiment with AC phase control
-

8. Full Bridge Rectifier (For AC Signal Processing)

- What is it?

A circuit that converts AC voltage into DC voltage.
 - Structure:
 - Four diodes arranged in a bridge configuration
 - What is it used for?
 - Creating a DC signal for zero-crossing detection
 - In the project:
 - Used to detect AC waveform zero points
-

9. Optocoupler (For Zero Crossing Detection)

- What is it?

A component that isolates and transfers signals using light.
 - Structure:
 - LED and phototransistor pair
 - What is it used for?
 - Detecting AC zero crossings for timing control
 - In the project:
 - Used to synchronize thyristor activation with AC waveform
-

10. Arduino Nano (For Microcontroller Control)

- What is it?

A compact microcontroller used for digital control.
- Structure:

- 14 digital I/O pins
 - Supports external interrupts
 - What is it used for?
 - Controlling thyristor activation timing
 - In the project:
 - Used to regulate phase angle control
-

11. Potentiometer (For Adjustable Delay)

- What is it?

A variable resistor used to adjust circuit parameters.
 - Structure:
 - Three terminals with adjustable resistance
 - What is it used for?
 - Modifying delay timing for phase control
 - In the project:
 - Used to adjust the timing of thyristor activation
-



Summary

This video explores thyristor fundamentals, phase angle control, and AC power regulation. It demonstrates how triacs manage AC waveforms, how optocouplers detect zero crossings, and how microcontrollers regulate power timing. The auto transformer ensures safe testing, while heat sinks prevent overheating in high-power applications.

VIDEO 21: Electronic Basics #21: OpAmp (Operational Amplifier)



List of Components Used

1. LM358 Dual Op-Amp IC
2. Resistors (Various Values: 10K Ω , 5.1K Ω , 1K Ω , 47K Ω , etc.)
3. Capacitors (For Signal Processing)
4. Electret Microphone (For Audio Input)
5. PT100 Temperature Sensor (For Analog Signal Amplification)
6. Speaker (For Audio Output)
7. Lab Bench Power Supply (For Voltage Testing)

8. Potentiometer (For Adjustable Gain Control)
 9. Voltage Divider Circuit (For Reference Voltage Setting)
 10. Comparator Circuit (For Signal Comparison)
-

Detailed Breakdown of Each Component

1. LM358 Dual Op-Amp IC

- What is it?
A widely used operational amplifier with two independent op-amps.
 - Structure:
 - 8-pin dual inline package (DIP)
 - Supports single and dual power supply
 - What is it used for?
 - Signal amplification, filtering, and voltage comparison
 - In the project:
 - Used to amplify signals from sensors and microphones
-

2. Resistors (Various Values: 10K Ω , 5.1K Ω , 1K Ω , 47K Ω , etc.)

- What are they?
Components that limit current flow and set reference values.
 - Structure:
 - Fixed resistance values
 - Used in voltage dividers and feedback loops
 - What are they used for?
 - Adjusting gain in op-amp circuits
 - Setting reference voltages
 - In the project:
 - Used to control amplification levels
-

3. Capacitors (For Signal Processing)

- What are they?
Components that store and release electrical energy.
 - Structure:
 - Electrostatic field storage
 - Various capacitance values
 - What are they used for?
 - Filtering and stabilizing signals
 - In the project:
 - Used to smooth amplified signals
-

4. Electret Microphone (For Audio Input)

- What is it?
A microphone that converts sound waves into electrical signals.
 - Structure:
 - Small diaphragm with an embedded capacitor
 - What is it used for?
 - Capturing audio signals for amplification
 - In the project:
 - Used to demonstrate op-amp signal amplification
-

5. PT100 Temperature Sensor (For Analog Signal Amplification)

- What is it?
A resistance-based temperature sensor.
 - Structure:
 - 100 Ω resistance at 0°C
 - Linear temperature response
 - What is it used for?
 - Measuring temperature in industrial applications
 - In the project:
 - Used to demonstrate op-amp-based signal amplification
-

6. Speaker (For Audio Output)

- What is it?
A device that converts electrical signals into sound waves.
 - Structure:
 - Coil and diaphragm mechanism
 - What is it used for?
 - Playing amplified audio signals
 - In the project:
 - Used to test op-amp audio amplification
-

7. Lab Bench Power Supply (For Voltage Testing)

- What is it?
A regulated power source for electronic circuits.
 - Structure:
 - Adjustable voltage output
 - What is it used for?
 - Providing stable voltage for op-amp circuits
 - In the project:
 - Used to test op-amp behavior under different voltage conditions
-

8. Potentiometer (For Adjustable Gain Control)

- What is it?
A variable resistor used to fine-tune circuit parameters.
 - Structure:
 - Three terminals
 - Adjustable resistance
 - What is it used for?
 - Modifying op-amp gain dynamically
 - In the project:
 - Used to adjust amplification levels
-

9. Voltage Divider Circuit (For Reference Voltage Setting)

- What is it?
A circuit that divides voltage using resistors.
 - Structure:
 - Two resistors in series
 - What is it used for?
 - Setting reference voltage for op-amp inputs
 - In the project:
 - Used to stabilize input signals
-

10. Comparator Circuit (For Signal Comparison)

- What is it?
A circuit that compares two voltage levels.
 - Structure:
 - Op-amp configured without feedback
 - What is it used for?
 - Detecting signal thresholds
 - In the project:
 - Used to demonstrate op-amp comparator functionality
-

Summary

This video explores operational amplifier fundamentals, covering signal amplification, voltage comparison, and filtering techniques. It demonstrates how op-amps process sensor signals, how voltage dividers set reference levels, and how comparators detect signal thresholds. The LM358 op-amp is used to amplify microphone and temperature sensor signals, while potentiometers adjust gain dynamically.

VIDEO 22: Electronic Basics #22: Transistor (BJT) as a Switch

List of Components Used

1. BC637 NPN Transistor
 2. BD535 High-Power Transistor
 3. TIP142 Darlington Transistor
 4. Resistors (Various Values: 10K Ω , 5.1K Ω , 1K Ω , etc.)
 5. High-Power LED (1W, 3.1V Forward Voltage)
 6. Light Bulb (6V, 21W)
 7. Heat Sink (For Thermal Management)
 8. Arduino Nano (For Microcontroller Control)
 9. Voltage Divider Circuit (For Reference Voltage Setting)
 10. MOSFET (For Switching High Current Loads)
-

Detailed Breakdown of Each Component

1. BC637 NPN Transistor

- What is it?
A general-purpose transistor used for switching and amplification.
 - Structure:
 - Three terminals: Emitter, Collector, Base
 - What is it used for?
 - Controlling small loads such as LEDs
 - In the project:
 - Used to switch a high-power LED on and off
-

2. BD535 High-Power Transistor

- What is it?
A high-current transistor capable of handling up to 8A.
- Structure:
 - Larger package for heat dissipation
- What is it used for?
 - Switching high-power loads such as light bulbs
- In the project:
 - Used to control a 6V, 21W light bulb

3. TIP142 Darlington Transistor

- What is it?
A transistor with two BJTs in series for higher current gain.
 - Structure:
 - High gain, low base current requirement
 - What is it used for?
 - Driving high-current loads with minimal base current
 - In the project:
 - Used to switch a 3.8A load with only 8.5mA base current
-

4. Resistors (Various Values: 10K Ω , 5.1K Ω , 1K Ω , etc.)

- What are they?
Components that limit current flow and set reference values.
 - Structure:
 - Fixed resistance values
 - What are they used for?
 - Controlling base current in transistor circuits
 - In the project:
 - Used to prevent excessive current flow through BJTs
-

5. High-Power LED (1W, 3.1V Forward Voltage)

- What is it?
A bright LED requiring controlled current for operation.
- Structure:
 - Requires a current-limiting resistor
- What is it used for?
 - Demonstrating transistor switching behavior
- In the project:
 - Used to test transistor switching circuits

6. Light Bulb (6V, 21W)

- What is it?
A resistive load used to test high-power switching.
 - Structure:
 - Filament-based incandescent bulb
 - What is it used for?
 - Demonstrating transistor behavior under high current
 - In the project:
 - Used to show heat dissipation and power control
-

7. Heat Sink (For Thermal Management)

- What is it?
A component used to dissipate excess heat.
 - Structure:
 - Metal fins for heat dissipation
 - What is it used for?
 - Preventing overheating in high-power circuits
 - In the project:
 - Used to cool the BD535 transistor under high current loads
-

8. Arduino Nano (For Microcontroller Control)

- What is it?
A compact microcontroller used for digital control.
- Structure:
 - 14 digital I/O pins
- What is it used for?
 - Controlling transistor activation timing
- In the project:
 - Used to regulate transistor switching behavior

9. Voltage Divider Circuit (For Reference Voltage Setting)

- What is it?
A circuit that divides voltage using resistors.
 - Structure:
 - Two resistors in series
 - What is it used for?
 - Setting reference voltage for transistor base control
 - In the project:
 - Used to stabilize input signals
-

10. MOSFET (For Switching High Current Loads)

- What is it?
A transistor used for fast electronic switching.
 - Structure:
 - Gate, Drain, and Source terminals
 - What is it used for?
 - Interrupting current flow to turn off the transistor
 - In the project:
 - Used to demonstrate transistor turn-off behavior
-

Summary

This video explores bipolar junction transistor fundamentals, covering switching applications, current gain, and thermal management. It demonstrates how BJTs regulate current, how Darlington transistors improve efficiency, and how heat sinks prevent overheating. The Arduino Nano is used for digital control, while MOSFETs provide high-speed switching.

VIDEO 23: Electronic Basics #23: Transistor (MOSFET) as a Switch

List of Components Used

1. IRLZ44N N-Channel MOSFET
 2. P-Channel MOSFET (Alternative for High-Side Switching)
 3. Resistors (Various Values: 10K Ω , 1.15 Ω , 470 Ω , etc.)
 4. High-Power LED (1W, 3.1V Forward Voltage)
 5. Light Bulb (Higher Current Load)
 6. Potentiometer (For Adjustable PWM Control)
 7. Arduino Nano (For Microcontroller Control)
 8. Oscilloscope (For Signal Analysis)
 9. MOSFET Driver IC (For High-Frequency Switching)
 10. Bootstrapping Circuit (For High-Side Switching)
-

Detailed Breakdown of Each Component

1. IRLZ44N N-Channel MOSFET

- What is it?
A power MOSFET used for switching high-current loads.
 - Structure:
 - Three terminals: Gate, Drain, Source
 - What is it used for?
 - Controlling loads efficiently with minimal power loss
 - In the project:
 - Used to switch LEDs and light bulbs with PWM control
-

2. P-Channel MOSFET (Alternative for High-Side Switching)

- What is it?
A MOSFET used for switching loads connected to ground.
- Structure:
 - Requires a pull-up resistor for proper operation
- What is it used for?
 - High-side switching applications
- In the project:
 - Used as an alternative to bootstrapping circuits

3. Resistors (Various Values: 10K Ω , 1.15 Ω , 470 Ω , etc.)

- What are they?
Components that limit current flow and set reference values.
 - Structure:
 - Fixed resistance values
 - What are they used for?
 - Controlling gate current and preventing unwanted activation
 - In the project:
 - Used to stabilize MOSFET switching behavior
-

4. High-Power LED (1W, 3.1V Forward Voltage)

- What is it?
A bright LED requiring controlled current for operation.
 - Structure:
 - Requires a current-limiting resistor
 - What is it used for?
 - Demonstrating MOSFET switching behavior
 - In the project:
 - Used to test PWM dimming functionality
-

5. Light Bulb (Higher Current Load)

- What is it?
A resistive load used to test high-power switching.
- Structure:
 - Filament-based incandescent bulb
- What is it used for?
 - Demonstrating MOSFET behavior under high current
- In the project:
 - Used to show heat dissipation and power control

6. Potentiometer (For Adjustable PWM Control)

- What is it?
A variable resistor used to fine-tune circuit parameters.
 - Structure:
 - Three terminals
 - Adjustable resistance
 - What is it used for?
 - Modifying PWM duty cycle dynamically
 - In the project:
 - Used to adjust LED brightness via MOSFET control
-

7. Arduino Nano (For Microcontroller Control)

- What is it?
A compact microcontroller used for digital control.
 - Structure:
 - 14 digital I/O pins
 - What is it used for?
 - Generating PWM signals for MOSFET control
 - In the project:
 - Used to regulate MOSFET switching behavior
-

8. Oscilloscope (For Signal Analysis)

- What is it?
A device used to visualize electrical signals over time.
- Structure:
 - Display screen showing voltage waveforms
- What is it used for?
 - Observing MOSFET switching characteristics
- In the project:
 - Used to analyze voltage spikes and oscillations

9. MOSFET Driver IC (For High-Frequency Switching)

- What is it?
A specialized IC used to drive MOSFETs efficiently.
 - Structure:
 - Provides high gate current for fast switching
 - What is it used for?
 - Reducing switching losses in high-frequency applications
 - In the project:
 - Used to improve MOSFET performance at 1MHz frequencies
-

10. Bootstrapping Circuit (For High-Side Switching)

- What is it?
A circuit used to provide proper gate drive voltage for high-side MOSFETs.
 - Structure:
 - Uses capacitors and diodes for voltage boosting
 - What is it used for?
 - Enabling MOSFETs to switch high-side loads
 - In the project:
 - Used to control loads connected to ground
-

Summary

This video explores MOSFET fundamentals, covering switching applications, PWM control, and high-frequency behavior. It demonstrates how MOSFETs improve efficiency, how gate resistors stabilize switching, and how bootstrapping circuits enable high-side control. The Arduino Nano generates PWM signals, while oscilloscopes analyze voltage spikes.

VIDEO 24: Electronic Basics #24: Stepper Motors and how to use them

List of Components Used

1. Hybrid Synchronous Stepper Motor
 2. Permanent Magnets (Rotor)
 3. Coils (Stator)
 4. Multimeter (For Coil Identification)
 5. Lab Bench Power Supply (For Coil Testing)
 6. H-Bridge Circuit (For Motor Control)
 7. Arduino Nano (For Microcontroller Control)
 8. A4988 Stepper Motor Driver IC
 9. 555 Timer Circuit (For Step Signal Generation)
 10. Potentiometer (For Adjustable Step Control)
-

Detailed Breakdown of Each Component

1. Hybrid Synchronous Stepper Motor

- **What is it?**
A motor that moves in discrete steps using electromagnetic fields.
 - **Structure:**
 - Rotor with permanent magnets
 - Stator with multiple coils
 - **What is it used for?**
 - Precise positioning in applications like 3D printers
 - **In the project:**
 - Used to demonstrate stepper motor control techniques
-

2. Permanent Magnets (Rotor)

- **What are they?**
Magnets that create the rotating magnetic field in a stepper motor.
- **Structure:**
 - Alternating polarity arrangement
- **What are they used for?**
 - Generating motion through electromagnetic interaction

- **In the project:**
 - Used to drive rotation in the motor
-

3. Coils (Stator)

- **What are they?**
Wound copper wires that generate electromagnetic fields.
 - **Structure:**
 - Arranged in multiple phases
 - **What are they used for?**
 - Creating rotating magnetic fields to drive the rotor
 - **In the project:**
 - Used to control motor movement
-

4. Multimeter (For Coil Identification)

- **What is it?**
A device used to measure electrical properties.
 - **Structure:**
 - Digital or analog display
 - **What is it used for?**
 - Identifying coil pairs in the stepper motor
 - **In the project:**
 - Used to determine coil connections before wiring
-

5. Lab Bench Power Supply (For Coil Testing)

- **What is it?**
A regulated power source for electronic circuits.
 - **Structure:**
 - Adjustable voltage output
 - **What is it used for?**
 - Testing coil activation in the stepper motor
 - **In the project:**
 - Used to verify coil polarity and function
-

6. H-Bridge Circuit (For Motor Control)

- **What is it?**
A circuit that allows bidirectional current flow through a motor.
 - **Structure:**
 - Four MOSFETs arranged in an H configuration
 - **What is it used for?**
 - Controlling stepper motor movement
 - **In the project:**
 - Used to switch coil polarity for stepper motor rotation
-

7. Arduino Nano (For Microcontroller Control)

- **What is it?**
A compact microcontroller used for digital control.
 - **Structure:**
 - 14 digital I/O pins
 - **What is it used for?**
 - Generating step signals for motor control
 - **In the project:**
 - Used to automate stepper motor movement
-

8. A4988 Stepper Motor Driver IC

- **What is it?**
A specialized IC used to drive stepper motors efficiently.
 - **Structure:**
 - Provides microstepping capabilities
 - **What is it used for?**
 - Reducing noise and improving step resolution
 - **In the project:**
 - Used to enable smooth motor movement
-

9. 555 Timer Circuit (For Step Signal Generation)

- **What is it?**
A circuit that generates pulse signals for motor control.
- **Structure:**
 - Adjustable frequency output
- **What is it used for?**
 - Creating step pulses for the motor driver
- **In the project:**

- Used to control stepper motor rotation without a microcontroller

10. Potentiometer (For Adjustable Step Control)

- **What is it?**
A variable resistor used to fine-tune circuit parameters.
 - **Structure:**
 - Three terminals
 - Adjustable resistance
 - **What is it used for?**
 - Modifying step frequency dynamically
 - **In the project:**
 - Used to adjust motor speed via step signal control
-

Summary

This video explores **stepper motor fundamentals**, covering **hybrid synchronous motors**, **microstepping**, and **driver circuits**. It demonstrates how **stepper motors** achieve precise **positioning**, how **H-bridge circuits** control movement, and how **microstepping** improves **smoothness**. The **Arduino Nano** generates **step signals**, while **oscilloscopes** analyze voltage behavior.

VIDEO 25: Electronic Basics #25: Servos and how to use them

List of Components Used

1. **Standard Servo Motor (MG996R, SG90, etc.)**
 2. **Potentiometer (For Position Feedback)**
 3. **DC Motor (Inside the Servo Housing)**
 4. **Gear System (For Torque Enhancement)**
 5. **KC5188 Control IC (For Position Control)**
 6. **H-Bridge Circuit (For Motor Direction Control)**
 7. **Arduino Nano (For Microcontroller Control)**
 8. **555 Timer Circuit (For PWM Signal Generation)**
 9. **Resistors (Various Values: 10K Ω , etc.)**
 10. **Diodes & Capacitors (For Signal Processing)**
-

Detailed Breakdown of Each Component

1. Standard Servo Motor (MG996R, SG90, etc.)

- **What is it?**
A motor with built-in control electronics for precise positioning.
 - **Structure:**
 - DC motor, gears, potentiometer, control IC
 - **What is it used for?**
 - Robotics, RC vehicles, and automation
 - **In the project:**
 - Used to demonstrate servo control techniques
-

2. Potentiometer (For Position Feedback)

- **What is it?**
A variable resistor that provides position feedback.
 - **Structure:**
 - Three terminals: VCC, Ground, Signal
 - **What is it used for?**
 - Determining the servo shaft position
 - **In the project:**
 - Used to provide feedback for precise control
-

3. DC Motor (Inside the Servo Housing)

- **What is it?**
A small motor that drives the servo mechanism.
 - **Structure:**
 - Rotates based on control signals
 - **What is it used for?**
 - Generating movement in the servo
 - **In the project:**
 - Used to drive the servo shaft
-

4. Gear System (For Torque Enhancement)

- **What is it?**
A set of gears that reduce speed and increase torque.
 - **Structure:**
 - Plastic or metal gears
 - **What is it used for?**
 - Improving servo strength and precision
 - **In the project:**
 - Used to enhance torque for heavy loads
-

5. KC5188 Control IC (For Position Control)

- **What is it?**
An integrated circuit that manages servo movement.
 - **Structure:**
 - Compares feedback voltage with target position
 - **What is it used for?**
 - Ensuring accurate servo positioning
 - **In the project:**
 - Used to regulate motor movement
-

6. H-Bridge Circuit (For Motor Direction Control)

- **What is it?**
A circuit that allows bidirectional current flow.
 - **Structure:**
 - Four MOSFETs arranged in an H configuration
 - **What is it used for?**
 - Controlling servo motor rotation
 - **In the project:**
 - Used to switch motor direction
-

7. Arduino Nano (For Microcontroller Control)

- **What is it?**
A compact microcontroller used for digital control.
- **Structure:**
 - 14 digital I/O pins
- **What is it used for?**
 - Generating PWM signals for servo control
- **In the project:**

- Used to automate servo movement
-

8. 555 Timer Circuit (For PWM Signal Generation)

- **What is it?**
A circuit that generates pulse signals for motor control.
 - **Structure:**
 - Adjustable frequency output
 - **What is it used for?**
 - Creating PWM signals for servo control
 - **In the project:**
 - Used to control servo movement without a microcontroller
-

9. Resistors (Various Values: 10K Ω , etc.)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Stabilizing servo control signals
 - **In the project:**
 - Used to prevent unwanted activation
-

10. Diodes & Capacitors (For Signal Processing)

- **What are they?**
Components used to filter and stabilize signals.
 - **Structure:**
 - Electrostatic field storage
 - **What are they used for?**
 - Ensuring smooth PWM signal generation
 - **In the project:**
 - Used to refine servo control signals
-



Summary

This video explores **servo motor fundamentals**, covering **position feedback**, **PWM control**, and **gear mechanisms**. It demonstrates how **servos achieve precise movement**, how **H-bridge circuits regulate direction**, and how **microcontrollers automate control**. The **Arduino Nano** generates **PWM signals**, while **555 timer circuits provide alternative control methods**.

VIDEO 26: Electronic Basics #26: 555 Timer IC

List of Components Used

1. NE555 Timer IC
2. TLC555 CMOS Timer IC (Alternative for Precision Timing)
3. Resistors (Various Values: 5K Ω , 10K Ω , etc.)
4. Capacitors (Various Values: 10nF, etc.)
5. Diodes (For Charging & Discharging Control)
6. Bipolar Junction Transistor (For Discharge Pin Functionality)
7. Potentiometer (For Adjustable Timing Control)
8. LED (For Output Indication)
9. Push Buttons (For Manual Triggering)
10. Oscilloscope (For Signal Analysis)

Detailed Breakdown of Each Component

1. NE555 Timer IC

- **What is it?**
A widely used integrated circuit for timing applications.
- **Structure:**
 - Contains comparators, flip-flop, and output driver
- **What is it used for?**
 - Generating precise time delays and oscillations
- **In the project:**
 - Used to demonstrate monostable, bistable, and astable configurations

2. TLC555 CMOS Timer IC (Alternative for Precision Timing)

- **What is it?**
A low-power version of the 555 timer with MOSFET-based output.
- **Structure:**
 - Provides near-supply voltage output

- **What is it used for?**
 - Achieving accurate 50% duty cycle PWM signals
 - **In the project:**
 - Used to improve timing accuracy and reduce power consumption
-

3. Resistors (Various Values: 5K Ω , 10K Ω , etc.)

- **What are they?**

Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Setting timing intervals in 555 circuits
 - **In the project:**
 - Used to control charge/discharge rates in astable mode
-

4. Capacitors (Various Values: 10nF, etc.)

- **What are they?**

Components that store and release electrical energy.
 - **Structure:**
 - Electrostatic field storage
 - **What are they used for?**
 - Stabilizing reference voltages and timing control
 - **In the project:**
 - Used to regulate pulse width in monostable mode
-

5. Diodes (For Charging & Discharging Control)

- **What are they?**

Semiconductor components that allow current flow in one direction.
 - **Structure:**
 - PN junction design
 - **What are they used for?**
 - Separating charge and discharge paths in astable mode
 - **In the project:**
 - Used to refine PWM signal generation
-

6. Bipolar Junction Transistor (For Discharge Pin Functionality)

- **What is it?**
A transistor used to discharge capacitors in 555 circuits.
 - **Structure:**
 - Three terminals: Collector, Base, Emitter
 - **What is it used for?**
 - Rapidly resetting timing cycles
 - **In the project:**
 - Used to discharge capacitors in monostable mode
-

7. Potentiometer (For Adjustable Timing Control)

- **What is it?**
A variable resistor used to fine-tune circuit parameters.
 - **Structure:**
 - Three terminals
 - Adjustable resistance
 - **What is it used for?**
 - Modifying pulse width dynamically
 - **In the project:**
 - Used to adjust timing intervals in astable mode
-

8. LED (For Output Indication)

- **What is it?**
A light-emitting diode used to visualize circuit behavior.
 - **Structure:**
 - Two terminals: Anode (+) and Cathode (-)
 - **What is it used for?**
 - Indicating pulse activation
 - **In the project:**
 - Used to show timing pulses in monostable mode
-

9. Push Buttons (For Manual Triggering)

- **What are they?**
Mechanical switches used to activate circuits.
- **Structure:**
 - Two terminals for signal activation

- **What are they used for?**
 - Manually triggering 555 timer circuits
 - **In the project:**
 - Used to initiate timing cycles in monostable mode
-

10. Oscilloscope (For Signal Analysis)

- **What is it?**

A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing pulse width and frequency
 - **In the project:**
 - Used to analyze PWM signals generated by the 555 timer
-

Summary

This video explores **555 timer fundamentals**, covering **monostable, bistable, and astable configurations**. It demonstrates how **timing circuits generate precise delays**, how **PWM signals are refined using diodes**, and how **CMOS versions improve efficiency**. The **Arduino Nano** generates **PWM signals**, while **oscilloscopes analyze pulse behavior**.

VIDEO 27: Electronic Basics #27: ADC (Analog to Digital Converter)

List of Components Used

1. **Arduino Nano (For ADC Demonstration)**
 2. **ADS7816 12-bit ADC IC**
 3. **Resistors (Various Values: 5K Ω , 10K Ω , etc.)**
 4. **Capacitors (For Signal Stabilization)**
 5. **Comparator Circuit (For Voltage Comparison)**
 6. **Digital-to-Analog Converter (DAC) (For Successive Approximation)**
 7. **Flash ADC Circuit (For High-Speed Conversion)**
 8. **Oscilloscope (For Signal Analysis)**
 9. **Voltage Reference Source (For ADC Calibration)**
 10. **Encoder Circuit (For Flash ADC Output Processing)**
-

Detailed Breakdown of Each Component

1. Arduino Nano (For ADC Demonstration)

- **What is it?**
A compact microcontroller used for digital control.
 - **Structure:**
 - 14 digital I/O pins
 - Built-in 10-bit ADC
 - **What is it used for?**
 - Sampling analog signals and converting them to digital values
 - **In the project:**
 - Used to demonstrate ADC functionality
-

2. ADS7816 12-bit ADC IC

- **What is it?**
A high-resolution ADC used for precise signal conversion.
 - **Structure:**
 - Successive approximation register (SAR) architecture
 - **What is it used for?**
 - Improving resolution in analog-to-digital conversion
 - **In the project:**
 - Used to enhance ADC accuracy beyond Arduino's built-in ADC
-

3. Resistors (Various Values: 5K Ω , 10K Ω , etc.)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Creating voltage dividers for ADC calibration
 - **In the project:**
 - Used to stabilize input signals for ADC processing
-

4. Capacitors (For Signal Stabilization)

- **What are they?**
Components that store and release electrical energy.
 - **Structure:**
 - Electrostatic field storage
 - **What are they used for?**
 - Filtering noise in ADC circuits
 - **In the project:**
 - Used to smooth sampled signals before conversion
-

5. Comparator Circuit (For Voltage Comparison)

- **What is it?**
A circuit that compares input voltage with reference levels.
 - **Structure:**
 - Two input terminals for voltage comparison
 - **What is it used for?**
 - Determining digital output based on analog input
 - **In the project:**
 - Used in successive approximation ADCs
-

6. Digital-to-Analog Converter (DAC) (For Successive Approximation)

- **What is it?**
A circuit that converts digital values back to analog signals.
 - **Structure:**
 - Uses resistor networks or charge redistribution
 - **What is it used for?**
 - Refining ADC accuracy in SAR-based conversions
 - **In the project:**
 - Used to generate reference voltages for ADC comparison
-

7. Flash ADC Circuit (For High-Speed Conversion)

- **What is it?**
A high-speed ADC that uses multiple comparators.
- **Structure:**
 - Parallel comparator array
- **What is it used for?**
 - Achieving rapid analog-to-digital conversion
- **In the project:**
 - Used to demonstrate fast ADC processing

8. Oscilloscope (For Signal Analysis)

- **What is it?**
A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing ADC sampling behavior
 - **In the project:**
 - Used to analyze reconstructed signals from ADC output
-

9. Voltage Reference Source (For ADC Calibration)

- **What is it?**
A stable voltage source used to set ADC reference levels.
 - **Structure:**
 - Fixed voltage output
 - **What is it used for?**
 - Ensuring accurate ADC readings
 - **In the project:**
 - Used to calibrate ADC input range
-

10. Encoder Circuit (For Flash ADC Output Processing)

- **What is it?**
A circuit that converts comparator outputs into binary values.
 - **Structure:**
 - Truth table-based logic processing
 - **What is it used for?**
 - Encoding ADC results into digital format
 - **In the project:**
 - Used to process flash ADC outputs efficiently
-

Summary

This video explores **ADC fundamentals**, covering **sampling rate, resolution, and conversion techniques**. It demonstrates how **successive approximation ADCs refine accuracy**, how **flash ADCs achieve high-speed conversion**, and how **voltage references ensure calibration**. The **Arduino Nano** samples **analog signals**, while **oscilloscopes analyze reconstructed waveforms**.

VIDEO 28: Electronic Basics #28: IGBT and when to use them

List of Components Used

1. RGP50B60PD1 N-Channel IGBT
2. MOSFET (For Comparison with IGBT)
3. Resistors (Various Values: 10K Ω , etc.)
4. Capacitors (For Gate Charge Management)
5. Light Bulb (For Load Testing)
6. TC4420 MOSFET/IGBT Driver IC
7. IR2113 High-Side Driver IC (For Bootstrap Operation)
8. Oscilloscope (For Signal Analysis)
9. Voltage Source (For Gate Control)
10. Comparator Circuit (For Voltage Drop Measurement)

Detailed Breakdown of Each Component

1. RGP50B60PD1 N-Channel IGBT

- **What is it?**
An insulated gate bipolar transistor used for high-power switching.
- **Structure:**
 - Combines MOSFET input with BJT output
- **What is it used for?**
 - Medium-frequency, high-voltage applications
- **In the project:**
 - Used to demonstrate switching behavior in power circuits

2. MOSFET (For Comparison with IGBT)

- **What is it?**
A transistor used for fast electronic switching.
- **Structure:**
 - Gate, Drain, and Source terminals
- **What is it used for?**
 - High-frequency applications above 200 kHz
- **In the project:**
 - Used to compare efficiency and switching speed with IGBT

3. Resistors (Various Values: 10K Ω , etc.)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Discharging gate charge in IGBT circuits
 - **In the project:**
 - Used to stabilize switching behavior
-

4. Capacitors (For Gate Charge Management)

- **What are they?**
Components that store and release electrical energy.
 - **Structure:**
 - Electrostatic field storage
 - **What are they used for?**
 - Managing gate charge for controlled switching
 - **In the project:**
 - Used to refine IGBT turn-on and turn-off behavior
-

5. Light Bulb (For Load Testing)

- **What is it?**
A resistive load used to test high-power switching.
 - **Structure:**
 - Filament-based incandescent bulb
 - **What is it used for?**
 - Demonstrating IGBT and MOSFET efficiency
 - **In the project:**
 - Used to measure power loss in switching circuits
-

6. TC4420 MOSFET/IGBT Driver IC

- **What is it?**
A specialized IC used to drive MOSFETs and IGBTs efficiently.
 - **Structure:**
 - Provides high gate current for fast switching
 - **What is it used for?**
 - Reducing switching losses in high-frequency applications
 - **In the project:**
 - Used to improve IGBT performance at higher frequencies
-

7. IR2113 High-Side Driver IC (For Bootstrap Operation)

- **What is it?**
A driver IC used for high-side switching applications.
 - **Structure:**
 - Provides voltage boost for gate control
 - **What is it used for?**
 - Enabling IGBTs to switch high-side loads
 - **In the project:**
 - Used to control high-voltage switching behavior
-

8. Oscilloscope (For Signal Analysis)

- **What is it?**
A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing IGBT and MOSFET switching characteristics
 - **In the project:**
 - Used to analyze voltage spikes and oscillations
-

9. Voltage Source (For Gate Control)

- **What is it?**
A regulated power source for electronic circuits.
- **Structure:**
 - Adjustable voltage output
- **What is it used for?**
 - Providing gate drive voltage for IGBT activation
- **In the project:**

- Used to test switching efficiency at different voltages

10. Comparator Circuit (For Voltage Drop Measurement)

- **What is it?**
A circuit that compares voltage levels in switching devices.
- **Structure:**
 - Two input terminals for voltage comparison
- **What is it used for?**
 - Measuring power loss in IGBT and MOSFET circuits
- **In the project:**
 - Used to analyze efficiency differences between transistors

Summary

This video explores **IGBT vs. MOSFET fundamentals**, covering **switching speed, power loss, and high-voltage applications**. It demonstrates how **IGBTs handle medium-frequency loads**, how **MOSFETs excel in high-frequency switching**, and how **driver ICs improve efficiency**. The **oscilloscope analyzes voltage behavior**, while **bootstrap circuits enable high-side switching**.

VIDEO 29: Electronic Basics #29: Solar Panel & Charge Controller

List of Components Used

1. Solar Cells (Individual Photovoltaic Units)
2. Multimeter (For Voltage Measurement)
3. Bypass Diodes (For Partial Shading Protection)
4. Blocking Diodes (For Reverse Current Prevention)
5. LEDs (For Load Testing)
6. Resistors (Various Values: 5K Ω , etc.)
7. DIY Power Logger (For Data Collection)
8. Micro SD Card (For Data Storage)
9. Charge Controllers (MPPT & PWM Types)
10. Excel (For Data Visualization & Analysis)

Detailed Breakdown of Each Component

1. Solar Cells (Individual Photovoltaic Units)

- **What are they?**
Semiconductor devices that convert sunlight into electrical energy.
 - **Structure:**
 - Thin silicon wafers with electrical contacts
 - **What are they used for?**
 - Generating voltage and current from sunlight
 - **In the project:**
 - Used to demonstrate solar panel wiring and efficiency
-

2. Multimeter (For Voltage Measurement)

- **What is it?**
A device used to measure electrical properties.
 - **Structure:**
 - Digital or analog display
 - **What is it used for?**
 - Measuring solar cell output voltage
 - **In the project:**
 - Used to verify voltage generation from solar cells
-

3. Bypass Diodes (For Partial Shading Protection)

- **What are they?**
Diodes placed in parallel with solar cells to prevent power loss.
 - **Structure:**
 - Semiconductor junction allowing current flow in one direction
 - **What are they used for?**
 - Maintaining power output when part of the panel is shaded
 - **In the project:**
 - Used to improve efficiency in real-world conditions
-

4. Blocking Diodes (For Reverse Current Prevention)

- **What are they?**
Diodes used to prevent reverse current flow in solar panel arrays.
- **Structure:**

- PN junction semiconductor
 - **What are they used for?**
 - Preventing energy loss when panels are connected in parallel
 - **In the project:**
 - Used to stabilize solar panel performance
-

5. LEDs (For Load Testing)

- **What are they?**
Light-emitting diodes used to visualize power output.
 - **Structure:**
 - Two terminals: Anode (+) and Cathode (-)
 - **What are they used for?**
 - Demonstrating solar panel voltage and current behavior
 - **In the project:**
 - Used to test solar panel efficiency under different conditions
-

6. Resistors (Various Values: 5K Ω , etc.)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Creating load conditions for solar panel testing
 - **In the project:**
 - Used to determine maximum power point (MPP)
-

7. DIY Power Logger (For Data Collection)

- **What is it?**
A custom-built device used to record electrical data.
 - **Structure:**
 - Microcontroller-based logging system
 - **What is it used for?**
 - Capturing voltage and current variations over time
 - **In the project:**
 - Used to analyze solar panel performance
-

8. Micro SD Card (For Data Storage)

- **What is it?**
A removable storage device used for logging data.
 - **Structure:**
 - Flash memory chip with data interface
 - **What is it used for?**
 - Storing solar panel test results
 - **In the project:**
 - Used to save voltage and current measurements
-

9. Charge Controllers (MPPT & PWM Types)

- **What are they?**
Devices used to regulate solar panel output for battery charging.
 - **Structure:**
 - Microcontroller-based regulation system
 - **What are they used for?**
 - Optimizing energy transfer to batteries
 - **In the project:**
 - Used to demonstrate efficient battery charging methods
-

10. Excel (For Data Visualization & Analysis)

- **What is it?**
A spreadsheet software used for organizing and analyzing data.
 - **Structure:**
 - Graphing and calculation tools
 - **What is it used for?**
 - Plotting solar panel efficiency curves
 - **In the project:**
 - Used to visualize maximum power point tracking
-

Summary

This video explores **solar panel fundamentals**, covering **wiring configurations**, **efficiency optimization**, and **charge controllers**. It demonstrates how **bypass diodes improve shading performance**, how **blocking diodes prevent reverse current**, and how **MPPT controllers**

maximize energy transfer. The **DIY power logger** records **real-time data**, while **Excel** visualizes **efficiency curves**.

VIDEO 30: Electronic Basics #30: Microcontroller (Arduino) Timers

List of Components Used

1. **ATmega328P Microcontroller (Arduino Uno)**
2. **Resistors (Various Values: 10KΩ, etc.)**
3. **Capacitors (For Signal Stabilization)**
4. **LEDs (For Timing Demonstration)**
5. **Push Buttons (For Interrupt Testing)**
6. **Oscilloscope (For Signal Analysis)**
7. **Potentiometers (For PWM Adjustment)**
8. **PWM Signal Generator (For Frequency Control)**
9. **External Clock Source (For Precision Timing)**
10. **EEPROM (For Storing Timer Settings)**

Detailed Breakdown of Each Component

1. ATmega328P Microcontroller (Arduino Uno)

- **What is it?**
A widely used microcontroller with built-in timers.
 - **Structure:**
 - 8-bit architecture with multiple peripherals
 - **What is it used for?**
 - Generating precise timing events and PWM signals
 - **In the project:**
 - Used to demonstrate timer configurations and interrupts
-

2. Resistors (Various Values: 10K Ω , etc.)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Stabilizing input signals for timer functions
 - **In the project:**
 - Used to prevent unwanted activation in timing circuits
-

3. Capacitors (For Signal Stabilization)

- **What are they?**
Components that store and release electrical energy.
 - **Structure:**
 - Electrostatic field storage
 - **What are they used for?**
 - Filtering noise in timer circuits
 - **In the project:**
 - Used to smooth PWM signals for precise control
-

4. LEDs (For Timing Demonstration)

- **What are they?**
Light-emitting diodes used to visualize timer behavior.
 - **Structure:**
 - Two terminals: Anode (+) and Cathode (-)
 - **What are they used for?**
 - Indicating timer-based events
 - **In the project:**
 - Used to show precise timing intervals
-

5. Push Buttons (For Interrupt Testing)

- **What are they?**
Mechanical switches used to activate circuits.
- **Structure:**
 - Two terminals for signal activation
- **What are they used for?**

- Manually triggering timer interrupts
 - **In the project:**
 - Used to test responsiveness of timer-based events
-

6. Oscilloscope (For Signal Analysis)

- **What is it?**

A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing PWM signal behavior
 - **In the project:**
 - Used to analyze frequency and duty cycle variations
-

7. Potentiometers (For PWM Adjustment)

- **What are they?**

Variable resistors used to fine-tune circuit parameters.
 - **Structure:**
 - Three terminals
 - Adjustable resistance
 - **What are they used for?**
 - Modifying PWM duty cycle dynamically
 - **In the project:**
 - Used to adjust frequency and pulse width
-

8. PWM Signal Generator (For Frequency Control)

- **What is it?**

A circuit that generates pulse-width modulation signals.
 - **Structure:**
 - Adjustable frequency output
 - **What is it used for?**
 - Creating precise timing signals for microcontroller applications
 - **In the project:**
 - Used to control LED brightness and alarm sounds
-

9. External Clock Source (For Precision Timing)

- **What is it?**
A stable clock signal used to improve timing accuracy.
 - **Structure:**
 - Quartz crystal oscillator
 - **What is it used for?**
 - Ensuring precise timing in microcontroller operations
 - **In the project:**
 - Used to stabilize timer-based events
-

10. EEPROM (For Storing Timer Settings)

- **What is it?**
A non-volatile memory used to store configuration data.
 - **Structure:**
 - Electrically erasable programmable memory
 - **What is it used for?**
 - Retaining timer settings across power cycles
 - **In the project:**
 - Used to save PWM frequency and duty cycle settings
-

Summary

This video explores **microcontroller timer fundamentals**, covering **interrupts**, **PWM generation**, and **precise timing events**. It demonstrates how **timers improve responsiveness**, how **PWM signals control brightness and sound**, and how **external clock sources enhance accuracy**. The **Arduino Nano** generates **timing signals**, while **oscilloscopes analyze frequency behavior**.

VIDEO 31: Electronic Basics #31: Schottky Diode & Zener Diode

List of Components Used

1. 1N5819 Schottky Diode
2. 1N4007 Standard Diode
3. Zener Diodes (Various Voltages: 5.1V, 15V, etc.)
4. Resistors (Various Values: 1K Ω , etc.)
5. Capacitors (For Signal Stabilization)

6. **MOSFET (For Gate Protection Example)**
 7. **Boost Converter Circuit (For Schottky Diode Demonstration)**
 8. **Oscilloscope (For Signal Analysis)**
 9. **Voltage Source (For Testing Diode Behavior)**
 10. **Comparator Circuit (For Voltage Clipping Demonstration)**
-

Detailed Breakdown of Each Component

1. 1N5819 Schottky Diode

- **What is it?**
A fast-switching diode with low forward voltage drop.
 - **Structure:**
 - Metal-semiconductor junction
 - **What is it used for?**
 - High-frequency rectification and reverse voltage protection
 - **In the project:**
 - Used to demonstrate efficiency in a boost converter circuit
-

2. 1N4007 Standard Diode

- **What is it?**
A general-purpose rectifier diode.
 - **Structure:**
 - PN junction semiconductor
 - **What is it used for?**
 - Basic rectification and voltage blocking
 - **In the project:**
 - Used to compare voltage drop with Schottky diodes
-

3. Zener Diodes (Various Voltages: 5.1V, 15V, etc.)

- **What are they?**
Diodes designed to regulate voltage by breaking down at a specific voltage.
- **Structure:**
 - Reverse-biased PN junction
- **What are they used for?**
 - Voltage regulation and overvoltage protection

- **In the project:**
 - Used to demonstrate voltage clipping and regulation
-

4. Resistors (Various Values: $1K\Omega$, etc.)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Controlling current in diode circuits
 - **In the project:**
 - Used to stabilize voltage regulation with Zener diodes
-

5. Capacitors (For Signal Stabilization)

- **What are they?**
Components that store and release electrical energy.
 - **Structure:**
 - Electrostatic field storage
 - **What are they used for?**
 - Filtering noise in diode circuits
 - **In the project:**
 - Used to smooth voltage fluctuations in rectifier circuits
-

6. MOSFET (For Gate Protection Example)

- **What is it?**
A transistor used for fast electronic switching.
 - **Structure:**
 - Gate, Drain, and Source terminals
 - **What is it used for?**
 - Protecting gate voltage from overvoltage damage
 - **In the project:**
 - Used to demonstrate Zener diode voltage clipping
-

7. Boost Converter Circuit (For Schottky Diode Demonstration)

- **What is it?**
A circuit that increases DC voltage using inductors and diodes.
 - **Structure:**
 - Coil, MOSFET, diode, and capacitor
 - **What is it used for?**
 - Demonstrating Schottky diode efficiency in high-frequency applications
 - **In the project:**
 - Used to show how Schottky diodes improve power conversion
-

8. Oscilloscope (For Signal Analysis)

- **What is it?**
A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing diode switching behavior
 - **In the project:**
 - Used to analyze rectification and voltage clipping
-

9. Voltage Source (For Testing Diode Behavior)

- **What is it?**
A regulated power source for electronic circuits.
 - **Structure:**
 - Adjustable voltage output
 - **What is it used for?**
 - Providing stable voltage for diode testing
 - **In the project:**
 - Used to verify diode voltage drop and breakdown behavior
-

10. Comparator Circuit (For Voltage Clipping Demonstration)

- **What is it?**
A circuit that compares voltage levels in diode applications.
- **Structure:**
 - Two input terminals for voltage comparison
- **What is it used for?**
 - Demonstrating Zener diode voltage regulation
- **In the project:**

- Used to analyze voltage clipping in AC signals
-

Summary

This video explores **Schottky and Zener diode fundamentals**, covering **voltage regulation, rectification, and efficiency improvements**. It demonstrates how **Schottky diodes reduce power loss**, how **Zener diodes regulate voltage**, and how **boost converters utilize fast-switching diodes**. The **oscilloscope analyzes diode behavior**, while **MOSFETs demonstrate voltage protection**.

VIDEO 32: Electronic Basics #32: Relays & Optocouplers

List of Components Used

1. **Electromechanical Relay (Commonly Found in Remote-Controlled Sockets)**
 2. **HX2272 IC (Remote Control Encoder)**
 3. **817 Optocoupler IC**
 4. **MOSFET (For DC Switching Comparison)**
 5. **TRIAC (For AC Switching Comparison)**
 6. **Flyback Diode (For Relay Coil Protection)**
 7. **Push Button (For Manual Relay Activation)**
 8. **Oscilloscope (For Signal Analysis)**
 9. **Voltage Source (For Testing Relay Behavior)**
 10. **Comparator Circuit (For Voltage Clipping Demonstration)**
-

Detailed Breakdown of Each Component

1. Electromechanical Relay (Commonly Found in Remote-Controlled Sockets)

- **What is it?**
A switch that uses an electromagnet to mechanically open or close electrical contacts.
- **Structure:**
 - Coil, anchor, and switch contacts
- **What is it used for?**
 - Switching high-power loads with electrical isolation
- **In the project:**
 - Used to demonstrate how remote-controlled sockets operate

2. HX2272 IC (Remote Control Encoder)

- **What is it?**
An integrated circuit used for encoding remote control signals.
 - **Structure:**
 - Digital encoding logic
 - **What is it used for?**
 - Sending control signals to activate relays remotely
 - **In the project:**
 - Used to trigger relay activation in remote sockets
-

3. 817 Optocoupler IC

- **What is it?**
A component that provides electrical isolation between input and output signals.
 - **Structure:**
 - Infrared LED and phototransistor
 - **What is it used for?**
 - Protecting microcontrollers from high-voltage circuits
 - **In the project:**
 - Used to demonstrate optocoupler-based switching
-

4. MOSFET (For DC Switching Comparison)

- **What is it?**
A transistor used for fast electronic switching.
 - **Structure:**
 - Gate, Drain, and Source terminals
 - **What is it used for?**
 - Comparing relay efficiency in DC applications
 - **In the project:**
 - Used to highlight differences between relays and MOSFETs
-

5. TRIAC (For AC Switching Comparison)

- **What is it?**
A semiconductor device used for AC power control.
- **Structure:**

- Three terminals: Gate, Main Terminal 1, Main Terminal 2
 - **What is it used for?**
 - Comparing relay efficiency in AC applications
 - **In the project:**
 - Used to demonstrate alternative AC switching methods
-

6. Flyback Diode (For Relay Coil Protection)

- **What is it?**

A diode used to prevent voltage spikes when a relay coil de-energizes.
 - **Structure:**
 - PN junction semiconductor
 - **What is it used for?**
 - Protecting transistors from high-voltage spikes
 - **In the project:**
 - Used to stabilize relay operation
-

7. Push Button (For Manual Relay Activation)

- **What is it?**

A mechanical switch used to trigger relay activation.
 - **Structure:**
 - Two terminals for signal activation
 - **What is it used for?**
 - Manually controlling relay switching
 - **In the project:**
 - Used to demonstrate relay activation without remote control
-

8. Oscilloscope (For Signal Analysis)

- **What is it?**

A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing relay coil voltage spikes
 - **In the project:**
 - Used to analyze switching behavior and voltage transients
-

9. Voltage Source (For Testing Relay Behavior)

- **What is it?**
A regulated power source for electronic circuits.
 - **Structure:**
 - Adjustable voltage output
 - **What is it used for?**
 - Providing stable voltage for relay coil activation
 - **In the project:**
 - Used to verify relay switching characteristics
-

10. Comparator Circuit (For Voltage Clipping Demonstration)

- **What is it?**
A circuit that compares voltage levels in switching applications.
 - **Structure:**
 - Two input terminals for voltage comparison
 - **What is it used for?**
 - Demonstrating optocoupler voltage isolation
 - **In the project:**
 - Used to analyze optocoupler-based switching behavior
-

Summary

This video explores **relay and optocoupler fundamentals**, covering **electromechanical switching, voltage isolation, and alternative switching methods**. It demonstrates how **relays handle high-power loads**, how **optocouplers provide electrical isolation**, and how **flyback diodes protect circuits from voltage spikes**. The **oscilloscope analyzes switching behavior**, while **MOSFETs and TRIACs offer alternative control methods**.

VIDEO 33: Electronic Basics #33: Strain Gauge/Load Cell and how to use them to measure weight

List of Components Used

1. Strain Gauge (120Ω Resistance)
2. Wheatstone Bridge Circuit (For Resistance Measurement)
3. Resistors (Precision 120Ω, 10-Turn Trimmers for Calibration)
4. Differential Op-Amp (For Signal Amplification)
5. HX711 ADC Module (For High-Resolution Weight Measurement)

6. **Load Cell (Pre-Calibrated Wheatstone Bridge Sensor)**
 7. **Microcontroller (Arduino for Data Processing)**
 8. **Protective Silicone Layer (For Strain Gauge Protection)**
 9. **Oscilloscope (For Signal Analysis)**
 10. **Voltage Source (For Wheatstone Bridge Excitation)**
-

Detailed Breakdown of Each Component

1. Strain Gauge (120Ω Resistance)

- **What is it?**
A sensor that measures strain by changing resistance when stretched or compressed.
 - **Structure:**
 - Thin plastic film with zig-zag resistance wire
 - **What is it used for?**
 - Measuring force or weight applied to an object
 - **In the project:**
 - Used to detect weight changes on the longboard
-

2. Wheatstone Bridge Circuit (For Resistance Measurement)

- **What is it?**
A circuit that converts small resistance changes into measurable voltage differences.
 - **Structure:**
 - Four resistors arranged in a bridge configuration
 - **What is it used for?**
 - Amplifying strain gauge resistance changes
 - **In the project:**
 - Used to convert strain gauge resistance into voltage signals
-

3. Resistors (Precision 120Ω, 10-Turn Trimmers for Calibration)

- **What are they?**
Components that limit current flow and set reference values.
- **Structure:**
 - Fixed resistance values and adjustable trimmers
- **What are they used for?**
 - Fine-tuning Wheatstone Bridge balance

- **In the project:**
 - Used to calibrate strain gauge readings
-

4. Differential Op-Amp (For Signal Amplification)

- **What is it?**

An operational amplifier that boosts small voltage differences.
 - **Structure:**
 - High-gain amplification circuit
 - **What is it used for?**
 - Increasing strain gauge signal strength for ADC conversion
 - **In the project:**
 - Used to amplify Wheatstone Bridge output voltage
-

5. HX711 ADC Module (For High-Resolution Weight Measurement)

- **What is it?**

A 24-bit analog-to-digital converter with an integrated amplifier.
 - **Structure:**
 - High-resolution ADC with gain settings
 - **What is it used for?**
 - Converting strain gauge signals into digital weight readings
 - **In the project:**
 - Used to process strain gauge data with high precision
-

6. Load Cell (Pre-Calibrated Wheatstone Bridge Sensor)

- **What is it?**

A metal sensor with built-in strain gauges for weight measurement.
 - **Structure:**
 - Aluminum profile with four strain gauges
 - **What is it used for?**
 - Measuring weight with pre-calibrated accuracy
 - **In the project:**
 - Used as an alternative to manually setting up strain gauges
-

7. Microcontroller (Arduino for Data Processing)

- **What is it?**
A programmable device used to process sensor data.
 - **Structure:**
 - Digital and analog I/O pins
 - **What is it used for?**
 - Handling strain gauge calculations and display output
 - **In the project:**
 - Used to convert strain gauge readings into weight values
-

8. Protective Silicone Layer (For Strain Gauge Protection)

- **What is it?**
A flexible coating used to shield strain gauges from damage.
 - **Structure:**
 - Silicone-based protective layer
 - **What is it used for?**
 - Preventing environmental damage to strain gauges
 - **In the project:**
 - Used to protect strain gauges on the longboard
-

9. Oscilloscope (For Signal Analysis)

- **What is it?**
A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing strain gauge signal behavior
 - **In the project:**
 - Used to analyze Wheatstone Bridge voltage output
-

10. Voltage Source (For Wheatstone Bridge Excitation)

- **What is it?**
A regulated power source for electronic circuits.
- **Structure:**
 - Adjustable voltage output
- **What is it used for?**
 - Providing stable voltage for Wheatstone Bridge operation
- **In the project:**

- Used to ensure accurate strain gauge readings



Summary

This video explores **strain gauge fundamentals**, covering **Wheatstone Bridge circuits**, **load cells**, and **high-resolution ADCs**. It demonstrates how **strain gauges detect force**, how **HX711 modules process weight data**, and how **protective coatings improve durability**. The **Arduino microcontroller converts strain gauge signals**, while **oscilloscopes analyze voltage behavior**.

VIDEO 34: Electronic Basics #34: Two-Position Controller & PID Controller



List of Components Used

1. Electromagnet
2. Hall Effect Sensor
3. Arduino Nano
4. MOSFET and driver
5. Levitation Setup Frame
6. Neodymium Magnet



Detailed Breakdown of Each Component

1. Electromagnet

- *What is it?* A device that generates a magnetic field when an electric current passes through it.
- *Structure:*
 - Consists of a coil of wire, usually wrapped around a metallic core.
- *What is it used for?*
 - Produces the upward force needed to levitate the magnet.
- *In the project:*
 - Levitates the magnet.

2. Hall Effect Sensor

- *What is it?* A sensor that detects the presence and strength of a magnetic field.

- *Structure:*
 - A semiconductor material that produces a voltage when exposed to a magnetic field.
- *What is it used for?*
 - Measures the position of the levitating magnet.
- *In the project:*
 - Provides feedback on the magnet's position to the control system.

3. Arduino Nano

- *What is it?* A small, complete, and breadboard-friendly microcontroller board.
- *Structure:*
 - Based on the ATmega328P microcontroller.
 - Has digital and analog input/output pins.
- *What is it used for?*
 - Processes the sensor data and controls the electromagnet.
- *In the project:*
 - Implements the control logic (two-position and PID) to adjust the electromagnet's power.

4. MOSFET and driver

- *What is it?* A Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is a type of transistor used to switch or amplify electronic signals. A driver is a circuit that provides the necessary current and voltage to control the MOSFET.
- *Structure:*
 - MOSFET: Three terminals - Gate, Drain, and Source.
- *What is it used for?*
 - MOSFET: Acts as an electronic switch.
 - Driver: Amplifies the control signal from the Arduino to drive the MOSFET.
- *In the project:*
 - Controls the power supplied to the electromagnet.

5. Levitation Setup Frame

- *What is it?* A physical structure to hold the electromagnet and sensor in place for levitation.
- *Structure:*
 - A frame made of metal sheets and rods.
- *What is it used for?*
 - Providing a stable platform for the magnetic levitation experiment.
- *In the project:*

- Supports all the components, ensuring the electromagnet is positioned correctly above the levitating magnet and the Hall sensor is below it for accurate measurements.

6. Neodymium Magnet

- *What is it?* A powerful permanent magnet.
- *Structure:*
 - Small, cylindrical bar magnets.
- *What is it used for?*
 - The object to be levitated.
- *In the project:*
 - It's the object that is levitated by the electromagnet, with its position being monitored by the Hall sensor.

Summary

The video demonstrates a magnetic levitation system. It uses an electromagnet to levitate a magnet, a Hall effect sensor to detect the magnet's position, an Arduino Nano to control the system, and a MOSFET with a driver to regulate the power to the electromagnet. The video compares a simple two-position controller with a PID controller, showing how the PID controller provides much more stable levitation.

VIDEO 35: Electronic Basics #35: Schmitt Trigger and when to use them

List of Components Used

1. **Operational Amplifier (For Comparator-Based Schmitt Trigger)**
2. **74HC14 Hex Inverting Schmitt Trigger IC**
3. **Resistors (Various Values for Hysteresis Control)**
4. **Capacitors (For Relaxation Oscillator Circuit)**
5. **Potentiometer (For Adjustable Threshold Voltage)**
6. **Push Button (For Debouncing Demonstration)**
7. **RC Network (For Reducing Bounce in Mechanical Switches)**
8. **Oscilloscope (For Signal Analysis)**
9. **Voltage Source (For Comparator Input Testing)**
10. **Microcontroller (For Digital Input Handling)**

Detailed Breakdown of Each Component

1. Operational Amplifier (For Comparator-Based Schmitt Trigger)

- **What is it?**
A high-gain amplifier used for voltage comparison.
 - **Structure:**
 - Two inputs (inverting & non-inverting)
 - Single output
 - **What is it used for?**
 - Creating a basic comparator circuit
 - **In the project:**
 - Used to demonstrate Schmitt trigger behavior with hysteresis
-

2. 74HC14 Hex Inverting Schmitt Trigger IC

- **What is it?**
A digital logic IC with built-in Schmitt trigger functionality.
 - **Structure:**
 - Six independent Schmitt trigger circuits
 - **What is it used for?**
 - Cleaning up noisy signals and creating stable transitions
 - **In the project:**
 - Used to demonstrate Schmitt trigger threshold voltages
-

3. Resistors (Various Values for Hysteresis Control)

- **What are they?**
Components that limit current flow and set reference values.
 - **Structure:**
 - Fixed resistance values
 - **What are they used for?**
 - Adjusting hysteresis voltage in Schmitt trigger circuits
 - **In the project:**
 - Used to fine-tune threshold voltages
-

4. Capacitors (For Relaxation Oscillator Circuit)

- **What are they?**
Components that store and release electrical energy.

- **Structure:**
 - Electrostatic field storage
 - **What are they used for?**
 - Creating oscillations in Schmitt trigger circuits
 - **In the project:**
 - Used to generate square wave signals
-

5. Potentiometer (For Adjustable Threshold Voltage)

- **What is it?**

A variable resistor used to fine-tune circuit parameters.
 - **Structure:**
 - Three terminals
 - Adjustable resistance
 - **What is it used for?**
 - Modifying Schmitt trigger threshold dynamically
 - **In the project:**
 - Used to adjust voltage levels for triggering
-

6. Push Button (For Debouncing Demonstration)

- **What is it?**

A mechanical switch used to trigger digital inputs.
 - **Structure:**
 - Two terminals for signal activation
 - **What is it used for?**
 - Demonstrating bouncing issues in mechanical switches
 - **In the project:**
 - Used to show how Schmitt triggers improve switch transitions
-

7. RC Network (For Reducing Bounce in Mechanical Switches)

- **What is it?**

A resistor-capacitor combination used for signal smoothing.
- **Structure:**
 - Series resistor and capacitor
- **What is it used for?**
 - Reducing noise in push-button signals
- **In the project:**
 - Used to stabilize mechanical switch transitions

8. Oscilloscope (For Signal Analysis)

- **What is it?**
A device used to visualize electrical signals over time.
 - **Structure:**
 - Display screen showing voltage waveforms
 - **What is it used for?**
 - Observing Schmitt trigger behavior
 - **In the project:**
 - Used to analyze hysteresis voltage transitions
-

9. Voltage Source (For Comparator Input Testing)

- **What is it?**
A regulated power source for electronic circuits.
 - **Structure:**
 - Adjustable voltage output
 - **What is it used for?**
 - Providing stable voltage for Schmitt trigger testing
 - **In the project:**
 - Used to verify threshold voltage behavior
-

10. Microcontroller (For Digital Input Handling)

- **What is it?**
A programmable device used to process digital signals.
 - **Structure:**
 - Digital and analog I/O pins
 - **What is it used for?**
 - Handling Schmitt trigger-like behavior in digital inputs
 - **In the project:**
 - Used to demonstrate built-in hysteresis in microcontroller inputs
-

Summary

This video explores **Schmitt trigger fundamentals**, covering **hysteresis voltage**, **signal stabilization**, and **digital logic applications**. It demonstrates how **Schmitt triggers clean up noisy signals**, how **push-button debouncing improves reliability**, and how **relaxation**

oscillators generate square waves. The oscilloscope analyzes voltage transitions, while microcontrollers handle digital input processing.

VIDEO 36: Electronic Basics #36: SPI and how to use it

List of Components Used

- DS3234 Real-Time Clock (RTC) IC
- Arduino (any board with SPI support)
- Hookup wires
- Oscilloscope (for verification)

Detailed Breakdown of Each Component

1. DS3234 Real-Time Clock (RTC) IC

- **What is it?**
A highly accurate real-time clock IC that uses the SPI protocol. It keeps track of time (seconds to year) and offers alarms and square wave outputs.
- **Structure:**
 - Package: SOIC-16
 - 8 functional pins used in this video:
 - **VCC** – Power supply (typically 3.3V or 5V)
 - **GND** – Ground
 - **CLK** – SPI Clock
 - **MISO** – Master In Slave Out
 - **MOSI** – Master Out Slave In
 - **SS/CS** – Slave Select / Chip Select
 - **SQW** – Square Wave Output
 - **/INT** – Interrupt output (used optionally)
- **What is it used for?**
 - Keeps track of time (calendar & clock)
 - Can output square waves
 - Can trigger alarms
 - Communicates over SPI (not I2C like other RTCs)
- **In the project (in the video):**
 - Used to **store and maintain real-time** with high accuracy
 - **Configured via SPI** from the Arduino
 - Outputs a **square wave signal** (e.g. 8.192 KHz) for timing verification via oscilloscope
 - Demonstrates **setting** and **reading** date/time values

2. Arduino (e.g., Uno, Nano, etc.)

- **What is it?**
A microcontroller board used to control and interface with other components.
 - **Structure:**
 - 14 digital I/O pins (in this case using SPI-specific ones):
 - **D13 (SCK)** → Clock
 - **D12 (MISO)** → Data from DS3234
 - **D11 (MOSI)** → Data to DS3234
 - **D10 (SS/CS)** → Selects DS3234 for SPI
 - Any other pin could serve as CS as long as configured in code
 - **What is it used for?**
 - Acts as **SPI Master**
 - Sends and receives commands/data to/from DS3234
 - Initializes time values and reads current time
 - Configures registers and triggers square wave output
 - **In the project (in the video):**
 - Interfaces with the DS3234 RTC over SPI
 - Sets time/date registers
 - Reads back data to verify RTC functionality
 - Outputs data for verification on oscilloscope
-

3. Hookup Wires

- **What is it?**
Basic jumper wires for making electrical connections on a breadboard or directly between components.
 - **Structure:**
 - Typically male-to-male, flexible wires
 - No polarity, just signal carriers
 - **What is it used for?**
 - Connecting SPI lines and power between Arduino and DS3234
 - **In the project (in the video):**
 - Used to physically connect VCC, GND, CLK, MISO, MOSI, and SS between Arduino and DS3234 RTC module
-

4. Oscilloscope

- **What is it?**
A test instrument that graphically displays electrical signals over time.

- **Structure:**
 - Probes for signal input
 - Display for waveform visualization
 - **What is it used for?**
 - Verifying square wave signal output from RTC
 - Measuring exact output frequency (e.g., 8.192 KHz)
 - **In the project (in the video):**
 - Used to confirm successful SPI communication
 - Verified the 8.192 KHz square wave output from the DS3234
-

Summary

The video introduces how to interface the **DS3234 RTC IC** with an **Arduino via SPI**. Instead of using I2C (which is more common with RTCs), this IC uses **SPI**, requiring pins for CLK, MISO, MOSI, and CS. The project shows how to send and receive data, set and read time/date, and verify output with an oscilloscope. This helps understand the SPI protocol's speed advantage and how it works in practice.

VIDEO 37: Electronic Basics #37: What is Impedance? (AC Resistance?)

List of Components Used

- Resistor (50 Ohm)
 - Capacitor (1 μ F)
 - Inductor (214 μ H)
 - Function Generator
 - Amplifier
 - LCR Meter
 - Current Clamp
-

Detailed Breakdown of Each Component

1. Resistor (50 Ohm)

- **What is it?** A passive component that resists the flow of electric current.
- **Structure:**
 - Two terminals (non-polarized)
 - Resistance measured in ohms (Ω)
 - Color-coded bands for value indication

- **What is it used for?**
 - Limits current flow
 - Controls voltage in a circuit
 - Dissipates power as heat
 - **In the project (video):**
 - Used to demonstrate ohmic resistance in AC circuits.
 - No phase shift occurs; current is proportional to voltage.
 - **Summary:** Controls current without affecting phase.
-

2. Capacitor (1 μ F)

- **What is it?** A passive component that stores electrical energy in an electric field.
 - **Structure:**
 - Two terminals (may be polarized in electrolytic capacitors)
 - Value in farads (F), often marked on the body
 - Capacitive reactance: $X_C = \frac{1}{2\pi fC}$
 - **What is it used for?**
 - Blocks DC while allowing AC
 - Stores and releases energy
 - Causes phase shift between current and voltage
 - **In the project (video):**
 - Shows **capacitive reactance**—current leads voltage.
 - Reactance decreases as frequency increases.
 - **Summary:** Reacts to AC frequency, current leads voltage.
-

3. Inductor (214 μ H)

- **What is it?** A passive component that stores energy in a magnetic field.
 - **Structure:**
 - Wire coil (air or core-based)
 - Two terminals
 - Inductive reactance: $X_L = 2\pi fL$
 - **What is it used for?**
 - Opposes changes in current
 - Filters signals, stores energy in power supplies
 - Causes phase shift between current and voltage
 - **In the project (video):**
 - Demonstrates **inductive reactance**—voltage leads current.
 - Reactance increases with frequency.
 - **Summary:** Opposes current change, voltage leads current.
-

4. Function Generator

- **What is it?** An electronic device that outputs repeating waveforms (sine, square, etc.).
 - **Structure:**
 - Output terminals
 - Control knobs or digital interface for frequency/amplitude
 - **What is it used for?**
 - Generates signal to test circuit response
 - **In the project (video):**
 - Produces sine waves at varying frequencies and amplitudes.
 - **Summary:** Supplies AC waveform for testing components.
-

5. Amplifier

- **What is it?** A circuit or device that boosts signal power.
 - **Structure:**
 - Input/output terminals
 - Gain control
 - **What is it used for?**
 - Amplifies low-power signals for use in circuits
 - **In the project (video):**
 - Boosts sine wave from function generator to power passive components.
 - **Summary:** Increases signal strength.
-

6. LCR Meter

- **What is it?** A test instrument to measure inductance (L), capacitance (C), and resistance (R).
 - **Structure:**
 - Probe connections
 - Digital display
 - **What is it used for?**
 - Measures component values and complex impedance
 - **In the project (video):**
 - Used to show real-world values (including parasitic elements) of components.
 - **Summary:** Measures key parameters and real behavior.
-

7. Current Clamp

- **What is it?** A device to measure current flow without direct contact.

- **Structure:**
 - Clamp jaw
 - Display or oscilloscope interface
 - **What is it used for?**
 - Measures AC or DC current in a conductor
 - **In the project (video):**
 - Used to observe phase shift between voltage and current.
 - **Summary:** Non-intrusive current measurement tool.
-



Summary

The video uses basic passive components to explain **impedance** in AC circuits. It demonstrates how resistors, inductors, and capacitors behave differently with AC signals, especially in terms of **reactance** and **phase shift**. Real-world measurements using tools like current clamps and LCR meters help illustrate theoretical vs. practical differences.

VIDEO 38: Electronic Basics #38: True, Reactive, Apparent & Deformed Power.



List of Components Used

- Energy Multimeter
- Transformer (mains voltage)
- Inductor
- Synchronous Motor (from microwave)
- Capacitors (0.068 microfarad x2 in parallel)
- Laptop Power Supply
- Oscilloscope
- Resistor (mentioned for comparison)



Detailed Breakdown of Each Component

1. Energy Multimeter

- **What is it?** A specialized multimeter that can measure electrical power and energy consumption over time for AC appliances.
- **Structure:**
 - Three probes for connection
 - Connects to mains socket adapter
 - Digital display showing voltage, current, power, and power factor
- **What is it used for?**
 - Measuring real power (watts)
 - Measuring apparent power (volt-amperes)
 - Measuring reactive power (VAR)

- Displaying power factor
- **In the project:** Used as the main measurement tool to demonstrate power factor concepts and measure different types of power in various components.

2. Transformer (Mains Voltage)

- **What is it?** An electromagnetic device that steps voltage up or down using magnetic coupling.
- **Structure:**
 - Primary winding (mains voltage input)
 - Secondary winding (lower voltage output)
 - Iron core for magnetic coupling
 - Input draws 24mA at 228V
- **What is it used for?**
 - Voltage transformation
 - Electrical isolation
 - Demonstrates inductive reactive power
- **In the project:** Used as the primary example to show how transformers draw reactive power due to their inductive nature, creating an 80-degree phase shift between voltage and current.

3. Inductor

- **What is it?** A passive component that stores energy in a magnetic field and opposes changes in current.
- **Structure:**
 - Coiled wire around a core
 - Two terminals
 - Creates 90-degree phase shift (current lagging voltage)
- **What is it used for?**
 - Energy storage in magnetic fields
 - Current filtering
 - Creating reactive power
- **In the project:** Used to demonstrate pure reactive power with 90-degree phase shift, showing how power oscillates between source and load with zero average power consumption.

4. Synchronous Motor (from Microwave)

- **What is it?** A small AC motor that rotates at a speed synchronized with the AC frequency.
- **Structure:**
 - Two terminals for mains connection
 - Rotating shaft
 - Internal windings creating inductive load
- **What is it used for?**

- Converting electrical energy to mechanical rotation
 - Timing applications in appliances
- **In the project:** Used to demonstrate real-world inductive load drawing 4.2 VA apparent power, 3 VAR reactive power, and 2.9W real power, showing practical power factor issues.

5. Capacitors (0.068 microfarad each)

- **What is it?** Passive components that store energy in an electric field.
- **Structure:**
 - Two parallel plates with dielectric
 - Two terminals
 - Connected in parallel for 0.136 μF total
 - Creates 90-degree phase shift (current leading voltage)
- **What is it used for?**
 - Energy storage in electric fields
 - Power factor correction
 - Filtering applications
- **In the project:** Used for power factor correction, reducing the motor's reactive power from original value to one-third by compensating inductive reactance with capacitive reactance.

6. Laptop Power Supply

- **What is it?** A switch-mode power supply that converts AC mains to DC for laptop operation.
- **Structure:**
 - AC input connector
 - DC output connector
 - Internal switching circuitry
 - Energy storage capacitors
- **What is it used for?**
 - Converting AC to regulated DC power
 - Providing stable power for electronic devices
- **In the project:** Used to demonstrate distortion power and harmonics, showing how modern switching power supplies create non-sinusoidal current waveforms that contribute to reactive power without phase shift.

7. Oscilloscope

- **What is it?** Electronic test instrument that displays voltage waveforms over time.
- **Structure:**
 - Multiple input channels
 - Display screen
 - Probe inputs
 - Math functions capability

- **What is it used for?**
 - Waveform analysis
 - Phase shift measurement
 - Harmonic analysis
 - Power calculation through math functions
- **In the project:** Used to visualize voltage and current waveforms, measure phase shifts, calculate power through math functions, and analyze current harmonics in the laptop power supply.

 **Summary** This educational video demonstrates power factor concepts using various components to show different types of electrical power. The transformer and motor illustrate inductive reactive power with phase shifts, while capacitors show how to correct power factor. The laptop power supply demonstrates harmonic distortion in modern switching supplies. All measurements were taken using an energy multimeter and oscilloscope to visualize the relationships between voltage, current, and power.

VIDEO 39: Controlling a BIG LED Matrix?! How Shift Registers work! || EB#39

Title: LED Matrix Control with Shift Registers

List of Components Used

- Custom LED Matrix (32x12 LEDs, 384 total)
- STP16CP05 LED Driver ICs (6 units)
- SN74LS15 Hex Schmitt Trigger Inverters (2 units)
- P-Channel MOSFETs (4 units)
- Pull-up Resistors
- Current Set Resistors
- Capacitors
- Male and Female Headers
- Arduino Nano
- Push Buttons (for testing)
- 10mm LEDs (individual testing)

Detailed Breakdown of Each Component

1. Custom LED Matrix (32x12)

- **What is it?** A custom-made PCB with 384 LEDs arranged in a 32x12 grid pattern.
- **Structure:**
 - 32 LEDs in x-direction, 12 LEDs in y-direction
 - Male and female headers for connections
 - Thick traces connecting components
 - Integrated driver ICs and passive components

- **What is it used for?**
 - Displaying visual patterns, text, and graphics
 - Creating large-scale LED displays
 - Educational demonstrations of multiplexing
- **In the project:** Main display element for showing scrolling text ("HI" and "COOL"), demonstrating shift register control and multiplexing techniques.

2. STP16CP05 LED Driver ICs (6 units)

- **What is it?** 16-bit constant current LED sink drivers with integrated shift registers.
- **Structure:**
 - 16 output pins per IC
 - Serial data input (SDI)
 - Clock input (CLK)
 - Latch input (LE)
 - Output enable pin (OE)
 - 16-bit shift register + 16-bit storage register
- **What is it used for?**
 - Controlling LED brightness with constant current
 - Reducing microcontroller pin requirements
 - Serial-to-parallel data conversion
- **In the project:** Controls LED cathodes through multiplexing, with each IC managing 16 LEDs. Six ICs provide 96 outputs to control 384 LEDs through 4:1 multiplexing.

3. SN74LS15 Hex Schmitt Trigger Inverters (2 units)

- **What is it?** Logic gate ICs that clean up digital signals and provide signal inversion.
- **Structure:**
 - 6 inverting Schmitt triggers per IC
 - 14-pin DIP package
 - VCC and GND pins
 - Input and output pins for each gate
- **What is it used for?**
 - Signal conditioning and noise reduction
 - Converting slow-changing signals to clean digital signals
 - Signal buffering and inversion
- **In the project:** Clean up data signals between Arduino and LED driver ICs, ensuring reliable digital communication.

4. P-Channel MOSFETs (4 units)

- **What is it?** Power switching transistors that conduct when gate voltage is lower than source voltage.
- **Structure:**
 - Three terminals: Gate, Source, Drain
 - TO-220 or similar package

- High current handling capability
- **What is it used for?**
 - High-side switching of power supplies
 - Controlling power to different circuit sections
 - Voltage level switching
- **In the project:** Switch power to different groups of LED lines (3 lines each) for multiplexing, controlled by Arduino digital pins.

5. Pull-up Resistors

- **What is it?** Resistors that pull signal lines to a high voltage level when not actively driven.
- **Structure:**
 - Standard resistor package (likely 10k Ω)
 - Two terminals
 - Connected between signal line and VCC
- **What is it used for?**
 - Ensuring defined logic levels
 - Preventing floating inputs
 - Proper MOSFET gate control
- **In the project:** Ensure MOSFET gates are properly pulled high when not actively driven low by Arduino pins.

6. Current Set Resistors

- **What is it?** Precision resistors that set the constant current output of LED drivers.
- **Structure:**
 - Low tolerance resistors
 - Connected to current setting pins of LED drivers
 - Determines LED brightness/current
- **What is it used for?**
 - Setting LED operating current
 - Protecting LEDs from overcurrent
 - Ensuring consistent LED brightness
- **In the project:** Set the constant current level for all LEDs to ensure uniform brightness across the matrix.

7. Arduino Nano

- **What is it?** Compact microcontroller board based on ATmega328P.
- **Structure:**
 - 14 digital I/O pins
 - 8 analog input pins
 - USB connector for programming
 - 5V and 3.3V power outputs
 - Timer/counter peripherals

- **What is it used for?**
 - Microcontroller applications
 - Sensor interfacing
 - Digital signal generation
 - Real-time control systems
- **In the project:** Controls the LED matrix by generating serial data, clock signals, and latch signals using timer interrupts. Also controls MOSFET switching for multiplexing.

8. Male and Female Headers

- **What is it?** Connector pins that allow removable connections between PCBs and wires.
- **Structure:**
 - Plastic housing with metal pins
 - Various pin counts and spacing
 - Male pins insert into female sockets
- **What is it used for?**
 - Creating removable connections
 - Interfacing between different PCBs
 - Prototyping and testing connections
- **In the project:** Provide connection points for power supply lines and control signals between the LED matrix PCB and external circuitry.

 **Summary** This project demonstrates controlling a large 384-LED matrix using shift registers and multiplexing. The STP16CP05 LED drivers act as both shift registers and constant current sources, reducing control pins from 384 to just 3 (data, clock, latch). P-channel MOSFETs enable 4:1 multiplexing by switching power to groups of LED rows. The Arduino generates precise timing using timer interrupts to create scrolling text displays. Schmitt trigger inverters ensure clean digital signals between the microcontroller and LED drivers.

VIDEO 40: How safe is contactless payment? || How does RFID & NFC work? || EB#40

Title: RFID and NFC Technology Explained

List of Components Used

- RC522 RFID Reader Module
- PN532 NFC/RFID Reader Module
- RDM6300 RFID Reader Module
- RFID Tags/Cards (various types)
- Arduino UNO Development Board
- Oscilloscope
- Girocard (contactless payment card)
- Smartphone with NFC capability
- Anti-skimming card holder

■ Detailed Breakdown of Each Component

1. RC522 RFID Reader Module

- **What is it?** A 13.56MHz RFID reader module based on the MFRC522 IC for reading and writing RFID tags.
- **Structure:**
 - SPI communication interface (MISO, MOSI, SCK, SDA pins)
 - 3.3V operating voltage
 - Built-in antenna coil on PCB
 - RST (reset) pin
 - IRQ (interrupt) pin
 - 8 connection pins total
- **What is it used for?**
 - Reading RFID tag UIDs and data
 - Writing data to compatible RFID tags
 - Access control systems
 - Inventory management
- **In the project:** Primary RFID reader for demonstrating basic RFID communication, reading tag information, and showing wireless energy transfer principles at 13.56MHz frequency.

2. PN532 NFC/RFID Reader Module

- **What is it?** A versatile NFC/RFID transceiver module that supports multiple communication protocols.
- **Structure:**
 - I2C, SPI, and UART communication options
 - 13.56MHz operating frequency
 - Built-in antenna
 - Multiple protocol support (ISO14443A/B, ISO18092)
 - DIP switch configuration
- **What is it used for?**
 - NFC communication between devices
 - Reading various RFID tag types
 - Peer-to-peer data exchange
 - Card emulation mode
- **In the project:** Used to attempt reading the Girocard and demonstrate more advanced NFC capabilities compared to basic RFID readers.

3. RDM6300 RFID Reader Module

- **What is it?** A low-frequency 125KHz RFID reader module for reading EM4100 compatible tags.
- **Structure:**
 - UART serial communication

- 125KHz operating frequency
 - External antenna connection
 - 5V power supply
 - Simple 4-pin interface (VCC, GND, TX, RX)
- **What is it used for?**
 - Reading low-frequency RFID tags
 - Animal identification systems
 - Simple access control
 - Asset tracking
- **In the project:** Represents the low-frequency RFID option (125KHz) as an alternative to high-frequency readers, showing different RFID frequency standards.

4. RFID Tags/Cards (Various Types)

- **What is it?** Passive devices containing a coil antenna and microchip that respond to RFID readers.
- **Structure:**
 - Antenna coil (copper traces)
 - IC chip (e.g., Mifare Classic 1K with 1KB storage)
 - Protective housing (plastic card or transparent casing)
 - No battery required (passive operation)
- **What is it used for?**
 - Storing identification data
 - Access control
 - Inventory tracking
 - Contactless payments
- **In the project:** Demonstrate RFID communication, wireless power transfer, and data storage. Show how tags receive power from reader's magnetic field and transmit data back.

5. Arduino UNO Development Board

- **What is it?** Microcontroller development board based on ATmega328P processor.
- **Structure:**
 - 14 digital I/O pins
 - 6 analog input pins
 - USB programming interface
 - 5V and 3.3V power outputs
 - SPI, I2C, UART communication
- **What is it used for?**
 - Prototyping and development
 - Interfacing with sensors and modules
 - Learning embedded programming
 - IoT applications

- **In the project:** Controls the RFID readers, processes tag data, runs example code from RFID libraries, and demonstrates practical RFID applications like LED control based on tag authentication.

6. Oscilloscope

- **What is it?** Electronic test instrument that displays voltage waveforms over time.
- **Structure:**
 - Multiple input channels with probes
 - Display screen for waveform visualization
 - Trigger controls
 - Time and voltage measurement capabilities
- **What is it used for?**
 - Analyzing electrical signals
 - Measuring frequency and amplitude
 - Troubleshooting circuits
 - Signal integrity analysis
- **In the project:** Used to observe the 13.56MHz carrier frequency from the RFID reader and visualize amplitude modulation when the tag communicates with the reader, proving the electromagnetic communication principle.

7. Girocard (Contactless Payment Card)

- **What is it?** A contactless payment card using NFC technology for secure financial transactions.
- **Structure:**
 - Embedded NFC chip (ISO 14443-4 compliant)
 - Antenna coil within card
 - Encrypted data storage
 - EMV payment application
- **What is it used for?**
 - Contactless payments
 - ATM transactions
 - Secure authentication
 - Banking services
- **In the project:** Test subject to demonstrate the security differences between basic RFID tags and secure NFC payment cards, showing that encrypted cards cannot be easily read by basic RFID readers.

8. Smartphone with NFC

- **What is it?** Mobile device with built-in NFC transceiver capability.
- **Structure:**
 - NFC antenna integrated in device
 - NFC controller chip
 - Operating system NFC support

- Apps for NFC functionality
- **What is it used for?**
 - Contactless payments (Google Pay, Apple Pay)
 - Data sharing between devices
 - Reading NFC tags
 - Access control applications
- **In the project:** Used with NFC Tools app to successfully read the Girocard information that basic RFID readers couldn't access, demonstrating advanced NFC capabilities and showing practical NFC applications.

9. Anti-skimming Card Holder

- **What is it?** A protective sleeve or wallet made with RF-blocking material to prevent unauthorized RFID/NFC reading.
- **Structure:**
 - Metal mesh or foil material
 - Card-sized protective sleeve
 - RF shielding properties
 - Blocks 13.56MHz and other RFID frequencies
- **What is it used for?**
 - Protecting payment cards from skimming
 - Preventing unauthorized RFID scanning
 - Privacy protection
 - Security enhancement
- **In the project:** Presented as a security solution for protecting contactless payment cards, demonstrating how RF shielding can block electromagnetic fields and prevent unwanted RFID communication.

 **Summary** This educational video explores RFID and NFC technologies using three different reader modules (RC522, PN532, RDM6300) with Arduino. It demonstrates wireless energy transfer, electromagnetic communication at 13.56MHz and 125KHz frequencies, and the security differences between basic RFID tags and encrypted NFC payment cards. The project shows practical applications from simple access control to secure contactless payments, highlighting both the convenience and security considerations of these technologies.

VIDEO 41: Does a DIY Audio Crossover make sense? How passive filters work! || EB#41

List of Components Used

- Resistor (2Ω, 10Ω, 4Ω)
- Capacitor (10 μF, others unspecified)
- Inductor (150 μH, others unspecified)
- Function Generator
- Oscilloscope

- Audio Crossover (Prebuilt & DIY)
 - Amplifier
 - 4Ω Resistor (used as dummy load)
 - Loudspeakers (Woofer and Tweeter)
-

Detailed Breakdown of Each Component

1. Resistors (2Ω , 10Ω , 4Ω dummy load)

- **What is it?**
A passive component that resists electric current, producing voltage drop and dissipating power as heat.
 - **Structure:**
 - 2 terminals
 - Fixed resistance value (e.g., 2Ω , 10Ω)
 - No polarity
 - Connected in series with capacitors/inductors in RC/LC filter networks
 - 4Ω resistor used to simulate loudspeaker load
 - **What is it used for?**
 - Voltage division
 - Damping audio signals
 - Simulating speaker resistance
 - Building RC low-pass/high-pass filters
 - **In the project:**
 - Used in filter experiments to demonstrate damping behavior
 - 4Ω resistor replaces the speaker during test phase
 - Essential in RC filter to determine cutoff frequency
 - **Short Summary:**
Controls current and helps form filters in crossover experiments.
-

2. Capacitors ($10\ \mu\text{F}$, various types)

- **What is it?**
A passive component that stores energy in an electric field; its impedance decreases with increasing frequency.
- **Structure:**
 - 2 terminals (no polarity unless electrolytic)
 - Value used: $10\ \mu\text{F}$
 - Connected in series with tweeter for high-pass filter
 - Used in RC filter for frequency selection

- **What is it used for?**
 - Filtering high frequencies (acts as a high-pass filter)
 - Forming cutoff frequencies in filters
 - Removing low-frequency noise
 - **In the project:**
 - Used to direct high-frequency signals to the tweeter
 - Combined with resistors/inductors in filter circuits
 - Demonstrates capacitive reactance and its impact on frequency response
 - **Short Summary:**

Filters out low frequencies to protect and optimize tweeter performance.
-

3. Inductors (150 μ H and others)

- **What is it?**

A passive component that stores energy in a magnetic field; its impedance increases with frequency.
 - **Structure:**
 - Wire coil (air-core or ferrite)
 - 2 terminals
 - 150 μ H used in testing
 - Connected in series with woofer for low-pass filter
 - **What is it used for?**
 - Filtering out high frequencies
 - Creating low-pass filters
 - Attenuating unwanted noise in circuits
 - **In the project:**
 - Used in series with the woofer to block high frequencies
 - Demonstrates inductive reactance in experiments
 - Part of LC and RLC filter combinations in the original crossover
 - **Short Summary:**

Passes bass while filtering treble, enabling low-pass filtering to the woofer.
-

4. Function Generator

- **What is it?**

A test instrument that generates various electrical waveforms over a wide range of frequencies.
- **Structure:**
 - Multiple waveform outputs (sine, square, etc.)
 - Frequency and amplitude control
 - Output terminals for signal injection
- **What is it used for?**

- Generating sine signals to test filter response
 - Injecting controlled signals for component testing
 - **In the project:**
 - Supplies test signals to the resistor, capacitor, and inductor networks
 - Used to analyze frequency response of filters
 - **Short Summary:**

Essential for simulating audio signals in circuit testing.
-

5. Oscilloscope

- **What is it?**

A tool for viewing and analyzing electrical signals over time.
 - **Structure:**
 - Display screen
 - Input probes
 - Measures voltage over time
 - **What is it used for?**
 - Viewing input/output voltages across filters
 - Verifying signal amplitude and waveform shape
 - **In the project:**
 - Used to compare how signals are affected by resistors, capacitors, inductors
 - Validates filter performance and cutoff points
 - **Short Summary:**

Visual tool for analyzing how filters modify audio signals.
-

6. Audio Crossover (Prebuilt and DIY)

- **What is it?**

An electronic circuit that splits audio signals into different frequency bands for appropriate speakers.
- **Structure:**
 - Passive components: resistors, capacitors, inductors
 - Two outputs: woofer and tweeter
 - Two filters: high-pass and low-pass
- **What is it used for?**
 - Directing low frequencies to woofer
 - Directing high frequencies to tweeter
 - Ensuring optimal performance of speaker components
- **In the project:**
 - Prebuilt crossover analyzed and reverse-engineered
 - DIY filters created with RC and LC circuits
 - Software (WinISD) used to simulate crossover response

- **Short Summary:**
Filters and splits audio signals to match speaker capabilities.
-

7. Amplifier

- **What is it?**
An electronic device that increases the power of an input signal.
 - **Structure:**
 - Input and output terminals
 - Power source
 - Gain control
 - **What is it used for?**
 - Boosting function generator output
 - Driving resistor dummy load in tests
 - **In the project:**
 - Necessary to deliver enough power to simulate real speaker conditions
 - **Short Summary:**
Increases weak audio signals for proper filter testing.
-

8. Loudspeakers (Woofer and Tweeter)

- **What is it?**
Electroacoustic transducers that convert electrical signals into sound.
 - **Structure:**
 - Woofer: large cone, handles low frequencies
 - Tweeter: small dome, handles high frequencies
 - Impedance: typically 4Ω – 8Ω
 - **What is it used for?**
 - Reproducing low (woofer) and high (tweeter) frequency sounds
 - Working with crossover filters to optimize sound
 - **In the project:**
 - Demonstrates why frequency separation is essential
 - Replaced with resistors during filter testing
 - **Short Summary:**
Converts filtered signals into audible sound, split by crossover.
-

Summary

This video explored how **audio crossovers** use **resistors, capacitors, and inductors** to split sound into low and high frequencies for **woofers and tweeters**. Through practical experiments

with **RC and LC filters**, the video showed how frequency-dependent behavior creates **high-pass** and **low-pass** filtering. It also reverse-engineered a commercial crossover to highlight the importance of proper tuning based on speaker response, concluding that using a generic replacement can harm audio quality.

VIDEO 42: Is it easy to create your own Transformer? Everything you need to know about Transformers! || EB#42

List of Components Used

- Iron Core (Electrical Steel Sheets)
- Primary and Secondary Coils (Copper Wire)
- 230V to 13.5V Transformer
- 10 Ω Resistor (used as load)
- 3.3 Ω Resistor (used as heavy load)
- Oscilloscope (for waveform viewing)
- Multimeter (for voltage/current/resistance)

Detailed Breakdown of Each Component

- **Iron Core (Electrical Steel Sheets)**
 - *What is it?* The magnetic core in transformers made from stacked steel laminations to reduce eddy currents.
 - *Structure:*
 - Laminated layers of electrical steel
 - High permeability, low conductivity
 - Used in E-I or toroidal form
 - *What is it used for?*
 - Guides and concentrates magnetic flux between coils
 - Minimizes energy losses
 - *In the project:*
 - Forms the magnetic path in the transformer
 - Demonstrates energy transfer between primary and secondary coils
 - *Summary:* The magnetic backbone of the transformer—boosting efficiency and performance.
-

- **Primary and Secondary Coils (Copper Windings)**
 - *What is it?* Coils of enameled copper wire responsible for voltage step-up/down via electromagnetic induction.
 - *Structure:*
 - Primary coil: connected to 230V input

- Secondary coil: outputs 13.5V at up to 1.6A
 - Windings ratio determines voltage transformation
 - *What is it used for?*
 - Create magnetic field (primary)
 - Induce voltage (secondary)
 - *In the project:*
 - Central to voltage transformation demonstration
 - Used to show how turn ratio affects output
 - *Summary:* The coil duo—creating and harvesting magnetic flux to change voltage safely.
-

- **230V to 13.5V Transformer**

- *What is it?* A complete transformer unit built with a steel core and primary/secondary coils.
 - *Structure:*
 - Input: 230V AC, 50Hz
 - Output: ~13.5V AC, up to 1.6A
 - Physical unit contains core + coils
 - *What is it used for?*
 - Safely steps down dangerous mains voltage to low-voltage AC
 - *In the project:*
 - Main subject of the video
 - Used to explain magnetic coupling, losses, and design principles
 - *Summary:* The star of the show—turning deadly mains into friendly volts.
-

- **10 Ω Resistor**

- *What is it?* A fixed resistor used to simulate a safe electrical load.
 - *Structure:*
 - 2 leads, resistive element
 - Rated for power dissipation based on current
 - *What is it used for?*
 - Draws ~1.4A to simulate medium load
 - *In the project:*
 - Used to demonstrate transformer under moderate load
 - Enables efficiency testing (~82%)
 - *Summary:* A safe sink for current—revealing how the transformer behaves under load.
-

- **3.3 Ω Resistor**

- *What is it?* A lower resistance load used to push the transformer to its limits.

- *Structure:*
 - Similar to 10 Ω resistor
 - *What is it used for?*
 - Simulates overload condition
 - *In the project:*
 - Shows transformer voltage drop and heat loss under stress
 - Efficiency drops to 72%
 - *Summary:* The tough test case—proving what happens beyond safe limits.
-

- **Oscilloscope**

- *What is it?* A device that visualizes voltage waveforms over time.
 - *Structure:*
 - Screen, probes, trigger and measurement system
 - *What is it used for?*
 - Displays sinewave input/output
 - Analyzes distortion and phase
 - *In the project:*
 - Used to validate transformer waveform characteristics
 - *Summary:* Like X-ray vision for voltage—watching electricity dance.
-

- **Multimeter**

- *What is it?* A handheld instrument for measuring voltage, current, resistance.
 - *Structure:*
 - Probes, selector dial, digital display
 - *What is it used for?*
 - Measures coil resistance, current flow, output voltage
 - *In the project:*
 - Verifies calculations vs real-world behavior
 - *Summary:* The trusty sidekick—double-checking every claim with numbers.
-

 Summary

The video breaks down how a transformer works—from magnetic flux to losses and efficiency—while explaining the significance of core material, coil design, and electrical behavior under different loads. It illustrates electromagnetic principles practically using test equipment and explains how each component affects power delivery and losses. A must-watch for understanding real-world transformer behavior and DIY design fundamentals.

VIDEO 43 : Controlling Mechanical 7-Segment Displays?! How RS-485 and UART works! || EB#43

List of Components Used

- Mechanical Seven-Segment Display
- Electromagnets
- ATmega32A Microcontroller
- High-Voltage Source Driver IC
- Darlington Transistor Array IC
- RJ11 Connectors
- RS-485 Interface
- MAX485 Breakout Board
- Arduino Nano
- ESP8266 Microcontroller

Detailed Breakdown of Each Component

- **Mechanical Seven-Segment Display**
 - *What is it?* A display module that uses moving plastic segments controlled magnetically.
 - *Structure:*
 - 7 movable plastic segments per digit
 - Embedded permanent magnets
 - Each segment controlled by an electromagnet
 - *What is it used for?*
 - Visually shows numeric values via mechanical movement
 - *In the project:*
 - Core visual component used to display YouTube subscriber count
 - *Summary:* Eye-catching mechanical digits that use magnetism instead of LEDs.
-

- **Electromagnets**
 - *What is it?* Coils of wire that generate magnetic fields when energized.
 - *Structure:*
 - Wound wire coils, one per segment
 - Controlled with polarity-switching circuits
 - *What is it used for?*
 - Push/pull plastic segments to display numbers
 - Maintain state without continuous power
 - *In the project:*
 - Switch segment positions for each display digit

- *Summary:* Magnetic muscles that move each digit into place.
-

- **ATmega32A Microcontroller**

- *What is it?* An 8-bit AVR microcontroller from Atmel.
 - *Structure:*
 - 40-pin DIP/MLF package
 - Interfaces: USART, SPI, I²C
 - GPIOs, timers, ADC
 - *What is it used for?*
 - Main brain of the display controller board
 - *In the project:*
 - Processes serial commands to drive the display
 - *Summary:* The coordinator commanding digits via serial signals.
-

- **High-Voltage Source Driver IC**

- *What is it?* An integrated circuit that supplies power to loads like electromagnets.
 - *Structure:*
 - Multiple high-side driver outputs
 - Handles high current loads
 - *What is it used for?*
 - Powers selected display's electromagnets one at a time
 - *In the project:*
 - Enables multiplexed control of multiple digit units
 - *Summary:* Smart power provider that energizes only what's needed.
-

- **Darlington Transistor Array IC**

- *What is it?* A chip with multiple high-gain transistor pairs.
 - *Structure:*
 - 7 or 8 transistor pairs per IC
 - Common emitters, pull-down resistors
 - *What is it used for?*
 - Sinks current from selected segment electromagnets
 - *In the project:*
 - Activates specific segments across all digits
 - *Summary:* The electrical bouncers deciding who gets grounded.
-

- **RJ11 Connectors**

- *What is it?* 6-pin telephone-style connectors.

- *Structure:*
 - Pinout provided by manufacturer
 - Carries RS-485 signals
 - *What is it used for?*
 - Serial interface connection between control board and MCU
 - *In the project:*
 - Entry point for RS-485 commands from Arduino/ESP8266
 - *Summary:* Communication ports for serial instructions.
-

- **RS-485 Interface**

- *What is it?* Electrical standard for differential data transmission.
 - *Structure:*
 - Two twisted-pair wires: A (non-inverting), B (inverting)
 - Full/half-duplex support
 - *What is it used for?*
 - Robust long-distance asynchronous communication
 - *In the project:*
 - Sends display commands to controller via UART
 - *Summary:* Noise-resistant link between brain and display.
-

- **MAX485 Breakout Board**

- *What is it?* A small board featuring the MAX485 RS-485 transceiver IC.
 - *Structure:*
 - 8-pin IC with TX/RX/DE/RE control
 - Converts UART to RS-485
 - *What is it used for?*
 - Interface between Arduino/ESP and RS-485 bus
 - *In the project:*
 - Enables microcontroller to talk to the display module
 - *Summary:* Translator that speaks UART on one side, RS-485 on the other.
-

- **Arduino Nano**

- *What is it?* A compact microcontroller board based on ATmega328P.
- *Structure:*
 - Mini USB, UART/SPI/I²C, 14 digital pins, 8 analog pins
- *What is it used for?*
 - Sends display values through RS-485 using USART
- *In the project:*
 - First test setup for verifying display protocol
- *Summary:* Small but mighty test driver for display control.

- **ESP8266 Microcontroller**

- *What is it?* A Wi-Fi-enabled microcontroller for IoT applications.
 - *Structure:*
 - GPIOs, UART, ADC
 - Built-in Wi-Fi stack
 - *What is it used for?*
 - Fetches live subscriber count from YouTube API
 - *In the project:*
 - Final brain sending real-time data to the display
 - *Summary:* The cloud-connected genius behind the subscriber counter.
-

 Summary

The video showcases a mechanical seven-segment display powered by electromagnets and smart multiplexing via ATmega32A. A combination of source drivers, transistor arrays, and RS-485 communication allow control of 10 digits through just a few wires. Using an Arduino Nano and later an ESP8266, the creator crafts a YouTube subscriber counter that visually ticks with each new sub—blending magnetic magic with microcontroller precision.

VIDEO 44 : Improving my electric longboard with a CAN Bus! What can the CAN Bus do? EB#44

 List of Components Used

- FS ESC (Electronic Speed Controller)
- CAN Bus Interface (CANH & CANL)
- 4-Pin Connector for CAN
- Electric Longboard Hub Motors
- VESC Tool Software
- Oscilloscope
- CAN Transceiver Circuitry

 Detailed Breakdown of Each Component

- **FS ESC (Electronic Speed Controller)**

- *What is it?* A motor controller specifically for electric skateboards and other hobby motors.
- *Structure:*
 - Multiple ports including PPM, USB, and 4-pin CAN port
 - CANH, CANL, 5V, and GND on CAN connector

- *What is it used for?*
 - Converts remote input to drive signals for BLDC motors
 - *In the project:*
 - Two FS ESCs drive individual hub motors; syncing them via CAN fixes motor misalignment
 - *Summary:* The motor's translator—turning control signals into motion and now working as a team via CAN.
-

- **CAN Bus Interface (CANH & CANL)**

- *What is it?* A two-wire serial communication standard used to link multiple control units.
 - *Structure:*
 - CANH (High) and CANL (Low) differential pair
 - Typically operates around 2.3V common, 2.9V high, 1.0V low
 - *What is it used for?*
 - Enables FS ESCs to sync motor behavior and communicate directly
 - *In the project:*
 - Used to connect both ESCs so they share control data, eliminating RPM mismatch
 - *Summary:* The chat line between ESCs—no more arguing, just synchronized spinning.
-

- **4-Pin Connector for CAN**

- *What is it?* Connector carrying 5V, GND, CANH, and CANL.
 - *Structure:*
 - Matches CAN port layout on ESC
 - Wired between ESCs
 - *What is it used for?*
 - Physically connects CAN interfaces for communication
 - *In the project:*
 - Wires soldered to create communication bridge between both ESCs
 - *Summary:* The physical handshake that made electrical harmony possible.
-

- **Electric Longboard Hub Motors**

- *What is it?* Brushless DC motors integrated into skateboard wheels.
- *Structure:*
 - Three-phase BLDC construction
 - Controlled via ESC
- *What is it used for?*
 - Drive wheels on electric longboard

- *In the project:*
 - Show RPM mismatch at low inputs; resolved with CAN synchronization
 - *Summary:* The wheels that needed a pep talk to move as one.
-

- **VESC Tool Software**

- *What is it?* Software for configuring and monitoring VESC-compatible ESCs.
 - *Structure:*
 - GUI for setting parameters like CAN mode, input method, RPM sync
 - *What is it used for?*
 - Configures multiple ESCs to communicate via CAN
 - *In the project:*
 - Enabled “Multiple VESCs over CAN” and adjusted CAN mode
 - *Summary:* The ESC whisperer—setting everything up for perfect partnership.
-

- **Oscilloscope**

- *What is it?* Instrument for visualizing electrical signals.
 - *Structure:*
 - Probes connected to CANH and CANL
 - *What is it used for?*
 - Measures signal voltages and verifies CAN bus behavior
 - *In the project:*
 - Confirmed proper voltage swing and data transfer over CAN lines
 - *Summary:* Showed the electrical language of CAN in full waveform glory.
-

- **CAN Transceiver Circuitry**

- *What is it?* Part of the ESC or external module that converts CAN protocol into logic levels.
 - *Structure:*
 - Integrated differential driver and receiver
 - *What is it used for?*
 - Allows microcontroller/ESC to send and receive CAN messages
 - *In the project:*
 - Built-in on ESC; responsible for encoding/decoding CAN frames
 - *Summary:* The gatekeeper that enables devices to talk CAN fluently.
-



Summary

The video explores synchronizing two electric longboard motors using the CAN bus built into their FS ESCs. By connecting CANH and CANL lines and configuring them with VESC Tool,

the motors spin in perfect sync even with only one signal input. The creator also breaks down the CAN protocol's structure, priority system, error checking, and why it's a powerful, robust communication method not just for vehicles—but also for microcontroller projects like this. Smooth torque, matched RPMs, and smarter control—thanks to just two wires.

VIDEO 45 : Building a Digital Music Player with I2S?! What is I2S! EB#45

List of Components Used

- ESP32 Microcontroller
- Micro SD Card Module
- MAX98357A I2S Amplifier Board
- INMP441 I2S Microphone Module
- Micro SD Card
- Speaker
- Oscilloscope

Detailed Breakdown of Each Component

- **ESP32 Microcontroller**
 - *What is it?* A powerful Wi-Fi/Bluetooth-enabled MCU with audio-capable peripherals.
 - *Structure:*
 - Multiple GPIO pins
 - 2× I2S peripherals
 - Integrated DAC (8-bit)
 - *What is it used for?*
 - Reads audio data from SD card via SPI
 - Sends audio to I2S amplifier
 - Reads digital mic input
 - *In the project:*
 - Acts as central controller for audio playback and capture
 - *Summary:* All-in-one controller that reads, routes, and plays digital audio.

- **Micro SD Card Module**
 - *What is it?* A board for reading microSD cards via SPI.
 - *Structure:*
 - Uses SPI (MISO, MOSI, SCK, CS)
 - Powered by 3.3V
 - *What is it used for?*

- Stores 16-bit/44.1kHz .wav audio files
 - *In the project:*
 - Supplies audio data to ESP32 via SPI
 - *Summary:* Tiny memory vault feeding pristine audio data.
-

- **MAX98357A I2S Amplifier Board**

- *What is it?* A mono class-D amplifier with built-in DAC and I2S interface.
 - *Structure:*
 - Pins: Vin, GND, SD, LRCLK (WS), BCLK (SCK), Din
 - Output: Speaker pins
 - Power: 3.3–5V
 - *What is it used for?*
 - Converts I2S PCM data into amplified analog audio
 - *In the project:*
 - Plays the “Stay Creative!” clip from ESP32
 - *Summary:* Digital audio in, amplified sound out—crisp and clean.
-

- **INMP441 I2S Microphone Module**

- *What is it?* A MEMS mic breakout with I2S digital output.
 - *Structure:*
 - Pins: VDD, GND, WS, SCK, SD, L/R
 - Outputs 16-bit PCM audio
 - *What is it used for?*
 - Sends microphone audio data to ESP32 over I2S
 - *In the project:*
 - Demonstrates I2S reading by visualizing mic data
 - *Summary:* Digitally whispers into the ESP32’s ear.
-

- **Micro SD Card**

- *What is it?* Removable flash storage containing audio files.
 - *Structure:*
 - FAT32 format with .wav file
 - 16-bit resolution, 44.1kHz sample rate
 - *What is it used for?*
 - Holds the “Stay Creative!” audio clip
 - *In the project:*
 - Audio source for playback demonstration
 - *Summary:* Stores sound in bits, ready to relive when called.
-

- **Speaker**

- *What is it?* An output transducer for generating audible sound.
- *Structure:*
 - Typically 8Ω impedance
 - Connected to MAX98357A output
- *What is it used for?*
 - Plays final analog signal from amplifier
- *In the project:*
 - Outputs voice clip audibly to the user
- *Summary:* The voice box of this sonic experiment.

- **Oscilloscope**

- *What is it?* An instrument for viewing voltage waveforms in real time.
- *Structure:*
 - Channels for multiple signal probes
 - Display for voltage vs. time
- *What is it used for?*
 - Monitors I2S signals (BCLK, WS, DATA)
- *In the project:*
 - Confirms signal integrity and timing of I2S lines
- *Summary:* Reveals the secret waveforms behind every sound byte.



Summary

The video shows how to play high-quality audio from a microSD card using an ESP32 and an I2S amplifier. While the ESP32's internal DAC is limited to 8-bit audio, the I2S bus enables streaming 16-bit PCM directly to a MAX98357A amplifier, producing great sound. It also dives into I2S theory with a microphone example, oscilloscope views, and code breakdown. Result: your ESP32 can literally say "Stay Creative!" with stellar audio quality.

VIDEO 46 : Does this old Induction Motor still work? || How do Asynchronous Motors work? EB#46



List of Components Used

- Asynchronous (Induction) Motor
- Motor Terminal Box
- Star and Delta Coil Configurations
- 3-Phase 400V AC Power Supply (CEE Cable)
- Multimeter
- Tachometer
- Capacitor (for single-phase asynchronous motor)

■ Detailed Breakdown of Each Component

- **Asynchronous (Induction) Motor**

- *What is it?* An AC motor that uses electromagnetic induction to generate motion—also known as an induction motor.
 - *Structure:*
 - Rotor: Squirrel cage structure made of conductive bars shorted at both ends
 - Stator: Three-phase winding coils embedded in laminated steel core
 - Terminal box: Six terminals (U1, U2, V1, V2, W1, W2)
 - Cooling fan on rotor shaft
 - *What is it used for?*
 - Converts 3-phase AC electrical energy into rotational mechanical energy
 - *In the project:*
 - Demonstrated functional analysis and repair, showing configuration and behavior under power
 - *Summary:* Reliable workhorse of the motor world—simple, rugged, and still spinning after ages.
-

- **Motor Terminal Box**

- *What is it?* Connection box for accessing coil terminals and configuring wiring mode.
 - *Structure:*
 - Six terminals grouped in three coil pairs
 - Can be connected in Delta or Star (Y) mode using metal jumpers
 - *What is it used for?*
 - Allows setting the coil configuration based on input voltage (230V or 400V)
 - *In the project:*
 - Configured in star mode for safe operation with 400V three-phase input
 - *Summary:* The configurable interface between grid power and torque.
-

- **Star and Delta Coil Configurations**

- *What is it?* Two ways of wiring the three stator coils to suit voltage requirements.
- *Structure:*
 - Star: One side of each coil joined; other ends to phases
 - Delta: Each coil connected between two phases
- *What is it used for?*
 - Star: Used for 400V phase-to-phase systems (each coil gets 230V)
 - Delta: Used for 230V phase-to-phase systems

- *In the project:*
 - Star configuration used to safely match motor coil rating
 - *Summary:* The method that determines whether your motor thrives or goes up in smoke.
-

- **3-Phase 400V AC Power Supply (CEE Cable)**

- *What is it?* High-voltage mains cable providing three-phase power used in European systems.
 - *Structure:*
 - 5 conductors: L1, L2, L3, Neutral, PE (Protective Earth)
 - 400V between any two phases, 230V from phase to neutral
 - *What is it used for?*
 - Powers the induction motor via L1, L2, L3 in star mode
 - *In the project:*
 - Safely delivered proper phase voltage to spin the motor
 - *Summary:* High-voltage juice—handle with care, wire with wisdom.
-

- **Multimeter**

- *What is it?* A tool to measure electrical properties—voltage, current, resistance.
 - *Structure:*
 - Leads, display, mode selector
 - *What is it used for?*
 - Verified coil continuity and identity in terminal box
 - *In the project:*
 - Helped identify coil pairs U1–U2, V1–V2, W1–W2
 - *Summary:* Trusty probe buddy for electrical detective work.
-

- **Tachometer**

- *What is it?* Device for measuring rotational speed (RPM) of a spinning object.
 - *Structure:*
 - Optical sensor + digital readout
 - *What is it used for?*
 - Confirmed motor speed below theoretical 3000 RPM due to slip
 - *In the project:*
 - Verified that rotor speed was appropriate for a 50 Hz asynchronous motor
 - *Summary:* Speed sniffer revealing the subtle truth about slip.
-

- **Capacitor (used in single-phase asynchronous motors)**

- *What is it?* A passive electrical component that stores electrical energy.
 - *Structure:*
 - Typically non-polarized or run capacitor in metal canister
 - *What is it used for?*
 - Creates phase shift in single-phase motors to simulate three-phase rotation
 - *In the project:*
 - Found in terminal box of single-phase water pump to produce rotating field
 - *Summary:* The clever trick to coax three-phase performance from two wires.
-



Summary

The video dives into the workings of the asynchronous induction motor—a quiet giant of the electrical world. It walks through the rotor-stator design, star/delta wiring, voltage ratings, and the creation of rotating magnetic fields without any permanent magnets. Using high-voltage power safely, it demonstrates how slip makes the motor “asynchronous,” and why this humble motor remains the global favorite: affordable, rugged, and easy to run straight off the grid.

VIDEO 47 : Building a Tube Amp! Does it produce better audio quality though? EB#47



List of Components Used

- 6J4 Triode Vacuum Tube
- Tube Amplifier Module
- Operational Amplifiers
- Class D Amplifier IC
- Headphone Amplifier IC
- High Voltage DC-DC Converter
- Function Generator
- BC637 NPN Transistor
- Multimeter
- Lab Bench Power Supply



Detailed Breakdown of Each Component

- **6J4 Triode Vacuum Tube**
 - *What is it?* A small signal vacuum tube triode used for audio pre-amplification.
 - *Structure:*
 - Heater: 6.3V rated (glows when active)
 - Anode (plate), Cathode, Grid
 - Max current: ~20mA

- *What is it used for?*
 - Provides warm-sounding analog pre-amplification
 - *In the project:*
 - Used as an audio signal preamp stage in the hybrid amplifier
 - *Summary:* Vintage amplifier heart—adds warm tone with characteristic distortion.
-

- **Tube Amplifier Module**

- *What is it?* A hybrid amplifier combining vacuum tube preamp with solid-state output stages.
 - *Structure:*
 - Metal case, replaceable tubes
 - Bluetooth & analog stereo input
 - Speaker & headphone output
 - *What is it used for?*
 - Drives speakers or headphones with pre-amplified signal
 - *In the project:*
 - Device under test for evaluating tube sound quality
 - *Summary:* Old meets new—a stylish throwback in a modern shell.
-

- **Operational Amplifiers**

- *What is it?* ICs that amplify low-power signals with high precision.
 - *Structure:*
 - Typically dual or quad op-amps per chip
 - Low voltage supply and high gain
 - *What is it used for?*
 - Buffers and filters input before Class D stage
 - *In the project:*
 - Found on PCB doing signal conditioning work
 - *Summary:* Quiet supporters making sure the signal's polished and clean.
-

- **Class D Amplifier IC**

- *What is it?* A high-efficiency power amp using pulse-width modulation to drive speakers.
- *Structure:*
 - Digital switching output
 - High power efficiency (>90%)
- *What is it used for?*
 - Drives speaker output
- *In the project:*
 - Final amplification stage after tube preamp

- *Summary:* Efficient powerhouse delivering sound to your speakers.
-

- **Headphone Amplifier IC**

- *What is it?* A dedicated IC for driving low-impedance headphones.
 - *Structure:*
 - Integrated amp with low output noise
 - Runs on low-voltage supply
 - *What is it used for?*
 - Cleanly powers headphones from line-level signal
 - *In the project:*
 - Drives 3.5mm headphone output
 - *Summary:* The personal sound booster for late-night listening.
-

- **High Voltage DC-DC Converter**

- *What is it?* A compact adjustable converter that steps up voltage for tube circuits.
 - *Structure:*
 - 12V input, adjustable up to ~150V output
 - Switching regulator (24.4kHz observed)
 - *What is it used for?*
 - Powers the anode of the 6J4 vacuum tube
 - *In the project:*
 - Used to build DIY class A amp circuit with tube
 - *Summary:* Little board, big volts—feeds the tube its lifeblood.
-

- **Function Generator**

- *What is it?* A device for generating sine, square, and other waveform signals.
 - *Structure:*
 - Frequency, amplitude, offset adjustable output
 - *What is it used for?*
 - Supplies test signal to input of amplifier circuits
 - *In the project:*
 - Used to test and compare signal output from tube vs. transistor circuits
 - *Summary:* Waveform wizard for clean and controlled testing.
-

- **BC637 NPN Transistor**

- *What is it?* A general-purpose BJT for small signal amplification.
- *Structure:*
 - 3 pins: Collector, Base, Emitter

- Max current: ~1A
 - *What is it used for?*
 - Built a class A amp circuit for comparison
 - *In the project:*
 - Served as a benchmark against the triode's output
 - *Summary:* The practical workhorse, good ol' reliable silicon.
-

- **Multimeter**

- *What is it?* A tool for measuring voltage, current, resistance.
 - *Structure:*
 - Digital display, probes, selector dial
 - *What is it used for?*
 - Checked output voltage from high voltage converter
 - *In the project:*
 - Verified safe and accurate DC supply to tube
 - *Summary:* The honest referee checking every electrical stat.
-

- **Lab Bench Power Supply**

- *What is it?* Adjustable DC power source for electronics prototyping.
 - *Structure:*
 - Controls for voltage/current
 - Accurate regulated output
 - *What is it used for?*
 - Powered tube heater (6.3V)
 - *In the project:*
 - Delivered precise voltage to warm up the vacuum tube
 - *Summary:* Silent partner ensuring glowing tubes and happy electrons.
-

Summary

This video explores whether vacuum tubes still hold their ground in modern audio. The creator examines a tube amplifier, builds a simple triode-based class A amp, and compares its behavior to a transistor alternative. While the tube produces more distortion and requires high-voltage power, some listeners find its sonic imperfections pleasant and "warm." Ultimately, tube preamps aren't objectively better—but they *do* offer nostalgic charm and analog character that some ears just love.

VIDEO 48 : The Best Protection for your Circuits? eFuse! Here is why they are awesome! EB#48

List of Components Used

- TPS259621 eFuse IC
- Arduino Nano
- Resistors (for voltage threshold and current limit)
- High Voltage DC Converter Module
- P-Channel MOSFET (for reverse protection)
- Perfboard
- Power Bank
- Power Supply or Battery Source

Detailed Breakdown of Each Component

- **TPS259621 eFuse IC**
 - *What is it?* A protection IC that combines overvoltage, undervoltage, overcurrent, and thermal safeguards.
 - *Structure:*
 - IN, OUT, GND pins
 - EN/UV pin for enabling and undervoltage threshold
 - OV pin for overvoltage threshold
 - ILM pin for current limit
 - Additional: FAULT, dVdt pins
 - *What is it used for?*
 - Protects circuits powered by external, possibly unsafe sources
 - *In the project:*
 - Key component in the DIY LiPo supercharger and Arduino projects
 - *Summary:* Tiny bouncer that lets in just the right voltage and current—and kicks out the rest.
-

- **Arduino Nano**
 - *What is it?* A compact microcontroller board based on the ATmega328P.
 - *Structure:*
 - USB, GPIO, analog and digital I/O
 - *What is it used for?*
 - Demonstrates if power protection is working through a blinking LED
 - *In the project:*
 - Acts as a test load for the eFuse IC
 - *Summary:* Trusty sidekick showing if power flows or fails.

- **Resistors (various values)**

- *What is it?* Passive components to control voltage thresholds and current limits.
 - *Structure:*
 - Pull-up/pull-down configs, divider networks
 - *What is it used for?*
 - Sets undervoltage at ~4V and overvoltage at ~5.7V
 - Limits output current to ~200mA
 - *In the project:*
 - Used in series with eFuse control pins for fine tuning
 - *Summary:* Precision tools dialing in just the right protection parameters.
-

- **High Voltage DC Converter Module**

- *What is it?* A step-up module originally used in a Nixie clock project.
 - *Structure:*
 - Input: 12V
 - Output: Adjustable (e.g., 100V)
 - *What is it used for?*
 - Powered eFuse and tube circuits in past builds
 - *In the project:*
 - Demonstrated power source setup before refining supply approach
 - *Summary:* Adjustable power backbone—hot and high-voltage.
-

- **P-Channel MOSFET**

- *What is it?* A transistor used for switching and reverse voltage protection.
 - *Structure:*
 - Source, Gate, Drain terminals
 - Works when gate is pulled low relative to source
 - *What is it used for?*
 - Prevents reverse polarity damage
 - *In the project:*
 - Complements eFuse for full protection
 - *Summary:* Guardian gatekeeper—won't let voltage sneak in the wrong way.
-

- **Perfboard**

- *What is it?* A prototyping board for soldering permanent circuits.
- *Structure:*
 - Grid of holes with copper pads
- *What is it used for?*

- Used to finalize the power-safe Arduino circuit
 - *In the project:*
 - Forms the physical base for the permanent build
 - *Summary:* The sturdy canvas for your protected electronics.
-

- **Power Bank**

- *What is it?* A portable 5V battery pack with built-in protection.
 - *Structure:*
 - USB output, internal Li-ion cells
 - *What is it used for?*
 - Initial power source recommendation
 - *In the project:*
 - Safe beginner-friendly supply
 - *Summary:* Portable, protected, and always there when a USB jack is needed.
-

- **Power Supply or Battery Source**

- *What is it?* Any external source that delivers power to your circuit.
 - *Structure:*
 - May include batteries, panels, adapters
 - *What is it used for?*
 - Demonstrates why protection is critical
 - *In the project:*
 - Simulates unsafe or varying inputs to trigger protection
 - *Summary:* The unpredictable outsider—the reason you build protection in the first place.
-



Summary

This video introduces the TPS259621 eFuse IC—a compact, configurable circuit protector that guards against undervoltage, overvoltage, overcurrent, and overheating. Through a practical Arduino Nano LED circuit, it showcases how to wire the IC with carefully chosen resistors to set voltage thresholds and current limits. Though it lacks built-in reverse voltage protection, a P-channel MOSFET neatly fills that gap. Affordable, solderable, and incredibly useful, this IC makes power protection modular and beginner-friendly—whether you're charging LiPo packs or building walkie-talkies on perfboard.

VIDEO 49: Everything you need to know when buying/using an Oscilloscope! EB#49

List of Components Used

- Oscilloscope
- Oscilloscope Passive Probe
- Function Generator
- Switched Mode Power Supply (prototype)
- Boost Converter
- Capacitor
- Current Shunt
- Current Clamp
- Differential Probe

Detailed Breakdown of Each Component

- Oscilloscope
 - What is it? An electronic test instrument that graphically displays varying electrical signals, typically as a two-dimensional plot of one or more electrical potential differences (voltage) as a function of time.
 - Structure:
 - **Channels:** Determines how many voltage and/or current signals can be viewed simultaneously (e.g., 2-channel, 4-channel).
 - **Bandwidth:** The frequency at which the input signal gets damped by -3dB (amplitude goes down to 70.7%). Higher bandwidth allows for more accurate measurement of high-frequency signals.
 - **Sampling Rate:** How many measurements the scope takes in a second (e.g., GSa/s - GigaSamples per second).
 - **Vertical Knob:** Adjusts the vertical voltage division (Volts per division).
 - **Horizontal Knob:** Adjusts the horizontal time division (seconds per division).
 - **Trigger Button/Settings:** Used to stabilize repetitive waveforms by capturing the waveform at a specific point (e.g., rising/falling edge, pulse width, pattern).
 - **Input Coupling (AC/DC):**
 - **DC Coupling:** Shows both AC and DC components of the signal.
 - **AC Coupling:** Adds a series capacitor to block the DC component, allowing only the AC signal to pass, useful for zooming in on ripple.

- **Measure Function:** Built-in features to automatically display parameters like frequency, peak-to-peak voltage, etc.
 - **Cursor Function:** Allows manual measurement of specific points on the waveform, useful for determining rise time, time differences, etc.
 - **Single Mode Capturing:** Captures a single event or non-repetitive signal when the trigger condition is met.
 - **Math Function:** Performs mathematical operations on waveforms.
 - **FFT (Fast Fourier Transform) Function:** Analyzes the frequency components of a signal.
- What is it used for?
 - Visualizes voltage and/or current values over time.
 - Essential for building, debugging, and repairing electronics by observing waveforms, rise times, frequencies, and anomalies.
 - Measures voltage, current (indirectly), frequency, rise/fall times, and ripple.
- In the project:
 - The main tool demonstrated in the video.
 - Used to visualize and analyze various electrical signals from different circuits (Switched Mode Power Supply, function generator, boost converter).
 - Demonstrates how to correctly set up and use an oscilloscope for basic and advanced measurements, including safe mains voltage measurements.
- Oscilloscope Passive Probe
 - What is it? An accessory for an oscilloscope that connects the circuit under test to the oscilloscope's input.
 - Structure:
 - **BNC Connector:** Connects to the oscilloscope's input.
 - **Alligator Clip (GND):** Connects to the ground (GND) reference of the circuit.
 - **Probe Tip:** Touches the point of the circuit to be measured.
 - **Scaling Factor Switch (x1/x10):**
 - **x1:** Passes the signal directly to the scope. Lower resistance, higher capacitance, lower bandwidth.
 - **x10:** Decreases the voltage signal by a factor of 10. Higher resistance, lower capacitance, higher bandwidth.
 - What is it used for?
 - To safely and effectively connect a circuit's signal to the oscilloscope for measurement and visualization.
 - The x10 setting is generally preferred for its higher bandwidth and lower loading effect on the circuit.

- In the project:
 - Used to connect various circuits (e.g., Switched Mode Power Supply, function generator) to the oscilloscope for waveform observation.
 - The video emphasizes the importance of using the correct scaling factor (x10) and safely connecting the ground clip, especially for mains voltage measurements.
- Function Generator
 - What is it? An electronic test equipment that generates various types of electrical waveforms over a wide range of frequencies.
 - Structure: Not detailed in the video, but typically has controls for waveform type (sine, square, triangle), frequency, and amplitude.
 - What is it used for?
 - To produce specific waveforms for testing electronic circuits.
 - Used to test the bandwidth and response of an oscilloscope.
 - In the project:
 - Used to produce a sine wave with variable frequency (e.g., 80MHz) to demonstrate the effect of oscilloscope bandwidth on signal amplitude at higher frequencies.
- Switched Mode Power Supply (prototype)
 - What is it? A type of electronic power supply that incorporates a switching regulator to convert electrical power efficiently. It converts AC mains voltage to a regulated DC output.
 - Structure: A crude but functional prototype, details not fully described, but implies components for AC-DC conversion, switching, and regulation.
 - What is it used for?
 - Efficiently converts high AC voltage (e.g., 230V) to a lower, regulated DC voltage (e.g., 5V).
 - In the project:
 - The initial project example shown at the beginning of the video.
 - Its 5V DC output was measured with the oscilloscope to confirm proper operation and to introduce the concept of safely working with mains voltage and oscilloscopes.
- Boost Converter
 - What is it? A DC-to-DC converter that steps up voltage from its input (source) to its output (load).
 - Structure: Not detailed in the video, but typically involves an inductor, capacitor, diode, and switching element.

- What is it used for?
 - To increase a DC voltage to a higher DC voltage.
 - Often used in applications where a higher voltage is needed from a lower voltage source.
- In the project:
 - Used as an example circuit to demonstrate measuring voltage ripple on its output while powering a load.
 - The oscilloscope's AC coupling and measure function were used to determine the peak-to-peak ripple voltage, indicating the converter's suitability for a given job.
- Capacitor
 - What is it? A passive two-terminal electrical component that stores electrical energy in an electric field.
 - Structure: Two conductive plates separated by a dielectric material. Has two terminals.
 - What is it used for?
 - To store and release electrical energy.
 - Used for filtering, smoothing, and timing applications in circuits.
 - In the project:
 - Used as an example to demonstrate "single mode capturing" on the oscilloscope, showing how to capture the charging curve of a capacitor, which is a non-repetitive event.
- Current Shunt
 - What is it? A precisely manufactured resistor used to measure electric current by producing a voltage drop proportional to the current flowing through it.
 - Structure: A low-value resistor.
 - What is it used for?
 - To indirectly measure current by converting it into a measurable voltage drop.
 - In the project:
 - Mentioned as one method for measuring current with an oscilloscope.
- Current Clamp
 - What is it? A type of electrical clamp meter that uses a jaw-like structure to measure current flowing through a conductor without making physical contact or breaking the circuit.

- Structure: A clamp that goes around a wire, typically with a BNC output for connecting to an oscilloscope or multimeter.
- What is it used for?
 - To non-invasively measure AC or DC current.
- In the project:
 - Recommended as an easy-to-use method for measuring current with an oscilloscope.
- Differential Probe
 - What is it? A type of oscilloscope probe that measures the voltage difference between two points, neither of which needs to be ground. It also provides isolation from the oscilloscope's ground.
 - Structure: Has two input leads (positive and negative) and an output for the oscilloscope, with internal isolation.
 - What is it used for?
 - To safely measure signals that are floating relative to ground or where neither point can be referenced to the oscilloscope's ground, such as mains voltage across live and neutral lines.
 - Provides isolation between the circuit under test and the oscilloscope, preventing ground loops and potential damage.
 - In the project:
 - Presented as the safe solution for measuring mains voltage with an oscilloscope, avoiding potential damage to the scope or tripping circuit breakers by isolating the measurement from the oscilloscope's ground.

Summary

The video provides a comprehensive guide to using an oscilloscope, explaining its purpose and essential features. It details how to choose an oscilloscope based on channels, bandwidth, and sampling rate, and how to effectively use passive probes with correct scaling factors. Key oscilloscope functions like triggering, voltage and time division, measure, and cursor functions are explained for analyzing waveforms. The video also covers advanced techniques such as AC/DC coupling for ripple measurement and single-mode capturing for non-repetitive signals. Crucially, it emphasizes safety when measuring mains voltage, highlighting the danger of direct grounding and introducing differential probes as a safe solution.

VIDEO 50: Probably the most used component nobody knows of! TL431 Guide! EB#50

-  **List of Components Used** • TL431 (Precise Programmable Reference) • Optocoupler • Resistors (Various values including 500Ω, 650Ω, 10kΩ) • Capacitor (100nF) • Comparator • Transistor • Zener Diode

Detailed Breakdown of Each Component • TL431

- **What is it?** A precise programmable reference IC used for voltage regulation and feedback control.
- **Structure:**
 - Three pins: Ref, Anode, Cathode
 - Operates between Vref and 36V
 - Requires around 1mA current
- **What is it used for?**
 - Acts as a reference voltage regulator
 - Functions as a comparator and adjustable zener diode
 - Used in feedback circuits of switched-mode power supplies
- **In the project:**
 - Used as a voltage reference to maintain output stability in the DIY switched-mode power supply
 - Serves as part of the feedback loop to regulate voltage conversion

• Optocoupler

- **What is it?** A component that isolates sections of a circuit while transmitting signals using light.
- **Structure:**
 - LED inside optocoupler sends a signal to a phototransistor
- **What is it used for?**
 - Provides electrical isolation between sections of a circuit
 - Helps transmit feedback signals while preventing interference
- **In the project:**
 - Used in the SMPS feedback loop to regulate the output voltage using the TL431

• Resistors

- **What are they?** Passive electrical components that limit current flow.
- **Structure:**
 - Various resistances including 500Ω, 650Ω, and 10kΩ
- **What is it used for?**
 - Voltage division
 - Current limiting for TL431 and optocoupler
- **In the project:**

- Adjusts reference voltage levels for the TL431 and ensures proper operation of feedback circuits

• Capacitor (100nF)

- **What is it?** Stores and releases electrical energy to filter voltage signals.
- **Structure:**
 - Two terminals: Positive and Negative
- **What is it used for?**
 - Stabilizes feedback loop
 - Helps prevent oscillations in voltage regulation
- **In the project:**
 - Used to fine-tune the performance of the feedback system

• Comparator

- **What is it?** A circuit that compares voltage signals and outputs a high or low signal.
- **Structure:**
 - Two inputs (positive & negative) and an output
- **What is it used for?**
 - Helps monitor voltage levels and switch signals accordingly
- **In the project:**
 - Used within the TL431 to detect voltage changes and regulate circuit behavior

• Transistor

- **What is it?** A semiconductor device that switches or amplifies signals.
- **Structure:**
 - Three terminals: Base, Collector, and Emitter
- **What is it used for?**
 - Helps regulate voltage changes
- **In the project:**
 - Works with the TL431 to control voltage flow in the feedback circuit

• Zener Diode

- **What is it?** A diode that maintains a stable voltage across its terminals.
- **Structure:**
 - Two terminals: Anode and Cathode
- **What is it used for?**
 - Voltage regulation
- **In the project:**
 - TL431 acts as an adjustable zener diode, improving voltage stability

 **Summary** This video explores the TL431 IC and its various applications, focusing on how it's used in switched-mode power supplies. The TL431 is demonstrated as a comparator,

reference voltage regulator, and adjustable zener diode, forming the core of the feedback loop in the DIY SMPS. The circuit also includes an optocoupler for isolation and various resistors, capacitors, and transistors to stabilize and regulate voltage.

VIDEO 51: This component can control tons of circuits! Digital Potentiometer Guide! EB#51

List of Components Used

- Digital Potentiometer (various models, including X9C102, X9C103, X9C104, X9C503, and MCP41HVX1)
- Arduino Microcontroller
- Real Time Clock (RTC) Module (DeadOn RTC DS3231)
- Trimpot (Trimmer Potentiometer)

Detailed Breakdown of Each Component

- **Digital Potentiometer**
 - **What is it?** An electronic component that functions like a mechanical potentiometer but is controlled digitally. Its resistance value can be adjusted using digital signals.
 - **Structure:**
 - Typically has 8 pins.
 - Key pins: INC, U/D, CS, Vcc, Vss, VH/RH, VW/RW, VL/RL.
 - Has 99 or 256 tap points.
 - **What is it used for?** To digitally control circuits by adjusting resistance. Used for controlling LED brightness and adjusting the output voltage of an LED driver.
 - **In the project:** Used to control the brightness of a high-power LED by adjusting the voltage output of an LED driver.
- **Arduino Microcontroller**
 - **What is it?** An open-source electronics platform for creating interactive projects.
 - **Structure:** Compact board (e.g., Arduino Nano) with digital and analog pins. Pins D6, D7, and D8 are used to control the digital potentiometer.
 - **What is it used for?** To send digital signals to the digital potentiometer.
 - **In the project:** Used to control the digital potentiometer, which regulates the LED driver's output.
- **Real Time Clock (RTC) Module**
 - **What is it?** A time-keeping device that keeps track of the current time.

- **Structure:** Module (e.g., DeadOn RTC DS3231) with pins for power (VCC, GND) and data communication (SDA, SCL).
- **What is it used for?** To keep accurate time and date information.
- **In the project:** Proposed to control LED brightness based on a schedule.
- **Trimpot (Trimmer Potentiometer)**
 - **What is it?** A small, adjustable resistor for fine-tuning circuits.
 - **Structure:** Three terminals with a screw for adjustment.
 - **What is it used for?** To manually adjust resistance in a circuit. Used to set the output voltage of the LED driver.
 - **In the project:** Replaced by the digital potentiometer to enable digital control.
- **MCP41HVX1 (High Voltage Digital Potentiometer)**
 - **What is it?** A specific model of digital potentiometer designed for higher voltages.
 - **Structure:** 7/8-bit single digital potentiometer with SPI interface. Supports up to +36V (+/-18V). Offers resistance options like 5 k Ω , 10 k Ω , 50 k Ω , and 100 k Ω . Has 127 or 256 taps.
 - **What is it used for?** To digitally control circuits that require higher voltage adjustments.
 - **In the project:** Presented as a suitable choice for controlling the high-power LED driver.

Summary

The video explains how digital potentiometers can be used to control circuits digitally, replacing manual trimpots. The Arduino is used to control the digital potentiometer, and an RTC module can be added for scheduled control. The MCP41HVX1 is highlighted for high-voltage applications.

VIDEO 52: Negative Voltages are more important than you think! So here is how to make them! EB#52

List of Components Used

- Boost Converter IC
- Schottky Diodes
- Capacitors (Various values)
- Center-Tapped Transformer
- Full Bridge Rectifier
- Linear Voltage Regulator
- Resistors (Voltage Divider)
- Operational Amplifier

- BUF634 Buffer IC
- TLE2426 IC
- Transistors (Discrete Rail Splitter)

Detailed Breakdown of Each Component

• Boost Converter IC

- **What is it?** A DC-DC converter that steps up voltage from a lower level to a higher level.
- **Structure:**
 - Input voltage pin
 - Output voltage pin
 - Feedback pin for regulation
 - Switching transistor inside
- **What is it used for?**
 - Converts a lower DC voltage into a higher DC voltage.
- **In the project:**
 - Used as the core of the charge pump circuit to generate a dual rail voltage.

• Schottky Diodes

- **What are they?** Diodes with low forward voltage drop and fast switching speed.
- **Structure:**
 - Two terminals: Anode and Cathode
- **What is it used for?**
 - Allows current flow in one direction while minimizing voltage loss.
- **In the project:**
 - Used in the charge pump circuit to create the negative voltage rail.

• Capacitors

- **What are they?** Components that store and release electrical energy.
- **Structure:**
 - Two terminals: Positive and Negative
- **What is it used for?**
 - Stabilizes voltage and reduces noise.
- **In the project:**
 - Used in the charge pump circuit to generate the negative voltage rail.
 - Used in the transformer-based power supply to smooth rectified voltage.

• Center-Tapped Transformer

- **What is it?** A transformer with a secondary winding split into two equal halves.
- **Structure:**
 - Primary winding (connected to mains voltage)

- Secondary winding with center tap (provides dual voltage output)
- **What is it used for?**
 - Converts AC mains voltage into dual rail AC voltage.
- **In the project:**
 - Used to generate a dual rail voltage with higher current capability.

• Full Bridge Rectifier

- **What is it?** A circuit made of four diodes that converts AC voltage into DC voltage.
- **Structure:**
 - Four diodes arranged in a bridge configuration
 - Two AC input terminals
 - Two DC output terminals (+ and -)
- **What is it used for?**
 - Converts AC voltage into DC voltage.
- **In the project:**
 - Used to rectify the AC voltage from the transformer into dual rail DC voltage.

• Linear Voltage Regulator

- **What is it?** A component that maintains a stable output voltage despite input fluctuations.
- **Structure:**
 - Input voltage pin
 - Output voltage pin
 - Ground pin
- **What is it used for?**
 - Provides a stable DC voltage output.
- **In the project:**
 - Used to regulate the dual rail voltage to $\pm 12V$.

• Resistors (Voltage Divider)

- **What are they?** Passive components that limit current and divide voltage.
- **Structure:**
 - Two terminals
- **What is it used for?**
 - Splits voltage into two equal parts to create a virtual ground.
- **In the project:**
 - Used in the simple rail splitter design to create dual rail voltage.

• Operational Amplifier

- **What is it?** A high-gain voltage amplifier used for signal processing.
- **Structure:**
 - Two input pins (Inverting and Non-inverting)

- One output pin
- **What is it used for?**
 - Stabilizes voltage and prevents fluctuations in virtual ground.
- **In the project:**
 - Used to improve the stability of the voltage divider-based rail splitter.

• BUF634 Buffer IC

- **What is it?** A high-speed buffer amplifier that increases current output.
- **Structure:**
 - Input pin
 - Output pin
 - Power supply pins
- **What is it used for?**
 - Provides higher current capability for voltage rail splitting.
- **In the project:**
 - Used to enhance the current output of the operational amplifier-based rail splitter.

• TLE2426 IC

- **What is it?** A precision voltage rail splitter IC.
- **Structure:**
 - Input voltage pin
 - Output voltage pin
 - Ground pin
- **What is it used for?**
 - Creates a stable virtual ground for dual rail voltage applications.
- **In the project:**
 - Used as an alternative to the Op Amp-based rail splitter.

• Transistors (Discrete Rail Splitter)

- **What are they?** Semiconductor devices used for switching and amplification.
- **Structure:**
 - Three terminals: Base, Collector, Emitter
- **What is it used for?**
 - Helps regulate voltage and current flow.
- **In the project:**
 - Used in a discrete transistor-based rail splitter design.



Summary

This video explores different methods of generating dual rail power supplies, which are essential for applications like audio amplifiers, sensors, and operational amplifiers. The charge pump method provides a simple way to create dual rail voltage but has limitations in current output and noise. The transformer-based method offers higher current capability but requires AC mains

voltage. Simpler rail splitter designs using resistors, operational amplifiers, and specialized ICs provide alternative solutions for low-power applications.

VIDEO 53: Mechanical Switches are Obsolete?! Switch to a Latch Circuit! EB#53

List of Components Used

- 74LS279 SR Latch IC
- NOR Gate IC
- 555 Timer IC
- P-Channel MOSFET
- NPN Bipolar Junction Transistors (BJTs)
- Capacitors (Various values)
- Resistors (Various values)
- Latching Relay

Detailed Breakdown of Each Component

• 74LS279 SR Latch IC

- **What is it?** A digital logic IC that implements an SR (Set-Reset) latch.
- **Structure:**
 - Inputs: Set (S) and Reset (R)
 - Outputs: Q (normal) and $\bar{Q}$ (inverted)
- **What is it used for?**
 - Stores a binary state until changed by an input signal.
- **In the project:**
 - Used to demonstrate basic latch functionality with push buttons.

• NOR Gate IC

- **What is it?** A digital logic gate that outputs LOW unless both inputs are LOW.
- **Structure:**
 - Two inputs
 - One output
- **What is it used for?**
 - Forms the basis of an SR latch circuit.
- **In the project:**
 - Used to manually construct an SR latch circuit on a breadboard.

• 555 Timer IC

- **What is it?** A widely used IC that includes an internal SR latch.
- **Structure:**
 - Multiple pins for timing and control

- **What is it used for?**
 - Generates pulses, delays, and oscillations.
- **In the project:**
 - Mentioned as an example of an IC that contains an SR latch.

• P-Channel MOSFET

- **What is it?** A transistor that switches power based on gate voltage.
- **Structure:**
 - Three terminals: Gate, Drain, Source
- **What is it used for?**
 - Controls power flow to a load.
- **In the project:**
 - Used in the toggle latch circuit to switch power on and off.

• NPN Bipolar Junction Transistors (BJTs)

- **What are they?** Semiconductor devices used for switching and amplification.
- **Structure:**
 - Three terminals: Base, Collector, Emitter
- **What is it used for?**
 - Controls current flow based on base voltage.
- **In the project:**
 - Used in the latch circuit to maintain the MOSFET state.

• Capacitors

- **What are they?** Components that store and release electrical energy.
- **Structure:**
 - Two terminals: Positive and Negative
- **What is it used for?**
 - Provides timing delays and stabilizes voltage.
- **In the project:**
 - Used to delay the reset action in the toggle latch circuit.

• Resistors

- **What are they?** Passive components that limit current and divide voltage.
- **Structure:**
 - Two terminals
- **What is it used for?**
 - Controls current flow and voltage levels.
- **In the project:**
 - Used to set bias points for transistors and MOSFETs.

• Latching Relay

- **What is it?** A relay that maintains its state after being triggered.
- **Structure:**
 - Coil and mechanical switching mechanism
- **What is it used for?**
 - Maintains a switched state without continuous power.
- **In the project:**
 - Used in corridor lighting to toggle power with a single button press.



Summary

This video explores latch circuits, which maintain a state until changed by an input signal. It covers digital logic implementations using SR latch ICs and NOR gates, as well as transistor-based latch circuits for power control. The toggle latch circuit uses a MOSFET and BJTs to switch power efficiently, while a latching relay demonstrates mechanical switching. The video also discusses applications like overcurrent protection and microcontroller self-deactivation.

VIDEO 54: The Best Protection for your Circuit is NOT a Fuse!.....but a Resettable Fuse? EB#54



List of Components Used

- Resettable Fuses (PPTC)
- Glass Fuses
- NTC Thermistors
- RGB LED Strip
- Constant Loads
- Big Power Resistor
- Trip Indicator Components



Detailed Breakdown of Each Component

- Resettable Fuses (PPTC)
 - What is it? A resettable fuse, also known as a polymeric positive temperature coefficient (PPTC) thermistor, multi-fuse, polyfuse, or polyswitch, is an electronic component used to protect circuits from overcurrent faults. Unlike traditional fuses, they reset themselves once the fault is removed.
 - Structure:
 - Available in both through-hole and surface-mount technology (SMT) packages.
 - PPTCs are made of a polymeric material with conductive particles. Under normal operating conditions, they have low resistance. When an

overcurrent occurs, the material heats up, causing its resistance to increase dramatically, limiting the current.

- What is it used for? To protect circuits from damage due to overcurrent or short circuits. They are used in various applications to provide self-resetting circuit protection.
- In the project: The video focuses on explaining how resettable fuses work and their advantages over traditional fuses. It demonstrates how they limit current during an overcurrent event and then reset when the fault is removed.
- Short summary: Resettable fuses are self-resetting circuit protection devices that limit current during a fault and return to a low-resistance state once the fault is cleared.
- Glass Fuses
 - What is it? A traditional type of overcurrent protection device that contains a thin wire or filament that melts and breaks the circuit when an overcurrent occurs.
 - Structure: Typically consists of a glass tube with metal end caps and a thin wire or filament inside.
 - What is it used for? To protect circuits from overcurrent conditions. They are a one-time use device and must be replaced after they blow.
 - In the project: Mentioned as a traditional alternative to resettable fuses. The video contrasts the behavior of glass fuses with resettable fuses, highlighting the advantage of resettable fuses in not needing replacement.
 - Short summary: Glass fuses are traditional, one-time-use overcurrent protection devices that break a circuit by melting a filament.
- NTC Thermistors
 - What is it? A Negative Temperature Coefficient (NTC) thermistor is a temperature-sensitive resistor whose resistance decreases as its temperature increases.
 - Structure: Typically made of semiconductor materials.
 - What is it used for? Temperature sensing and control.
 - In the project: Briefly mentioned as related to PPTCs, but commonly used in 3D printers for temperature measurement.
 - Short summary: NTC thermistors are temperature-sensitive resistors used for temperature measurement and control, with resistance decreasing as temperature increases.
- RGB LED Strip
 - What is it? A flexible circuit board populated with multiple RGB (Red, Green, Blue) LEDs.
 - Structure: Contains multiple LEDs and resistors connected in series, often with a flexible adhesive backing.

- What is it used for? To produce colored light in various applications.
 - In the project: Used as a practical example of a circuit that normally draws current and needs protection. The video demonstrates how a resettable fuse can protect the LED strip from overcurrent.
 - Short summary: An RGB LED strip is a flexible light source used to produce colored light, and the video uses it as an example of a circuit requiring overcurrent protection.
- Constant Loads
 - What is it? A device or circuit that draws a consistent amount of current.
 - Structure: Not specifically detailed in the video, but generally refers to a component or combination of components that presents a stable electrical resistance or draws a consistent current from a power source.
 - What is it used for? To simulate a circuit for testing purposes, by providing a predictable current draw.
 - In the project: Used to simulate various current draws for testing the behavior of the resettable fuse under different load conditions.
 - Short summary: Constant loads are used in the video to mimic a circuit for testing the resettable fuse's response to different current levels.
- Big Power Resistor
 - What is it? A resistor designed to dissipate a significant amount of power as heat.
 - Structure: Typically a larger-sized resistor, often ceramic or wire-wound, designed to handle high power ratings. It might have a heatsink or be designed for direct mounting to a heatsink.
 - What is it used for? To dissipate electrical energy as heat, often used as a load in testing power supplies or circuits where significant current needs to be drawn.
 - In the project: Used as a heavy load to investigate the trip current and response time of the resettable fuse.
 - Short summary: A big power resistor is used as a substantial electrical load to test the resettable fuse's behavior under high-current conditions.
- Trip Indicator Components
 - What is it? Components that are used to visually indicate when the resettable fuse has tripped (entered its high-resistance state).
 - Structure: The video doesn't specify exact components but shows them wired in parallel with the fuse. This could imply LEDs, small light bulbs, or other visual indicators along with current-limiting resistors if needed.
 - What is it used for? To provide a clear visual feedback mechanism, making it easy to observe the fuse's protection state.

- In the project: Added to the test circuit to clearly show when the resettable fuse transitions from its low-resistance (normal) state to its high-resistance (tripped) state.
- Short summary: Trip indicator components provide a visual signal when the resettable fuse has activated due to an overcurrent, confirming its protective action.

Summary

The video provides an in-depth explanation of resettable fuses (PPTCs), highlighting their self-resetting nature as a key advantage over traditional glass fuses. It delves into their working principle, explaining how their resistance increases dramatically under overcurrent conditions to protect a circuit. The video demonstrates the practical application of these fuses, using an RGB LED strip as an example of a circuit requiring protection. It also shows testing methodologies with constant loads and a big power resistor to observe the fuse's trip characteristics, enhanced by trip indicator components for clear visual feedback.

VIDEO 55: The Most Important Circuit for our Electrical Future?! (PFC) EB#55

List of Components Used

- Power Factor Correction (PFC) IC
- Boost Converter Circuit
- Full Bridge Rectifier
- Capacitors (Various values)
- Inductors (Various values)
- MOSFET (Switching Element)
- Resistors (Various values)
- Development Board (for PFC testing)

Detailed Breakdown of Each Component

• Power Factor Correction (PFC) IC

- **What is it?** An integrated circuit designed to regulate power factor by shaping input current.
- **Structure:**
 - Input voltage monitoring
 - Current sensing
 - MOSFET control for switching
- **What is it used for?**
 - Improves power factor by reducing reactive power and harmonics.
- **In the project:**

- Used to regulate input current waveform to match mains voltage, reducing power grid inefficiencies.

• Boost Converter Circuit

- **What is it?** A DC-DC converter that steps up voltage from a lower level to a higher level.
- **Structure:**
 - Inductor
 - Switching MOSFET
 - Diode
 - Capacitor
- **What is it used for?**
 - Converts varying AC voltage into a stable DC voltage.
- **In the project:**
 - Forms the core of the active PFC circuit to stabilize voltage and improve efficiency.

• Full Bridge Rectifier

- **What is it?** A circuit made of four diodes that converts AC voltage into DC voltage.
- **Structure:**
 - Four diodes arranged in a bridge configuration
 - Two AC input terminals
 - Two DC output terminals (+ and -)
- **What is it used for?**
 - Converts AC voltage into DC voltage.
- **In the project:**
 - Used before the PFC circuit to rectify mains AC voltage.

• Capacitors

- **What are they?** Components that store and release electrical energy.
- **Structure:**
 - Two terminals: Positive and Negative
- **What is it used for?**
 - Stabilizes voltage and reduces noise.
- **In the project:**
 - Used in the boost converter and PFC circuit to smooth voltage fluctuations.

• Inductors

- **What are they?** Components that store energy in a magnetic field.
- **Structure:**
 - Coil of wire
- **What is it used for?**

- Helps regulate current flow in switching circuits.
- **In the project:**
 - Used in the boost converter to store and transfer energy efficiently.

• MOSFET (Switching Element)

- **What is it?** A transistor that switches power based on gate voltage.
- **Structure:**
 - Three terminals: Gate, Drain, Source
- **What is it used for?**
 - Controls power flow in high-frequency switching circuits.
- **In the project:**
 - Used in the boost converter to regulate voltage conversion.

• Resistors

- **What are they?** Passive components that limit current and divide voltage.
- **Structure:**
 - Two terminals
- **What is it used for?**
 - Controls current flow and voltage levels.
- **In the project:**
 - Used for voltage sensing and feedback control in the PFC circuit.

• Development Board (for PFC testing)

- **What is it?** A pre-built circuit board designed for testing PFC ICs.
- **Structure:**
 - PFC IC
 - Boost converter components
 - Input/output terminals
- **What is it used for?**
 - Allows easy testing of PFC circuits before full implementation.
- **In the project:**
 - Used to test active PFC functionality before integrating into a power supply.

Summary

This video explores Power Factor Correction (PFC) circuits, which improve electrical efficiency by reducing reactive power and harmonics. Passive PFC methods use capacitors and inductors to correct phase shifts, while active PFC circuits employ boost converters and specialized ICs to shape input current. The video demonstrates testing an active PFC circuit using a development board, showing how it improves power factor and reduces grid inefficiencies.

VIDEO 56: These 3 Cent Components are actually USEFUL?! (Color Ring Inductor) EB#56

List of Components Used

- Color Ring Inductor
- SMD Inductor
- Power Inductor (Ferrite Core)
- LCR Meter
- MOSFET (Switching Element)
- Function Generator
- Resistors (Various values)
- Capacitors (Various values)

Detailed Breakdown of Each Component

• Color Ring Inductor

- **What is it?** A type of inductor with a wire wound around a ferrite core, identified using color bands.
- **Structure:**
 - Copper wire wound around a ferrite core
 - Color-coded rings indicating inductance value
- **What is it used for?**
 - Used in signal filtering, oscillators, and low-power applications.
- **In the project:**
 - Tested in a boost converter circuit to evaluate its performance under load.

• SMD Inductor

- **What is it?** A surface-mount inductor used in compact electronic circuits.
- **Structure:**
 - Ferrite core with wire winding inside a small package
- **What is it used for?**
 - Used in power supplies and RF circuits for filtering and energy storage.
- **In the project:**
 - Compared with color ring inductors to analyze performance differences.

• Power Inductor (Ferrite Core)

- **What is it?** A high-current inductor used in power electronics.
- **Structure:**
 - Large ferrite core with thick copper winding
- **What is it used for?**
 - Stores energy in magnetic fields for power conversion circuits.
- **In the project:**

- Used as the reference inductor in the boost converter test.

• LCR Meter

- **What is it?** A device used to measure inductance, capacitance, and resistance.
- **Structure:**
 - Digital display with test probes
- **What is it used for?**
 - Measures inductance values of different inductors.
- **In the project:**
 - Used to verify inductance values of color ring inductors.

• MOSFET (Switching Element)

- **What is it?** A transistor used for high-speed switching in power circuits.
- **Structure:**
 - Three terminals: Gate, Drain, Source
- **What is it used for?**
 - Controls current flow in switching circuits.
- **In the project:**
 - Used in the saturation current test circuit.

• Function Generator

- **What is it?** A device that generates electrical waveforms for testing circuits.
- **Structure:**
 - Adjustable frequency and amplitude settings
- **What is it used for?**
 - Provides pulse signals for testing inductors.
- **In the project:**
 - Used to drive the MOSFET in the saturation current test.

• Resistors

- **What are they?** Passive components that limit current and divide voltage.
- **Structure:**
 - Two terminals
- **What is it used for?**
 - Controls current flow and voltage levels.
- **In the project:**
 - Used in the saturation current measurement circuit.

• Capacitors

- **What are they?** Components that store and release electrical energy.
- **Structure:**

- Two terminals: Positive and Negative
- **What is it used for?**
 - Stabilizes voltage and reduces noise.
- **In the project:**
 - Used in the boost converter circuit to smooth voltage fluctuations.

Summary

This video explores color ring inductors, comparing them with other types like SMD and ferrite-core inductors. The inductors were tested in a boost converter circuit, revealing limitations in current handling and saturation properties. A custom test circuit was built to measure saturation current, showing that color ring inductors saturate at lower currents than standard power inductors. While useful for low-power applications, they are not ideal for high-current circuits.

VIDEO 57: Ground is MORE IMPORTANT than you think! EB#57

List of Components Used

- Soil (as a conductor)
- Metal Plates (for soil conductivity test)
- AC Power Cable (Live, Neutral, and Ground wires)
- Multimeter
- Circuit Breaker (for safety testing)
- Residual Current Breaker (RCB)
- Ground Rod (for earth grounding)
- Electrostatic Charge Generator
- Solar Panels (for grounding demonstration)
- PCB Ground Layer

Detailed Breakdown of Each Component

- **Soil (as a conductor)**
 - **What is it?** Natural earth material that can conduct electricity under certain conditions.
 - **Structure:**
 - Contains minerals, moisture, and organic matter that affect conductivity.
 - **What is it used for?**
 - Used in grounding systems to dissipate excess charge.
 - **In the project:**
 - Demonstrated as a conductor by lighting a bulb using metal plates inserted into the soil.
- **Metal Plates (for soil conductivity test)**
 - **What are they?** Conductive surfaces used to increase contact area with soil.
 - **Structure:**

- Large surface area to improve conductivity.
- **What is it used for?**
 - Helps transfer electrical charge into the soil.
- **In the project:**
 - Used to improve soil conductivity for lighting a bulb.

- **AC Power Cable (Live, Neutral, and Ground wires)**

- **What is it?** Standard electrical cable used in household appliances.
- **Structure:**
 - Live wire (brown)
 - Neutral wire (blue)
 - Protective Earth (PE) wire (green-yellow)
- **What is it used for?**
 - Provides power and safety grounding for appliances.
- **In the project:**
 - Used to demonstrate the role of the ground wire in preventing electrical hazards.

- **Multimeter**

- **What is it?** A device used to measure voltage, current, and resistance.
- **Structure:**
 - Digital display with test probes.
- **What is it used for?**
 - Measures electrical properties in circuits.
- **In the project:**
 - Used to check continuity between ground and appliance chassis.

- **Circuit Breaker (for safety testing)**

- **What is it?** A device that interrupts excessive current flow to prevent damage.
- **Structure:**
 - Switch mechanism with overload protection.
- **What is it used for?**
 - Prevents electrical fires and equipment damage.
- **In the project:**
 - Demonstrated how a short circuit between live and ground trips the breaker.

- **Residual Current Breaker (RCB)**

- **What is it?** A safety device that detects leakage current and shuts off power.
- **Structure:**
 - Monitors current difference between live and neutral.
- **What is it used for?**
 - Protects against electric shocks by disconnecting power when leakage occurs.
- **In the project:**

- Demonstrated how an RCB trips when current flows through the ground wire.

- **Ground Rod (for earth grounding)**

- **What is it?** A metal rod driven into the earth to provide a low-resistance path for electrical grounding.
- **Structure:**
 - Copper or galvanized steel rod.
- **What is it used for?**
 - Dissipates excess charge into the earth.
- **In the project:**
 - Showcased how electrical systems connect to earth for safety.

- **Electrostatic Charge Generator**

- **What is it?** A device that generates static electricity for testing.
- **Structure:**
 - High-voltage generator with discharge terminals.
- **What is it used for?**
 - Demonstrates electrostatic buildup and discharge.
- **In the project:**
 - Used to show how grounding prevents static electricity hazards.

- **Solar Panels (for grounding demonstration)**

- **What are they?** Devices that convert sunlight into electrical energy.
- **Structure:**
 - Photovoltaic cells with metal frames.
- **What is it used for?**
 - Generates electricity while requiring grounding for safety.
- **In the project:**
 - Demonstrated how grounding prevents electrostatic buildup on solar panels.

- **PCB Ground Layer**

- **What is it?** A dedicated layer in printed circuit boards for electrical grounding.
- **Structure:**
 - Continuous copper plane on the PCB bottom layer.
- **What is it used for?**
 - Provides a common reference potential for circuit components.
- **In the project:**
 - Explained how PCB ground layers function similarly to earth grounding.



Summary

This video explores the concept of electrical grounding, demonstrating how soil can conduct electricity and explaining the role of ground wires in AC power systems. It covers safety

mechanisms like circuit breakers and residual current breakers, the importance of earth grounding for electrostatic discharge, and how grounding is applied in PCB designs. The experiments highlight why grounding is essential for electrical safety and stability.

VIDEO 58: This Component solves "All" Motor Problems?! (Motor Encoder) EB#58

List of Components Used

- AS5600 Magnetic Encoder
- Rotary Encoder
- Optical Encoder
- AMT102 Capacitive Encoder
- BLDC Motor
- Stepper Motor
- DC Motor
- Solo Uno Motor Driver
- Microcontroller (Arduino/Raspberry Pi)
- Function Generator
- Oscilloscope
- Resistors (Various values)
- Capacitors (Various values)

Detailed Breakdown of Each Component

• AS5600 Magnetic Encoder

- **What is it?** A contactless magnetic rotary position sensor.
- **Structure:**
 - Uses Hall effect sensors to detect magnet position.
 - Provides 12-bit resolution (4096 steps).
- **What is it used for?**
 - Measures precise angular position of rotating objects.
- **In the project:**
 - Used to track motor shaft position for the smart knob mock-up.
 - Provides high-resolution feedback for precise control.
 - More details available on [ams AS5600 Position Sensor](#).

• Rotary Encoder

- **What is it?** A mechanical device that detects rotation using conductive contacts.
- **Structure:**
 - Two output pins (A and B) generate quadrature signals.
- **What is it used for?**
 - Measures rotation direction and position.
- **In the project:**
 - Compared with the AS5600 to highlight differences in resolution and accuracy.

• Optical Encoder

- **What is it?** A sensor that detects rotation using light and a patterned disc.
- **Structure:**
 - Light source and photodetectors.
- **What is it used for?**
 - Provides precise rotational feedback.
- **In the project:**
 - Used to explain encoder principles and compare with magnetic encoders.

• AMT102 Capacitive Encoder

- **What is it?** A rotary encoder that uses capacitive sensing instead of magnetic or optical methods.
- **Structure:**
 - Capacitive plates detect rotation.
- **What is it used for?**
 - Provides high-resolution position feedback.
- **In the project:**
 - Used with a BLDC motor for precise positioning.

• BLDC Motor

- **What is it?** A brushless DC motor that requires electronic commutation.
- **Structure:**
 - Stator coils and rotor magnets.
- **What is it used for?**
 - Provides high-speed and high-torque motion.
- **In the project:**
 - Controlled using an encoder to mimic stepper motor behavior.

• Stepper Motor

- **What is it?** A motor that moves in discrete steps for precise positioning.
- **Structure:**
 - Multiple coils energized in sequence.
- **What is it used for?**
 - Used in CNC machines and 3D printers.
- **In the project:**
 - Compared with BLDC motors to highlight encoder advantages.

• DC Motor

- **What is it?** A simple motor that runs on direct current.
- **Structure:**
 - Brushes and commutator.
- **What is it used for?**

- Provides continuous rotation.
- **In the project:**
 - Used in a robot where encoders help maintain straight movement.

• Solo Uno Motor Driver

- **What is it?** A high-power motor driver with encoder support.
- **Structure:**
 - MOSFET-based switching circuit.
- **What is it used for?**
 - Controls BLDC motors with precision.
- **In the project:**
 - Used to drive the BLDC motor with encoder feedback.

• Microcontroller (Arduino/Raspberry Pi)

- **What is it?** A programmable device for controlling electronics.
- **Structure:**
 - CPU, memory, I/O pins.
- **What is it used for?**
 - Reads encoder data and controls motors.
- **In the project:**
 - Used to process encoder signals and adjust motor behavior.

• Function Generator

- **What is it?** A device that generates electrical waveforms.
- **Structure:**
 - Adjustable frequency and amplitude.
- **What is it used for?**
 - Provides test signals for circuits.
- **In the project:**
 - Used to analyze encoder output signals.

• Oscilloscope

- **What is it?** A tool for visualizing electrical signals.
- **Structure:**
 - Display screen and probe inputs.
- **What is it used for?**
 - Measures voltage waveforms.
- **In the project:**
 - Used to observe encoder signal patterns.

• Resistors

- **What are they?** Passive components that limit current.

- **Structure:**
 - Two terminals.
- **What is it used for?**
 - Controls voltage levels.
- **In the project:**
 - Used in encoder circuits for signal conditioning.

• Capacitors

- **What are they?** Components that store and release electrical energy.
- **Structure:**
 - Two terminals.
- **What is it used for?**
 - Filters noise and stabilizes voltage.
- **In the project:**
 - Used in motor driver circuits.



Summary

This video explores motor encoders, comparing mechanical rotary encoders with magnetic and capacitive alternatives. The AS5600 magnetic encoder provides high-resolution feedback, enabling precise motor control. The AMT102 capacitive encoder is used with a BLDC motor to achieve stepper-like positioning. The Solo Uno motor driver integrates encoder feedback for accurate speed and torque control. Encoders are essential for robotics, automation, and interactive input devices like the smart knob mock-up.

VIDEO 59: This \$0.70 Component SAVES your Circuit?! (Surge Protection) EB#59



List of Components Used

- TVS Diode (Transient Voltage Suppressor)
- MOV (Metal Oxide Varistor)
- GDT (Gas Discharge Tube)
- Insulation Tester
- Electrostatic Discharge (ESD) Generator
- Microcontroller (Test Circuit)
- Thermal Fuse
- Surge Protection Device (Cheap vs. Expensive)



Detailed Breakdown of Each Component

• TVS Diode (Transient Voltage Suppressor)

- **What is it?** A diode designed to protect circuits from transient voltage spikes.
- **Structure:**
 - Two terminals (Anode and Cathode)

- Clamps voltage at a specific threshold
- **What is it used for?**
 - Protects sensitive electronics from voltage surges.
- **In the project:**
 - Used to protect a microcontroller from high-voltage pulses.
 - Successfully prevented damage from repeated shocks.

• MOV (Metal Oxide Varistor)

- **What is it?** A voltage-dependent resistor that absorbs excess energy.
- **Structure:**
 - Two terminals
 - Non-conductive at normal voltage, conductive at high voltage
- **What is it used for?**
 - Protects circuits from large energy surges.
- **In the project:**
 - Used in the test circuit to absorb high-energy pulses.
 - Found in AC power supplies for surge protection.

• GDT (Gas Discharge Tube)

- **What is it?** A component filled with inert gas that ionizes under high voltage.
- **Structure:**
 - Two terminals
 - Gas-filled chamber
- **What is it used for?**
 - Provides high-voltage surge protection.
- **In the project:**
 - Tested but found unsuitable for low-voltage microcontroller protection.

• Insulation Tester

- **What is it?** A device that generates high voltage to test insulation resistance.
- **Structure:**
 - Adjustable voltage output (125V–1000V)
- **What is it used for?**
 - Checks insulation integrity in electrical systems.
- **In the project:**
 - Attempted to generate controlled overvoltage pulses but was too safe for destruction testing.

• Electrostatic Discharge (ESD) Generator

- **What is it?** A device that creates high-voltage arcs to simulate electrostatic discharge.
- **Structure:**
 - Adjustable arc gap

- **What is it used for?**
 - Simulates real-world electrostatic discharge events.
- **In the project:**
 - Used to generate 15,000V pulses to test protection components.

• Microcontroller (Test Circuit)

- **What is it?** A small computing device used for embedded applications.
- **Structure:**
 - Processor, memory, I/O pins
- **What is it used for?**
 - Controls electronic devices.
- **In the project:**
 - Used as a test subject for overvoltage protection experiments.

• Thermal Fuse

- **What is it?** A fuse that blows when excessive heat is generated.
- **Structure:**
 - Temperature-sensitive material
- **What is it used for?**
 - Provides additional protection in surge suppression circuits.
- **In the project:**
 - Found in AC power supply surge protection setups.

• Surge Protection Device (Cheap vs. Expensive)

- **What is it?** A device that protects electrical systems from voltage surges.
- **Structure:**
 - MOVs, thermal fuses, mechanical disconnect mechanisms
- **What is it used for?**
 - Prevents damage from lightning strikes and power grid fluctuations.
- **In the project:**
 - Compared a cheap surge protector with an industrial-grade one.
 - Found that the expensive version had better safety mechanisms.

Summary

This video explores different overvoltage protection components, including TVS diodes, MOVs, and GDTs. The TVS diode effectively protected a microcontroller from transient voltage spikes, while the MOV absorbed larger energy bursts. The GDT was found unsuitable for low-voltage applications. The video also compared cheap and expensive surge protection devices, revealing that industrial-grade models offer superior safety features.

VIDEO 60: Is This the NEW GOLDEN Standard for Communication? (I3C) EB#60

List of Components Used

- BOS1921 Piezo Haptic Driver IC
- I3C Communication Protocol
- I2C Communication Protocol
- Piezo Actuator
- Capacitors (Various values, including 100μF)
- Inductor (for power regulation)
- Microcontroller (for testing I3C/I2C)
- Development Board (Expensive reference board)
- Oscilloscope (Keysight HD3)
- Thermal Camera (for troubleshooting)

Detailed Breakdown of Each Component

• BOS1921 Piezo Haptic Driver IC

- **What is it?** A specialized IC designed to drive piezo actuators for haptic feedback.
- **Structure:**
 - Supports I3C and I2C communication
 - Requires external capacitors and an inductor for power regulation
- **What is it used for?**
 - Generates precise vibration patterns for haptic feedback.
- **In the project:**
 - Used to create a custom breakout board for affordable haptic feedback integration.

• I3C Communication Protocol

- **What is it?** A next-generation serial communication protocol designed to replace I2C.
- **Structure:**
 - Two-wire interface (SDA and SCL)
 - Push-pull drivers instead of open-drain
 - Supports dynamic addressing and in-band interrupts
- **What is it used for?**
 - Enables faster and more efficient communication between ICs.
- **In the project:**
 - Investigated as an alternative to I2C for controlling the BOS1921 IC.
 - Compared with I2C to determine practical advantages.

• I2C Communication Protocol

- **What is it?** A widely used serial communication protocol for embedded systems.
- **Structure:**

- Two-wire interface (SDA and SCL)
 - Uses pull-up resistors for signal integrity
- **What is it used for?**
 - Enables communication between microcontrollers and peripherals.
- **In the project:**
 - Used as a fallback communication method for the BOS1921 IC.
 - Compared with I3C to evaluate speed and efficiency differences.

• Piezo Actuator

- **What is it?** A device that converts electrical signals into mechanical vibrations.
- **Structure:**
 - Thin ceramic material that expands and contracts with voltage changes
- **What is it used for?**
 - Provides haptic feedback in touch interfaces.
- **In the project:**
 - Connected to the breakout board to test vibration patterns.

• Capacitors (Various values, including 100μF)

- **What are they?** Components that store and release electrical energy.
- **Structure:**
 - Two terminals: Positive and Negative
- **What is it used for?**
 - Stabilizes voltage and filters noise.
- **In the project:**
 - Used in the breakout board for power regulation.
 - Incorrect polarity caused initial failure, later corrected.

• Inductor (for power regulation)

- **What is it?** A coil that stores energy in a magnetic field.
- **Structure:**
 - Wire wound around a core
- **What is it used for?**
 - Helps regulate power delivery in circuits.
- **In the project:**
 - Positioned close to the BOS1921 IC for optimal performance.

• Microcontroller (for testing I3C/I2C)

- **What is it?** A programmable device used to control electronic components.
- **Structure:**
 - CPU, memory, I/O pins
- **What is it used for?**
 - Interfaces with sensors and actuators.

- **In the project:**
 - Used to test communication with the BOS1921 IC via I3C and I2C.

- **Development Board (Expensive reference board)**

- **What is it?** A pre-built circuit board designed for testing the BOS1921 IC.
- **Structure:**
 - Includes all necessary components for haptic feedback testing
- **What is it used for?**
 - Provides a reference design for custom breakout boards.
- **In the project:**
 - Used to validate the functionality of the custom breakout board.

- **Oscilloscope (Keysight HD3)**

- **What is it?** A tool for visualizing electrical signals.
- **Structure:**
 - High-resolution display with advanced signal analysis features
- **What is it used for?**
 - Measures voltage waveforms and signal integrity.
- **In the project:**
 - Used to analyze I3C and I2C communication signals.

- **Thermal Camera (for troubleshooting)**

- **What is it?** A device that detects heat signatures in electronic components.
- **Structure:**
 - Infrared sensor with digital display
- **What is it used for?**
 - Identifies overheating components in circuits.
- **In the project:**
 - Used to diagnose incorrect capacitor placement in the breakout board.

Summary

This video explores the creation of a custom breakout board for the BOS1921 haptic driver IC, comparing I3C and I2C communication protocols. I3C offers higher speed, dynamic addressing, and in-band interrupts, but requires specialized microcontrollers. The breakout board successfully replicated the functionality of an expensive development board, enabling affordable haptic feedback integration.

VIDEO 61: The World's Simplest Audio Amp just got BETTER?! (MOSFET Amp) EB#61

List of Components Used

- Bipolar Junction Transistor (BJT)
- Loudspeaker
- TIP142 Darlington Transistor
- Resistors (various power ratings and resistances)
- Heatsink
- Electrolytic Capacitors (10uF, 4700uF)
- IRFZ44N MOSFET
- Potentiometer
- Function Generator
- Oscilloscope
- nRF54L Series System-on-Chips (SoC)

Detailed Breakdown of Each Component

- Bipolar Junction Transistor (BJT)
 - What is it? A Bipolar Junction Transistor (BJT) is a three-terminal semiconductor device used for amplifying or switching electronic signals and electrical power.
 - Structure: Has three terminals: Base (B), Collector (C), and Emitter (E). A small current applied to the Base controls a larger current flow between the Collector and Emitter.
 - What is it used for? Amplification of signals (e.g., in audio amplifiers) and switching applications.
 - In the project: This was the core active component in the "original" simple audio amplifier design. The video shows the limitations of a standard BJT for driving a loudspeaker due to its current handling capacity and voltage drop. It is later replaced by a Darlington transistor and then a MOSFET for comparison.
 - Short summary: A BJT is a semiconductor device that amplifies or switches signals, where a small base current controls a larger collector-emitter current.
- Loudspeaker
 - What is it? An electroacoustic transducer that converts an electrical audio signal into corresponding sound waves.

- Structure: Consists of a permanent magnet, a voice coil, and a diaphragm (cone). The electrical signal causes the voice coil to move in the magnetic field, which in turn moves the diaphragm to produce sound.
- What is it used for? To convert electrical audio signals into audible sound.
- In the project: Used as the output device for all amplifier circuits demonstrated, allowing the sound quality and loudness to be evaluated.
- Short summary: A loudspeaker converts electrical audio signals into sound waves, serving as the output for the amplifiers in the video.
- TIP142 Darlington Transistor
 - What is it? A Darlington transistor (or Darlington pair) is a compound structure consisting of two bipolar transistors configured in such a way that the current amplified by the first transistor is further amplified by the second. This results in a very high current gain.
 - Structure: Internally consists of two BJTs connected where the emitter of the first transistor is connected to the base of the second transistor, and their collectors are joined. It has a Base, Collector, and Emitter pin, similar to a single BJT, but with a higher overall current gain (H_{fe}). The TIP142 specifically is an NPN power Darlington transistor.
 - What is it used for? Applications requiring very high current gain or where a small input current needs to control a large output current, such as power amplifiers, motor drivers, and power regulators.
 - In the project: Used as an upgrade from the single BJT to handle more current (up to 10A) for driving a proper loudspeaker. The video demonstrates its higher voltage drop compared to a single BJT and its use in a Class A amplifier configuration.
 - Short summary: A Darlington transistor is a pair of BJTs connected for extremely high current gain, used in the video to boost the audio amplifier's power output.
- Resistors (various power ratings and resistances)
 - What is it? Passive two-terminal electrical components that implement electrical resistance as a circuit element.
 - Structure: Can be carbon film, metal film, wire-wound, etc., with two leads for connection. They are rated by their resistance value (Ohms) and power dissipation capacity (Watts).
 - What is it used for? To limit current, divide voltage, bias transistors, and dissipate unwanted energy as heat in electronic circuits.
 - In the project:
 - **R_c Resistor (Collector Resistor):** Used in the Darlington amplifier circuit to set the quiescent operating point and provide a path for the collector current. Its value and power rating were recalculated to handle the increased current and voltage.

- **Emitter Resistor:** Used to provide current stabilization and feedback for the transistor, its value and power rating were also recalculated for the higher current.
 - **Biasing Resistors:** Used to set the DC operating point (bias) of the BJT and Darlington transistors, allowing a small base current (for BJTs/Darlington) or gate voltage (for MOSFETs) to control a larger current. Their values were adjusted for the Darlington transistor's higher current gain.
- Short summary: Resistors are fundamental components used to limit current, divide voltage, and bias active components like transistors in the amplifier circuits.
- Heatsink
 - What is it? A passive heat exchanger that transfers heat generated by an electronic or mechanical device to a fluid medium, often air or¹ a liquid coolant.
 - Structure: Typically made of a highly thermally conductive metal, such as aluminum alloys or copper, often with fins or a large surface area to maximize heat dissipation into the surrounding air.
 - What is it used for? To dissipate excess heat from electronic components (like transistors) to prevent overheating and thermal damage, ensuring stable and reliable operation.
 - In the project: Added to the Darlington transistor and later the MOSFET because these components were getting too hot due to increased current flow and power dissipation in the amplifier circuit, indicating their crucial role in thermal management.
 - Short summary: A heatsink is a metal component used to dissipate excess heat from power-hungry components like transistors, preventing them from overheating.
- Electrolytic Capacitors (10uF, 4700uF)
 - What is it? A type of capacitor that uses an electrolyte to achieve a larger capacitance per unit volume than other capacitor types. They are polarized, meaning they must be connected with the correct polarity.
 - Structure: Typically cylindrical with two leads, one marked for polarity (usually negative). They are rated by capacitance (Farads) and maximum voltage.
 - What is it used for? Filtering, smoothing, energy storage, and signal coupling/decoupling in DC circuits. In audio circuits, they are often used as coupling capacitors to block DC while allowing AC audio signals to pass, or for power supply filtering.
 - In the project:
 - **10uF output capacitor:** Initially used as the output coupling capacitor.
 - **4700uF output capacitor:** Replaced the 10uF capacitor to provide a "massive boost" in audio quality and loudness, indicating its role in improving low-frequency response and power delivery to the loudspeaker.

- Short summary: Electrolytic capacitors are polarized components used for filtering and signal coupling; a larger one was used in the project's output stage to improve audio quality.
- IRFZ44N MOSFET
 - What is it? An N-channel power Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) commonly used for switching and amplification in power electronics. Unlike BJTs, MOSFETs are voltage-controlled devices.
 - Structure: Has three terminals: Gate (G), Drain (D), and Source (S). A voltage applied to the Gate controls the current flow between the Drain and Source. The IRFZ44N is a common TO-220 package power MOSFET.
 - What is it used for? High-speed switching applications, power supplies, motor control, and power amplification, due to their low ON-resistance and high current handling capability.
 - In the project: Explored as an alternative to the Darlington transistor for the audio amplifier. The video investigates its characteristics for amplification, noting its very low voltage drop when fully turned on. The challenge of biasing a MOSFET for linear amplification is highlighted.
 - Short summary: The IRFZ44N is a power MOSFET, a voltage-controlled transistor explored in the video as a potentially more efficient amplifier than the Darlington due to its low ON-resistance.
- Potentiometer
 - What is it? A three-terminal resistor with a sliding or rotating contact that forms an adjustable voltage divider.
 - Structure: Has three terminals: two end terminals connected to the resistive track, and a third terminal connected to the wiper (the movable contact).
 - What is it used for? To manually vary resistance or to produce a variable voltage output. In audio applications, they are commonly used for volume control.
 - In the project: Used in the MOSFET amplifier circuit to find the "sweet point" (optimal DC offset voltage) for the gate, allowing the audio signal to be amplified without clipping.
 - Short summary: A potentiometer is a variable resistor used in the MOSFET amplifier to manually adjust the bias voltage for optimal audio amplification.
- Function Generator
 - What is it? An electronic test equipment that generates various types of electrical waveforms over a wide range of frequencies.
 - Structure: Contains circuitry to produce standard waveforms like sine, square, triangle, and sometimes arbitrary waveforms. It has controls for waveform type, frequency, amplitude, and DC offset.
 - What is it used for? To provide a precise input signal for testing electronic circuits and systems, such as amplifiers, filters, and digital logic.

- In the project: Used to generate a sine wave (20Hz to 20kHz) to test the linearity and distortion of the MOSFET and Darlington amplifiers, specifically in conjunction with the oscilloscope's FFT function.
- Short summary: A function generator is a test instrument used to create sine waves for analyzing the distortion and performance of the audio amplifiers.
- Oscilloscope
 - What is it? An electronic test instrument that graphically displays varying electrical signals, typically as a two-dimensional plot of one or more electrical potential differences (voltage) as a function of time.
 - Structure: Features include input channels (probes), a display screen, controls for vertical scale (Volts/Div), horizontal scale (Time/Div), triggering, and various measurement and analysis functions (e.g., FFT).
 - What is it used for? To visualize, measure, and analyze electrical signals for debugging, design, and repair of electronic circuits. It helps in understanding waveform shape, frequency, amplitude, and distortion.
 - In the project: Used extensively to visualize the audio waveforms, analyze clipping in the Darlington amplifier, confirm proper biasing, and critically, to perform Fast Fourier Transform (FFT) analysis to measure harmonic distortions in both the Darlington and MOSFET amplifiers.
 - Short summary: An oscilloscope is a vital tool used to visualize and analyze electrical signals, crucial in the project for debugging amplifier performance and measuring audio distortion.
- nRF54L Series System-on-Chips (SoC)
 - What is it? A series of ultra-low power System-on-Chips (SoC) from Nordic Semiconductor, featuring an integrated microcontroller and multi-protocol radio.
 - Structure: Integrates a microcontroller (processor), memory, and wireless communication capabilities (Bluetooth LE, Thread, Matter, Zigbee) onto a single chip. Specific pin configurations are not detailed in the video, as it's a sponsored mention.
 - What is it used for? Designing low-power wireless applications requiring communication protocols like Bluetooth LE, Thread, Matter, and Zigbee. Used in IoT devices, wearables, smart home devices, etc.
 - In the project: Mentioned as a sponsor and a modern component, but not actively used in the amplifier build itself. It's presented as a solution for wireless connectivity in future projects.
 - Short summary: The nRF54L Series is a low-power SoC with an integrated microcontroller and multi-protocol radio, advertised as a solution for wireless connectivity in modern electronics.

Summary

The video explores upgrading a simple BJT-based audio amplifier to drive a loudspeaker, comparing a Darlington transistor with a MOSFET. Initially, a TIP142 Darlington transistor replaces the BJT, requiring recalculation of resistor values and the addition of a heatsink due to increased current and power dissipation. The video then experiments with an IRFZ44N MOSFET, highlighting its lower voltage drop but also the challenge of linear biasing, solved with a potentiometer. Throughout the process, an oscilloscope is crucial for analyzing waveforms and detecting clipping. Finally, a detailed comparison using a function generator and the oscilloscope's FFT function reveals that while the MOSFET amplifier can be louder, BJTs/Darlington transistors generally offer better linearity and lower total harmonic distortion for audio amplification.

VIDEO 62: Not a Microcontroller!...This is Better?! (PLC) EB#62

List of Components Used

- Arduino Opta PLC
- Digital Expansion Module
- Conveyor Belt System
- Industrial Switch Board (Start/Stop Buttons)
- LEDs (System Status Indicators)
- DC Motor (for Conveyor Belt)
- Capacitive Proximity Sensor
- Inductive Proximity Sensor
- Power Supply (5V for Motor, 24V for PLC)

Detailed Breakdown of Each Component

• Arduino Opta PLC

- **What is it?** A programmable logic controller (PLC) designed for industrial automation.
- **Structure:**
 - Supports IEC 61131-3 programming languages
 - 4 relay outputs (250V AC, 10A)
 - Digital and analog inputs (0-24V digital, 0-10V analog)
- **What is it used for?**
 - Controls industrial automation systems with high reliability.
- **In the project:**
 - Used to automate a conveyor belt system with sensors and logic control.
 - Programmed using Function Block Diagram (FBD) in the Arduino PLC IDE.
 - More details available on [Arduino Opta Hardware](#).

• Digital Expansion Module

- **What is it?** An add-on module that extends the PLC's input/output capabilities.
- **Structure:**
 - Additional digital inputs and outputs
- **What is it used for?**
 - Expands the number of controllable devices in an automation system.
- **In the project:**
 - Used to integrate more sensors and actuators into the conveyor belt system.

• Conveyor Belt System

- **What is it?** A mechanical system that moves objects along a defined path.
- **Structure:**
 - Belt driven by a DC motor
- **What is it used for?**
 - Automates material handling in industrial applications.
- **In the project:**
 - Controlled by the PLC to move objects and detect metal contaminants.

• Industrial Switch Board (Start/Stop Buttons)

- **What is it?** A panel with physical buttons for manual control.
- **Structure:**
 - Two push buttons (Start and Stop)
- **What is it used for?**
 - Allows manual control of the conveyor belt system.
- **In the project:**
 - Connected to the PLC as input signals to start and stop the conveyor.

• LEDs (System Status Indicators)

- **What are they?** Light-emitting diodes used for visual feedback.
- **Structure:**
 - Red and green indicator lights
- **What is it used for?**
 - Displays system status (running, stopped, error).
- **In the project:**
 - Controlled by the PLC to indicate conveyor belt operation.

• DC Motor (for Conveyor Belt)

- **What is it?** A motor that runs on direct current to drive mechanical systems.
- **Structure:**
 - 5V power supply
- **What is it used for?**
 - Moves the conveyor belt at a controlled speed.
- **In the project:**
 - Powered by the PLC to transport objects along the belt.

• Capacitive Proximity Sensor

- **What is it?** A sensor that detects objects based on their dielectric properties.
- **Structure:**
 - Detects non-metallic objects
- **What is it used for?**
 - Identifies objects passing on the conveyor belt.
- **In the project:**
 - Used to detect general objects moving along the conveyor.

• Inductive Proximity Sensor

- **What is it?** A sensor that detects metallic objects using electromagnetic fields.
- **Structure:**
 - Detects only metal objects
- **What is it used for?**
 - Identifies metallic contaminants in industrial processes.
- **In the project:**
 - Used to detect metal objects and trigger conveyor reversal.

• Power Supply (5V for Motor, 24V for PLC)

- **What is it?** A regulated power source for electronic components.
- **Structure:**
 - 5V for motor
 - 24V for PLC and sensors
- **What is it used for?**
 - Provides stable voltage for system operation.
- **In the project:**
 - Ensures reliable power delivery to all components.

**Summary**

This video explores the use of the Arduino Opta PLC to automate a conveyor belt system, comparing PLCs with microcontrollers. The PLC was programmed using Function Block Diagram (FBD) in the Arduino PLC IDE, allowing easy logic implementation. Sensors detect objects and trigger conveyor movement, while LEDs indicate system status. The video highlights the advantages of PLCs in industrial automation, including reliability, standardized inputs/outputs, and live debugging.